Absorption cutoff and stationary singularities for rounded Gaussian random dynamical systems

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Aug 13, 2026

Abstract

We study Gaussian random dynamical systems with coordinatewise tanh nonlinearity and nearest-grid rounding. Gaussian symmetry reduces the dynamics to an exact Markov chain for the normalized squared radius. Rounding makes the origin absorbing, and the total variation distance to the absorbing equilibrium equals the survival probability of the absorption time. At fixed width, we prove an absorption cutoff with Gaussian profile as the mesh tends to zero; its threshold is determined by the mean log-radius increment. At fixed precision, we prove a large-dimension absorption cutoff under global contraction of the rounded squared-radius mean map, and we obtain metastability when this map has a positive-drift component. In the supercritical regime, we prove a large-dimension cutoff to a nonzero invariant law and show that, at fixed dimension, its mass near the repelling origin has a power-law asymptotic. All six main results are formalized in Lean 4 on top of Mathlib. The production library contains no `sorry` and declares no custom axiom. Each result is also independently checked against a restatement written in Mathlib vocabulary alone.

Keywords: cutoff; absorption; finite precision; Gaussian Markov chains; metastability; renewal tails; random neural networks.

2020 Mathematics Subject Classification. Primary 60J05, 60K05; secondary 60G50, 37H15, 68T07.

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1. Introduction

We study cutoff for nonlinear Markov chains obtained by rounding random iterated maps to a finite grid. The grid makes exact absorption possible. In the absorbing regimes considered below, the total variation distance to equilibrium equals the survival probability of the absorption time. The cutoff problem therefore reduces to sharp first passage for a random radius.

Random neural-network dynamics give one instance of this problem. The scale of the weights is usually chosen so that activations and gradients do not collapse or grow uncontrollably with depth. This principle underlies the Xavier–Glorot initialization [GB10] and its rectifier-specific analogue [HZRS15]. In an ideal real-valued model, signal propagation is commonly formulated through moments, correlations, or mean-field recursions [PLR+16] and [SGGSD17]. Finite-precision arithmetic instead restricts numerical values to a finite set of machine numbers [Hig02]. The zero state may then be reached exactly, so signal loss can occur abruptly as depth increases. This exact absorption gives a finite-precision mechanism for the loss of long-term signal associated with vanishing gradients in recurrent dynamics [BSF94]. In mean-field feedforward theory, ordered and chaotic phases correspond to vanishing and exploding gradients, respectively [SGGSD17].

We consider the cutoff phenomenon arising in the finite-precision neural-network dynamics studied in joint work with Karlsson [AK22]. In that model one iterates random fully connected layers whose weight matrices have independent entries distributed uniformly on $[-N^{-1/2}, N^{-1/2}]$:

$$X_{t+1}^{(N)} = \tanh\left(W_{t+1}^{(N)} X_t^{(N)}\right), \qquad X_t^{(N)} \in [-1,1]^N, \tag{1.1}$$

where tanh acts coordinatewise. The cited simulations were performed at finite precision, so the numerical state space is finite. The origin is absorbing. If the weights have a continuous law, the unrounded vector chain started away from zero does not hit the origin at any fixed deterministic time. Convergence to δ_0 is therefore not visible in total variation. The rounded vector chain can be absorbed exactly, and its total variation distance to δ_0 equals the survival probability of the absorption time. At the fixed rounding precision 0.001, the numerical experiments in dimensions 1 and 2 show a sharp transition in the depth [AK22]. The results below quantify how the cutoff location changes with the width and the mesh size. We use cutoff in the standard sense of an abrupt transition in distance to equilibrium along a sequence of Markov chains [Dia96]; see also [LP17].

1.1. Gaussian rounded model.

We replace the bounded weights in the motivating simulations by independent Gaussian weights with gain parameter $A > 0$:

$$(\mathsf{W}_{t,A}^{(N)})_{ij} \sim \mathcal{N}(0, A^2/N), \qquad 1 \leqslant i,j \leqslant N, \quad t \geqslant 1. \tag{1.2}$$

For $x \in [-1,1]^N$, the unrounded vector chain is

$$X_0^{(N)} = x, \qquad X_{t+1}^{(N)} = \tanh\left(\mathsf{W}_{t+1,A}^{(N)} X_t^{(N)}\right). \tag{1.3}$$

If the present state has normalized squared radius $q = N^{-1}\|x\|_2^2$, then one preactivation coordinate has variance $A^2 q$. Since $\tanh u = u + O(u^3)$ at the origin, the linearized large-width small-signal variance is multiplied by A^2. Below the corresponding threshold $A = 1$, small signals contract toward the absorbing state; above it, the origin is linearly unstable. Boundedness of tanh prevents unbounded growth and produces a nonzero attracting radius.

For a mesh size $\varrho > 0$, let Q_ϱ denote coordinatewise nearest-grid rounding to $\varrho\mathbb{Z}^N$, with ties rounded toward the grid point of smaller absolute value. The rounded vector chain is

$$Y_0^{(N)} = Q_\varrho x, \qquad Y_{t+1}^{(N)} = Q_\varrho\left(\tanh\left(\mathsf{W}_{t+1,A}^{(N)} Y_t^{(N)}\right)\right), \tag{1.4}$$

and we write $P_{\varrho,A,N}$ for its rounded vector transition kernel. The unrounded and rounded updates are examples of iterated random functions; see Diaconis and Freedman [DF99] for invariant laws and convergence under average contraction. The point 0 is absorbing. With

$$\tau_\varrho^{(N)} := \inf\{t \geqslant 0 : Y_t^{(N)} = 0\}, \tag{1.5}$$

the total variation distance to the absorbing equilibrium is exactly

$$\|P_{\varrho,A,N}^t(Q_\varrho x, \cdot) - \delta_0\|_{\mathrm{TV}} = \mathbb{P}(\tau_\varrho^{(N)} > t). \tag{1.6}$$

Thus finite-precision cutoff to δ_0 is an absorption-time question. The full construction, including the rounding convention, is given in Section 2.

The Gaussian replacement is convenient for analysis. It reduces the dynamics to the normalized squared radius

$$Z_t^{(N)} := N^{-1}\|X_t^{(N)}\|_2^2$$

which is itself a Markov chain. If $G_1, \ldots, G_N$ are independent standard normal variables, its transition is given by

$$Z_{t+1}^{(N)} \overset{d}{=} \frac{1}{N}\sum_{i=1}^{N} \tanh^2\Big(A\sqrt{Z_t^{(N)}}\, G_i\Big). \tag{1.7}$$

We denote the unrounded squared-radius transition kernel by $K_{A,N}$. The unrounded squared-radius chain contains both contraction by the activation function and random expansion by the weights.

The stability threshold depends on the scale and asymptotic regime. For a standard normal variable G, the unrounded squared-radius mean map is defined as

$$V_A(q) := \mathbb{E}\tanh^2(A\sqrt{q}\, G). \tag{1.8}$$

Since $\tanh u = u + O(u^3)$ near zero, we have $V_A(q) = A^2 q + O(q^2)$ for $q > 0$ small, so the large-width variance threshold is $A = 1$. Strict concavity of V_A gives a unique nonzero attracting fixed point q_* when $A > 1$.

At fixed width, the relevant small-radius quantity is the mean one-step change in the logarithmic radius. If χ_N denotes the Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$, the radius multiplier near zero is $(A/\sqrt{N})\chi_N$, and the corresponding log-radius increment is $\log((A/\sqrt{N})\chi_N)$. The fixed-width threshold is therefore

$$A_c(N) = \sqrt{\frac{N}{2}}\, \exp\Big\{-\frac{1}{2}\psi\Big(\frac{N}{2}\Big)\Big\}, \tag{1.9}$$

where ψ is the digamma function. Equivalently, $A_c(N)$ is the solution of $\mathbb{E}\log((A/\sqrt{N})\chi_N) = 0$. It is strictly larger than one for finite N and decreases to one as $N \to \infty$.

At fixed mesh size, the dynamics are governed by the rounded squared-radius mean map

$$V_{A,\varrho}(h) := \mathbb{E}\Big[Q_1\Big(\varrho^{-1}\tanh(\varrho A\sqrt{h}\, G)\Big)^2\Big]. \tag{1.10}$$

where Q_1 is nearest-integer rounding, with ties rounded toward the integer of smaller absolute value. Since $|\tanh z| \leqslant |z|$, this map is bounded above by the unit-grid lattice comparison map

$$L_A(h) := \mathbb{E}[Q_1(A\sqrt{h}\, G)^2].$$

The lattice-subcritical threshold is

$$A_{\mathrm{lat}}^2 := \inf_{\alpha>0} \frac{\alpha^2}{\mathbb{E}[Q_1(\alpha G)^2]}. \tag{1.11}$$

The condition $A < A_{\mathrm{lat}}$ is equivalent to $L_A(h) < h$ for every $h > 0$. In this case, both L_A and $V_{A,\varrho}$ contract globally toward zero. The rightmost positive-drift component on which $V_{A,\varrho}(h) > h$ produces a rounded metastable well. Thus the relevant threshold is determined by whether the linearized radius or rounded squared-radius chain drifts toward zero or supports a persistent nonzero state.

The cutoff results concern the limits $\varrho \downarrow 0$ at fixed width and $N \to \infty$ at fixed gain. In the subcritical fixed-width problem, the relevant distance is

$$\|P_{\varrho,A,N}^t(Q_\varrho x, \cdot) - \delta_0\|_{\mathrm{TV}}.$$

In the supercritical large-dimension problem, the origin is repelling and the unrounded vector chain has a nonzero invariant law. The relevant distance is then total variation distance to this law. Here and below, nonzero means that the invariant probability law assigns zero mass to the absorbing state; mixtures with δ_0 are excluded.

Definition 1.1 (Total variation cutoff)**.** Let $\mathcal{I} \subset \mathbb{R}$, and let $r_* \in [-\infty, \infty]$ belong to the closure of $\mathcal{I}$ in the extended real line. For each $r \in \mathcal{I}$, let P_r be a Markov kernel on a measurable space E_r, let π_r be an invariant probability measure, and let $x_r \in E_r$. Set

$$d_r(t) := \|P_r^t(x_r, \cdot) - \pi_r\|_{\mathrm{TV}}, \qquad t \in \mathbb{N}_0. \tag{1.12}$$

For $u \in \mathbb{R}$, write $\lfloor u \rfloor_+ := \max\{0, \lfloor u \rfloor\}$. Let $t_r \to \infty$ and $a_r = o(t_r)$ with $a_r > 0$. We say that $(P_r, x_r, \pi_r)_{r \in \mathcal{I}}$ has total variation cutoff at time t_r with window a_r if

$$\lim_{c\to\infty} \liminf_{r\to r_*} d_r(\lfloor t_r - c a_r \rfloor_+) = 1, \qquad \lim_{c\to\infty} \limsup_{r\to r_*} d_r(\lfloor t_r + c a_r \rfloor_+) = 0. \tag{1.13}$$

A function $F : \mathbb{R} \to [0, 1]$ is a cutoff profile on the scale a_r if, for every $c \in \mathbb{R}$,

$$d_r(\lfloor t_r + c a_r \rfloor_+) \to F(c), \qquad r \to r_*, \tag{1.14}$$

where $F(c) \to 1$ as $c \to -\infty$ and $F(c) \to 0$ as $c \to \infty$.

When no window is specified, cutoff at time t_r means that, for every fixed $\delta \in (0, 1)$,

$$d_r(\lfloor (1-\delta) t_r \rfloor_+) \to 1, \qquad d_r(\lfloor (1+\delta) t_r \rfloor_+) \to 0, \qquad r \to r_*. \tag{1.15}$$

Equivalently, there exists $b_r = o(t_r)$ with $b_r > 0$ such that the family has cutoff at time t_r with window b_r.

Definition 1.2 (Mixing time)**.** For $\varepsilon \in (0, 1)$, define

$$t_{\mathrm{mix}}^{(r)}(\varepsilon) := \inf\{t \in \mathbb{N}_0 : \ d_r(t) \leqslant \varepsilon\}, \tag{1.16}$$

where the infimum is ∞ if the set is empty.

Remark 1.3. The function d_r is nonincreasing. Hence cutoff with window a_r implies that, for every fixed $\varepsilon \in (0, 1)$,

$$t_{\mathrm{mix}}^{(r)}(\varepsilon) = t_r + O_\varepsilon(a_r), \qquad r \to r_*. \tag{1.17}$$

In particular, $t_{\mathrm{mix}}^{(r)}(\varepsilon)/t_r \to 1$. In the window-free convention, this becomes

$$t_{\mathrm{mix}}^{(r)}(\varepsilon) = t_r + o(t_r). \tag{1.18}$$

If $a_r \to \infty$ and the profile F is continuous and strictly decreasing at the level ε, let c_ε be the unique solution of $F(c_\varepsilon) = \varepsilon$. Then

$$t_{\mathrm{mix}}^{(r)}(\varepsilon) = t_r + c_\varepsilon a_r + o(a_r).$$

1.2. Main results. The notation and detailed hypotheses are given in Section 2. We write Φ_{G} for the distribution function of a standard Gaussian random variable.

For fixed $N \geqslant 1$, $0 < A < A_c(N)$, deterministic $x_0 \in \mathbb{R}^N \backslash \{0\}$, and $0 < \varrho < R_0 := \|x_0\|_2$, set

$$L_\varrho := \log \frac{R_0}{\varrho}, \qquad \gamma_{A,N} := \log \frac{A_c(N)}{A}, \qquad \sigma_N^2 := \mathrm{Var}(\log \chi_N) = \frac{1}{4}\psi_1\left(\frac{N}{2}\right),$$

where $\psi_1 = \psi'$ is the trigamma function. For $a \in \mathbb{R}$, define the nonnegative integer

$$t_\varrho(a) := \max\left\{0, \left\lfloor \frac{L_\varrho}{\gamma_{A,N}} + a\, \frac{\sigma_N}{\gamma_{A,N}^{3/2}} \sqrt{L_\varrho} \right\rfloor\right\}.$$

Theorem 1.4 (Vanishing-mesh cutoff). *Assume $0 < A < A_c(N)$. Then, for every fixed $a \in \mathbb{R}$,*

$$\|P^{t_\varrho(a)}_{\varrho,A,N}(Q_\varrho x_0, \cdot) - \delta_0\|_{\mathrm{TV}} = \mathbb{P}(\tau^{(N)}_\varrho > t_\varrho(a)) \to \Phi_{\mathrm{G}}(-a), \qquad \varrho \downarrow 0. \tag{1.19}$$

In particular, the rounded vector chain has total variation cutoff to δ_0 at time $L_\varrho/\gamma_{A,N}$ with window $\sigma_N \gamma_{A,N}^{-3/2}\sqrt{L_\varrho}$ and Gaussian profile $a \mapsto \Phi_{\mathrm{G}}(-a)$.

Remark 1.5. At fixed width, the cutoff window is determined by first-passage fluctuations of the log-radius increments $\log((A/\sqrt{N})\chi_N)$. At fixed mesh and growing dimension, coordinate averaging suppresses the one-step fluctuations, and the cutoff is determined by the rounded squared-radius mean map $V_{A,\varrho}$.

For $A > 0$, $\varrho \in (0, 1)$, and deterministic $x_N \in [-1, 1]^N$, define

$$\bar{h}_{0,N} := \frac{1}{N\varrho^2}\|Q_\varrho x_N\|_2^2, \qquad \bar{h}_{t+1,N} := V_{A,\varrho}(\bar{h}_{t,N}),$$

and

$$\bar{t}_N := \inf\left\{t \geqslant 0 : \bar{h}_{t,N} \leqslant \frac{1}{\log N}\right\}.$$

Theorem 1.6 (Subcritical dimension cutoff). *Assume $A < A_{\mathrm{lat}}$ and that the rounded initial radii are macroscopic:*

$$\liminf_{N\to\infty} \bar{h}_{0,N} > 0.$$

Then $\bar{t}_N \to \infty$ and

$$\|P^{\bar{t}_N - 1}_{\varrho,A,N}(Q_\varrho x_N, \cdot) - \delta_0\|_{\mathrm{TV}} \to 1, \qquad \|P^{\bar{t}_N + 2}_{\varrho,A,N}(Q_\varrho x_N, \cdot) - \delta_0\|_{\mathrm{TV}} \to 0.$$

Thus the rounded vector chain has total variation cutoff to δ_0 at the exact deterministic rounded-map center $\bar{t}_N$, with a bounded window.

Remark 1.7. Put

$$b_\varrho := \varrho^{-1}\operatorname{arctanh}(\varrho/2), \qquad \kappa_{A,\varrho} := \frac{b_\varrho^2}{2A^2}.$$

At radius $h = (\log N)^{-1}$, the expected number of coordinates outside the zero bin is

$$2N\mathbb{P}\left(G > \frac{b_\varrho\sqrt{\log N}}{A}\right) \asymp \frac{N^{1-\kappa_{A,\varrho}}}{\sqrt{\log N}}.$$

Thus, when $\kappa_{A,\varrho} < 1$, entrance below $(\log N)^{-1}$ does not uniformly imply absorption in the next step. One further deterministic iterate is $o((\log N)^{-1})$, and the following update absorbs with probability tending to one. The upper time improves from $\bar{t}_N + 2$ to $\bar{t}_N + 1$ if

$$\limsup_{N\to\infty}(\log N)\bar{h}_{\bar{t}_N,N} < \kappa_{A,\varrho}.$$

Remark 1.8. The macroscopic assumption in Theorem 1.6 is sufficient but not necessary. Continuity and positivity of $V_{A,\varrho}$ imply that it gives $\bar{t}_N \to \infty$. On the other hand, lattice-subcritical contraction yields $\bar{h}_{t,N} \leqslant C_\varrho\theta_A^t$ for some $\theta_A \in (0, 1)$, and hence $\bar{t}_N = O(\log\log N)$. The detailed theorem assumes only $\bar{t}_N \to \infty$ and therefore also permits initial radii approaching the zero bin.

Remark 1.9. The lattice condition in Theorem 1.6 gives global contraction of the rounded squared-radius mean map. When it fails, the map may have a positive-drift component on which it points away from zero. The rounded vector chain then remains near the upper endpoint of the rightmost such component for exponentially long times, although zero remains absorbing.

For $\varrho \in (0,1)$, set

$$M_\varrho := \max_{|u|\leqslant \varrho^{-1}} Q_1(u)^2, \qquad \mathcal{O}_{A,\varrho} := \{h \in (0, M_\varrho) : V_{A,\varrho}(h) > h\},$$

and call its connected components positive-drift components. Define the fixed-precision existence threshold by

$$A_{\mathrm{ex}}(\varrho)^2 := \inf_{\alpha>0} \frac{\alpha^2}{\mathbb{E}\Big[Q_1(\varrho^{-1}\tanh(\varrho\alpha G))^2\Big]}.$$

Write

$$\mathscr{H}_t^{(N)} := \frac{1}{N\varrho^2}\|Y_t^{(N)}\|_2^2.$$

Theorem 1.10 (Metastability)**.** *Assume $A > A_{\mathrm{ex}}(\varrho)$, and let (h_-, h_+) be the rightmost positive-drift component of $\mathcal{O}_{A,\varrho}$. Fix a compact set $B \subset (h_-, h_+]$ and deterministic initial states $Y_0^{(N)}$ satisfying $\mathscr{H}_0^{(N)} \in B$ for every $N \geqslant 1$. Then there are $T \in \mathbb{N}_0$ and $c > 0$ such that*

$$\mathbb{P}\Big(\tau_\varrho^{(N)} > T + \lfloor \exp(cN)\rfloor\Big) \to 1.$$

For each fixed N, however, absorption still occurs almost surely from every deterministic rounded initial state.

Remark 1.11. The metastable well is created by the interaction between the squared-radius mean map and the rounding lattice. For each fixed N, the rounded vector chain is finite and eventually absorbs at zero. The unrounded vector chain has no corresponding metastable well; its persistent nonzero state is the supercritical invariant law.

The preceding discussion separates two time scales in the supercritical regime. For fixed $\varrho > 0$ and N, rounding makes the origin absorbing, so that the nonzero rounded state is only metastable and absorption eventually occurs. When the mesh is small, however, the probability of reaching the zero rounding bin on the relaxation time scale is negligible: the chain remains at radii of order one while it approaches the positive attracting fixed point. Thus rounding controls the eventual absorption and the exponentially long metastable time scale, whereas it does not determine the leading relaxation to the nonzero state in the small-mesh, large-dimension regime. We therefore study this relaxation through the unrounded chain. Its nonzero invariant law is a genuine equilibrium, and the next theorem gives the resulting supercritical dimension cutoff.

Let $A > 1$, and let q_* and $\mu_A = V_A'(q_*) \in (0,1)$ be the nonzero attracting fixed point and multiplier of V_A. For $q_0 \in (0,1]\backslash\{q_*\}$, let $C(q_0) \neq 0$ be determined by

$$V_A^t(q_0) - q_* = C(q_0)\mu_A^t + o(\mu_A^t),$$

and set

$$t_N(q_0) := \frac{\frac{1}{2}\log N + \log|C(q_0)|}{|\log \mu_A|}. \tag{1.20}$$

For $c \in \mathbb{R}$, define the nonnegative integer

$$n_N(c) := \max\{0, \lfloor t_N(q_0) + c\rfloor\}. \tag{1.21}$$

Theorem 1.12 (Radial supercritical dimension-cutoff)**.** *Let $x_N \in [-1,1]^N$ be deterministic and suppose*

$$q_N := N^{-1}\|x_N\|_2^2 \to q_0.$$

Then, for all sufficiently large N, the unrounded squared-radius transition kernel $K_{A,N}$ has a unique nonzero invariant law $\nu_{A,N}$, and

$$\lim_{c\to\infty} \liminf_{N\to\infty} \|K_{A,N}^{n_N(-c)}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}} = 1, \tag{1.22}$$

while
$$\lim_{c\to\infty}\limsup_{N\to\infty}\|K_{A,N}^{n_N(c)}(q_N,\cdot)-\nu_{A,N}\|_{\mathrm{TV}}=0. \tag{1.23}$$

Corollary 1.13 (Vector supercritical dimension-cutoff)**.** *Under the assumptions of Theorem 1.12, the unrounded vector chain has a unique nonzero invariant law $\pi_{A,N}$ for all sufficiently large N, and*
$$\lim_{c\to\infty}\liminf_{N\to\infty}\|P_{A,N}^{n_N(-c)}(x_N,\cdot)-\pi_{A,N}\|_{\mathrm{TV}}=1, \tag{1.24}$$
while
$$\lim_{c\to\infty}\limsup_{N\to\infty}\|P_{A,N}^{n_N(c)}(x_N,\cdot)-\pi_{A,N}\|_{\mathrm{TV}}=0. \tag{1.25}$$

At the cutoff time, the deterministic bias has reached the stationary fluctuation scale $N^{-1/2}$. The supercritical window has order one. Since the chain is observed at integer times, the offset $n_N(c)$ depends on the lattice position of $t_N(q_0)+c$. Consequently, a nonconstant continuous profile on $\mathbb{R}$, and in particular a Gaussian profile, cannot arise on this scale. A sharper profile statement would have to retain the integer offset and may require subsequences along which the fractional part of $t_N(q_0)$ converges.

Remark 1.14. The dimension cutoff does not describe the invariant law near the repelling origin. At fixed N, this behavior is governed by the radius multiplier and therefore by the threshold $A_c(N)$.

Theorem 1.15 (Stationary singularity in the supercritical regime)**.** *Fix $N\geqslant 1$ and assume $A>A_c(N)$. Let $\pi_{A,N}$ be the nonzero invariant law of the unrounded vector chain. Let $\beta_{A,N}\in(0,N)$ be the unique positive solution of*
$$A^{-\beta}N^{\beta/2}2^{-\beta/2}\frac{\Gamma((N-\beta)/2)}{\Gamma(N/2)}=1.$$
Here $\bar\sigma_N$ denotes normalized surface measure on $\mathbb{S}^{N-1}$. Then there is a constant $c_{A,N}>0$ such that, for every Borel set $B\subseteq\mathbb{S}^{N-1}$ satisfying $\bar\sigma_N(\partial B)=0$,
$$s^{-\beta_{A,N}}\pi_{A,N}\left(0<\|x\|_2\leqslant s,\ \frac{x}{\|x\|_2}\in B\right)\longrightarrow c_{A,N}\bar\sigma_N(B)\qquad \text{as } s\downarrow 0.$$
In particular,
$$\pi_{A,N}(0<\|x\|_2\leqslant s)\sim c_{A,N}s^{\beta_{A,N}}\qquad \text{as } s\downarrow 0.$$

1.3. Previous work and novelty. The closest neural-network literature studies signal propagation through randomly initialized deep networks by tracking deterministic recursions for variances, covariances, or correlations. The loss of long-term dependence caused by vanishing gradients in recurrent networks was isolated in [BSF94]. Xavier–Glorot initialization uses variance calculations to control forward activations and backward gradients [GB10], while He et al. derive the corresponding rectifier-specific scaling [HZRS15]. In the large-width limit, signal propagation is described by scalar or low-dimensional moment and correlation recursions. Their stability and the resulting order-to-chaos transition are studied in [PLR+16] and [SGGSD17]. We instead study the absorption time created by finite-precision rounding and its concentration as the mesh or width varies.

The three thresholds correspond to different limits. The unrounded squared-radius mean map V_A has threshold $A=1$; the fixed-width mean log-radius increment $\mathbb{E}\log((A/\sqrt{N})\chi_N)$ vanishes at $A_c(N)$; and the fixed-precision rounded map $V_{A,\varrho}$ has threshold A_{lat}. The radial reduction permits a direct comparison of these limits.

The identity (1.6) places the rounded vector chain in the classical absorption-cutoff setting. Martínez and Ycart study chains in which a distinguished state is made absorbing and the distance to the absorbing point mass is exactly the survival function of the absorption time [MY01]. In their general framework, the initial states eventually leave every finite set, and cutoff is related to concentration of the absorption times [MY01]. Explicit moment criteria are developed separately for self-adjoint tree-type birth-and-death chains [MY01]. Here the absorbing state is created by finite-precision rounding of a nonlinear random dynamical system, the family is indexed by ϱ or N, and the concentration estimates come from logarithmic-radius random walks and fluid-limit bounds.

The general cutoff literature provides several complementary perspectives. For background on finite Markov chains and mixing times, see the monograph of Levin and Peres [LP17]. Chen and Saloff-Coste give L^p-cutoff criteria for ergodic Markov semigroups when $1 < p < \infty$, together with reversible max-L^p extensions, and emphasize that $p = 1$ behaves differently [CSC08]. Diaconis and Saloff-Coste [DSC06] give a necessary-and-sufficient criterion for separation cutoff in finite continuous-time birth-and-death chains started from 0. Ding, Lubetzky, and Peres prove that the product condition is necessary and sufficient for total-variation cutoff for irreducible birth-and-death families in lazy discrete time or continuous time. The cutoff location is the larger of the two expected hitting times from the endpoints to a stationary median [DLP10]. More generally, for finite irreducible lazy reversible families, Basu, Hermon, and Peres characterize total-variation cutoff by concentration of the worst-case hitting times of sets of stationary mass at least α, for some $\alpha \in (0, 1/2]$ [BHP17]. For finite irreducible chains with symmetric support and nonnegative Ollivier or Bakry–Émery curvature, Salez proves cutoff under the strengthened product-type condition $t_{\mathrm{mix}}(\varepsilon) \gg (t_{\mathrm{rel}} \log \Delta)^2$, where Δ is the inverse of the smallest positive transition probability, using entropic concentration [Sal24]. These results concern convergence for ergodic chains or semigroups, whereas the rounded dynamics have an absorbing state and fall outside those hypotheses. They nevertheless place our total-variation statements in the broader theory of cutoff.

In a recent preprint, Pedrotti and Salez introduce a local Poincaré constant along the evolving laws and obtain a universal upper bound on the mixing-window width [PS26]. This perspective is particularly relevant when the transition is controlled by the law before equilibrium is reached. We do not apply their criterion here, because its standing assumption $\mu_t \ll \pi$ for every $t > 0$ fails: before absorption is certain, the rounded law assigns positive mass outside the support of the invariant law $\pi = \delta_0$ [PS26].

Cutoff for small random perturbations of deterministic dynamics is closer in spirit to the present work. Under global strong-coercivity and growth assumptions, Barrera and Jara prove total-variation window thermalisation for small Brownian perturbations from a fixed nonzero initial condition and characterize when a cutoff profile exists [BJ20]. Under their moment, regularity, and nondegeneracy hypotheses, Barrera, Högele, and Pardo prove total-variation window cutoff from nonzero initial data for coercive nonlinear Langevin systems driven by small strongly locally layered stable pure-jump Lévy noise with index $\alpha \in (3/2, 2)$, and characterize profile cutoff under an additional geometric condition [BHP21a]. For generalized Ornstein–Uhlenbeck systems with a stable drift matrix, fixed nonzero initial data, and possibly degenerate small Lévy noise satisfying a finite-moment assumption, they prove related results in renormalized Wasserstein distance [BHP21b]. In recent joint work with Barrera, we analyze, regime by regime, cutoff, profile cutoff, and absence of cutoff for one-dimensional Langevin–Kolmogorov dynamics with monomial convex potentials and small Brownian noise, using a fixed finite Galerkin truncation of the χ^2-distance [AB26]. This is complementary to the present work: the former uses a spectral Fokker–Planck formulation, whereas we study discrete-time total-variation absorption and metastability. In the neural-network setting, together with Karlsson we observed a numerical cutoff at fixed precision in one- and two-dimensional examples [AK22]. The present paper gives rigorous absorption cutoffs for that mechanism, separates the mesh and dimension limits, and adds the metastability and stationary singularity results.

The contrast between cutoff and metastable escape is studied by Barrera, Bertoncini, and

Fernández for discrete-time birth-and-death chains under suitable strong-drift and regularity hypotheses. At the level of hitting times, downhill motion concentrates around its mean, whereas the normalized uphill escape time is asymptotically exponential [BBF09]. In our model, cutoff and metastable persistence likewise occur in different parameter regimes. The rounded metastable well is related to the general theory of metastability [BdH15] and its absorbing behavior to quasi-stationarity [CMSM13] and [CV23], but we prove only exponential persistence, not a sharp exit-time law or a quasi-stationary limit. The stationary singularity is related to Kesten–Goldie renewal theory for random recursions [Kes73], [Gol91], and [BDM16]. In the present Gaussian rank-one setting the angular variable is refreshed in one step, so the Markov-additive renewal problem reduces to a scalar renewal calculation. To the best of our knowledge, these absorption cutoffs, the rounded metastability estimate, and the stationary singularity have not previously been proved for this random neural-network model.

1.4. Proof overview and organization. For the fixed-width cutoff, we couple the rounded vector chain to the unrounded vector chain until the radius reaches the grid scale. The logarithmic radius is then a perturbed random walk, and a first-passage central limit theorem gives the Gaussian profile. A final finite-grid estimate turns entrance at the grid scale into absorption.

At fixed precision and growing dimension, the rounded squared-radius chain concentrates around the iterates of $V_{A,\varrho}$. In the contractive regime, the orbit enters the zero-bin scale, and two further steps give absorption. On the rightmost positive-drift component of $V_{A,\varrho}(h) > h$, Hoeffding bounds keep the radius near the attracting endpoint for exponentially many steps.

In the supercritical regime, the unrounded deterministic orbit first reaches the $N^{-1/2}$ invariant-law fluctuation scale. Negative moments localize the squared-radius chain away from zero. Concentration, a one-step total-variation comparison, and synchronous contraction then give the cutoff bounds.

For the stationary singularity, we write the invariant equation in log-polar coordinates. Gaussian isotropy refreshes the angular variable in one step, and the radial equation becomes a scalar renewal equation with Cramér root $\beta_{A,N}$.

The common notation and scalar estimates are in Section 2. Sections 3 and 4 prove the two absorption cutoffs, and Section 5 treats metastability. The supercritical cutoff and stationary singularity are proved in Sections 6 and 7. The auxiliary estimates are in Appendices A and B.

1.5. Formalization. All six headline results in Section 1.2 have been formalized in Lean 4 on top of Mathlib. The Lean development is publicly available at `https://github.com/BennyAvelin/AbsorptionCutoff`.[1] The production library contains no `sorry` and declares no custom axiom. A machine-checked axiom audit shows that the seven declarations below use no axiom beyond Mathlib's three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`). The fixed-dimensional absorption clause of the metastability theorem is exposed separately for an additional audit, giving the following correspondence.

Result	Lean declaration
Theorem 1.4	`rounded_gaussian_nearest_cutoff`
Theorem 1.6	`subcritical_dimension_cutoff`
Theorem 1.10, persistence	`rounded_qualitative_metastability`
Theorem 1.10, absorption	`rounded_fixed_dimension_absorption`
Theorem 1.12	`gaussian_process_cutoff`
Corollary 1.13	`gaussian_vector_cutoff`
Theorem 1.15	`nd_power_singularity`

[1] The repository is pinned to Lean `v4.32.0` and Mathlib `v4.32.0`.

Reading a formalization is itself work, and a reader should not have to trust a seventy-thousand-line development in order to trust a theorem statement. Each of the six paper-facing results is therefore checked by the Lean comparator against a restatement written in Mathlib vocabulary alone; the fixed-dimensional absorption clause of the metastability theorem receives a separate seventh check. Each challenge file imports only Mathlib and contains one intentional statement-level `sorry`; these files are not part of the production library. The corresponding solution proves that exact statement from the development. The comparator verifies that each challenge–solution pair has the same elaborated statement, that the solution uses no axiom beyond the three above, and that the Lean kernel accepts the proof. It does not certify that the restatement expresses the intended mathematics. That comparison remains a human task, and the restatements are written to make it direct.

Where the formalized statement is packaged differently from the statement printed here, the difference is recorded result by result in the formalization. After unfolding the standing notation and the equivalent regime formulations recorded there, the hypotheses printed here imply the hypotheses of the corresponding Lean declaration, and its conclusion implies the printed conclusion. Thus no printed result is proved in Lean only under an additional hypothesis; in several cases the Lean conclusion is stronger.

Our use of autoformalization is inspired by the recent Lean formalizations of De Giorgi–Nash–Moser theory by Armstrong and Kempe [AK26a], and of coarse-graining theory for elliptic equations by Armstrong and Kuusi [AK26b]. The Lean development was produced by autoformalization. The proofs were written from the arguments of this paper by large language models, OpenAI's ChatGPT and Anthropic's Claude, driven by agentic coding harnesses under the author's supervision, rather than typed by hand. The kernel-level guarantee does not rest on the models having behaved well. Every definition, statement and proof in the production library is checked by the Lean kernel against Mathlib, and the paper-facing statements and their axiom dependencies are checked by the comparator as described above.

2. Gaussian squared-radius chains and common scalar estimates

Rounding makes the origin absorbing. We introduce the vector and radius chains below, while common stopping, product, and invariant-law estimates are proved in Appendix A.

We write $\|\cdot\|_2$ for the Euclidean norm on vectors. If B is a matrix, then

$$\|B\|_{\ell^2\to\ell^2} := \sup_{\|u\|_2=1} \|Bu\|_2.$$

Fix $A > 0$ and $N \geqslant 1$. Throughout the paper the Gaussian weights are

$$(\mathsf{W}^{(N)}_{t,A})_{ij} \sim \mathcal{N}(0, A^2/N), \qquad 1 \leqslant i,j \leqslant N, \quad t \geqslant 1, \tag{2.1}$$

independently over all indices. The unrounded vector chain is

$$X^{(N)}_0 = x, \qquad X^{(N)}_{t+1} = \tanh\Big(\mathsf{W}^{(N)}_{t+1,A} X^{(N)}_t\Big). \tag{2.2}$$

For $\varrho > 0$, let $Q_\varrho : \mathbb{R}^N \to \varrho\mathbb{Z}^N$ be coordinatewise nearest-grid rounding, with ties rounded toward the grid point of smaller absolute value. Let $Q_1 : \mathbb{R} \to \mathbb{Z}$ be the corresponding scalar unit-grid map. Then

$$Q_\varrho(x)_i = \varrho Q_1(x_i/\varrho), \qquad Q_1(u) = 0 \iff |u| \leqslant \frac{1}{2}, \qquad |Q_1(u) - u| \leqslant \frac{1}{2}.$$

The rounded vector chain is

$$Y^{(N)}_0 = Q_\varrho x, \qquad Y^{(N)}_{t+1} = Q_\varrho\Big(\tanh\Big(\mathsf{W}^{(N)}_{t+1,A} Y^{(N)}_t\Big)\Big). \tag{2.3}$$

We denote its rounded vector transition kernel by $P_{\varrho,A,N}$ and its absorption time by

$$\tau_\varrho^{(N)} := \inf\{t \geqslant 0 : Y_t^{(N)} = 0\}. \tag{2.4}$$

The point 0 is absorbing, and hence δ_0 is invariant. With the convention

$$\|\mu - \nu\|_{\mathrm{TV}} := \sup_{B \in \mathcal{B}(\mathbb{R}^N)} |\mu(B) - \nu(B)|,$$

we have the exact identity

$$\|P^t_{\varrho,A,N}(Q_\varrho x, \cdot) - \delta_0\|_{\mathrm{TV}} = \mathbb{P}(\tau_\varrho^{(N)} > t). \tag{2.5}$$

Thus cutoff to δ_0 is equivalent to concentration of the absorption time.

2.1. Gaussian radius observables. By Gaussian rotational invariance, the conditional law of the next preactivation vector depends on the present state only through its Euclidean norm. Set

$$R_t^{(N)} := \|X_t^{(N)}\|_2, \qquad Z_t^{(N)} := \frac{1}{N}\|X_t^{(N)}\|_2^2.$$

Since $R_t^{(N)} = \sqrt{N Z_t^{(N)}}$, both processes are Markov. We call $R^{(N)}$ the unrounded radius chain and $Z^{(N)}$ the unrounded squared-radius chain. Conditionally on $X_t^{(N)}$, the vector $\mathsf{W}_{t+1,A}^{(N)} X_t^{(N)}$ has the same law as $(A R_t^{(N)}/\sqrt{N}) G_{t+1}$, where G_{t+1} is a standard Gaussian vector in $\mathbb{R}^N$. Equivalently, given $Z_t^{(N)} = q$,

$$Z_{t+1}^{(N)} \overset{d}{=} \frac{1}{N} \sum_{i=1}^N \tanh^2(A\sqrt{q}\, G_{t+1,i}). \tag{2.6}$$

The chain $Z^{(N)}$ takes values in $[0,1]$.

For $g = (g_1, \ldots, g_N) \in \mathbb{R}^N$, define

$$F_{A,N}(q,g) := \frac{1}{N} \sum_{i=1}^N \tanh^2(A\sqrt{q}\, g_i), \qquad 0 \leqslant q \leqslant 1. \tag{2.7}$$

Let $K_{A,N}$ be the unrounded squared-radius transition kernel on $[0,1]$ given by

$$K_{A,N}(q,B) = \int_{\mathbb{R}^N} \mathbb{1}_{\{F_{A,N}(q,g) \in B\}}\, \mathcal{G}_N(\mathrm{d}g), \tag{2.8}$$

where $\mathcal{G}_N$ is standard Gaussian measure on $\mathbb{R}^N$. Thus, with $G_{t+1} \sim \mathcal{N}(0, I_N)$,

$$Z_{t+1}^{(N)} = F_{A,N}(Z_t^{(N)}, G_{t+1}). \tag{2.9}$$

Equivalently, conditionally on $Z_t^{(N)} = q$, $Z_{t+1}^{(N)} \overset{d}{=} F_{A,N}(q, G_{t+1})$. For fixed g, the map $q \mapsto F_{A,N}(q,g)$ is nondecreasing.

The unrounded squared-radius mean map is

$$V_A(q) := \mathbb{E} \tanh^2(A\sqrt{q}\, G), \qquad G \sim \mathcal{N}(0,1). \tag{2.10}$$

For fixed q, the conditional moments are

$$\mathbb{E}\Big[Z_{t+1}^{(N)} \mid Z_t^{(N)} = q\Big] = V_A(q), \qquad \mathrm{Var}\Big(Z_{t+1}^{(N)} \mid Z_t^{(N)} = q\Big) = \frac{1}{N} \mathrm{Var}\big(\tanh^2(A\sqrt{q}\, G)\big). \tag{2.11}$$

Moreover, near zero the unrounded squared-radius mean map expands as

$$V_A(q) = A^2 q - 2A^4 q^2 + O(q^3), \qquad q \downarrow 0. \tag{2.12}$$

Conditionally on $Z_t^{(N)} = q$, define the centered one-step fluctuation by

$$\zeta_{t+1}^{(N)} = \frac{1}{N} \sum_{i=1}^{N} \big[\tanh^2(A\sqrt{q}\, G_{t+1,i}) - V_A(q)\big], \tag{2.13}$$

so, with respect to the natural filtration of the unrounded squared-radius chain,

$$Z_{t+1}^{(N)} = V_A(Z_t^{(N)}) + \zeta_{t+1}^{(N)}. \tag{2.14}$$

The summands in (2.13) are independent, centered, and each lies in an interval of length at most one. Hence

$$\mathbb{E}\Big[\zeta_{t+1}^{(N)} \mid \mathcal{F}_t\Big] = 0, \qquad \operatorname{Var}\Big(\zeta_{t+1}^{(N)} \mid \mathcal{F}_t\Big) \leqslant \frac{1}{4N}, \tag{2.15}$$

and Hoeffding's inequality gives, for every $u > 0$,

$$\mathbb{P}\Big(\Big|\zeta_{t+1}^{(N)}\Big| > u \,\Big|\, \mathcal{F}_t\Big) \leqslant 2\exp(-2Nu^2). \tag{2.16}$$

The rounded vector chain admits an exact scalar reduction. Define its normalized squared radius by

$$\mathscr{H}_t^{(N)} := \frac{1}{N\varrho^2} \|Y_t^{(N)}\|_2^2. \tag{2.17}$$

For $h \geqslant 0$ and $g \in \mathbb{R}$, put

$$U_{A,\varrho}(h,g) := Q_1\Big(\varrho^{-1}\tanh(\varrho A\sqrt{h}\, g)\Big)^2. \tag{2.18}$$

Using the Gaussian representation above together with the coordinatewise rounding rule gives

$$\mathscr{H}_{t+1}^{(N)} \stackrel{d}{=} \frac{1}{N} \sum_{i=1}^{N} U_{A,\varrho}(h, G_{t+1,i}), \tag{2.19}$$

where the variables $G_{t+1,i}$ are independent standard Gaussians. Thus $\mathscr{H}^{(N)}$ is the rounded squared-radius chain. Its absorption time satisfies

$$\tau_\varrho^{(N)} = \inf\{t \in \mathbb{N}_0 : \mathscr{H}_t^{(N)} = 0\}. \tag{2.20}$$

A coordinate lies in the zero bin precisely when its unrounded value has absolute value at most $\varrho/2$.

The rounded squared-radius mean map is

$$V_{A,\varrho}(h) := \mathbb{E}\bigg[Q_1\Big(\varrho^{-1}\tanh(\varrho A\sqrt{h}\, G)\Big)^2\bigg], \qquad h \geqslant 0. \tag{2.21}$$

With respect to the natural filtration, define

$$\widehat{\zeta}_{t+1}^{(N)} := \mathscr{H}_{t+1}^{(N)} - V_{A,\varrho}(\mathscr{H}_t^{(N)}). \tag{2.22}$$

If $B_\varrho := \max_{|u| \leqslant \varrho^{-1}} Q_1(u)^2$, then $0 \leqslant U_{A,\varrho}(h,G) \leqslant B_\varrho$, and hence

$$\mathbb{E}[\widehat{\zeta}_{t+1}^{(N)} \mid \mathcal{F}_t] = 0, \qquad \mathbb{E}[(\widehat{\zeta}_{t+1}^{(N)})^2 \mid \mathcal{F}_t] \leqslant \frac{B_\varrho^2}{4N}. \tag{2.23}$$

Moreover, Hoeffding's inequality yields

$$\mathbb{P}\Big(\Big|\widehat{\zeta}_{t+1}^{(N)}\Big| > u \,\Big|\, \mathcal{F}_t\Big) \leqslant 2\exp\Big(-\frac{2Nu^2}{B_\varrho^2}\Big), \qquad u > 0. \tag{2.24}$$

2.2. Thresholds and mean maps. The derivative $V_A'(0) = A^2$ gives the large-dimension threshold $A = 1$. Recall that χ_N denotes the Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$. At fixed width, the radius multiplier near zero is

$$\ell_{A,N} := \frac{A}{\sqrt{N}}\chi_N. \tag{2.25}$$

The corresponding log-radius increment is $\log \ell_{A,N}$, whose negative mean is

$$\gamma_{A,N} := -\mathbb{E}\log \ell_{A,N}.$$

The mean log-radius increment vanishes at

$$A = A_c(N) = \sqrt{\frac{N}{2}}\exp\Big(-\frac{1}{2}\psi\Big(\frac{N}{2}\Big)\Big), \tag{2.26}$$

where ψ is the digamma function. Thus $A < A_c(N)$ is equivalent to $\gamma_{A,N} > 0$, and

$$\gamma_{A,N} = \log\frac{A_c(N)}{A}, \qquad \operatorname{Var}(\log \ell_{A,N}) = \operatorname{Var}(\log \chi_N) = \frac{1}{4}\psi_1\Big(\frac{N}{2}\Big). \tag{2.27}$$

Here $\psi_1 = \psi'$ is the trigamma function. Moreover,

$$\frac{F_{A,N}(q,G)}{q} \to A^2\frac{\chi_N^2}{N} \qquad \text{as } q \downarrow 0 \tag{2.28}$$

almost surely. In particular, $A_c(N) > 1$ for finite N and $A_c(N) \downarrow 1$ as $N \to \infty$.

The unit-grid lattice comparison map, profile, and contraction threshold are

$$L_A(h) := \mathbb{E}[Q_1(A\sqrt{h}\,G)^2], \tag{2.29}$$

and

$$m(\alpha) := \mathbb{E}[Q_1(\alpha G)^2], \qquad A_{\mathrm{lat}}^2 := \inf_{\alpha>0}\frac{\alpha^2}{m(\alpha)}. \tag{2.30}$$

Thus $L_A(h) = m(A\sqrt{h})$. The condition $A < A_{\mathrm{lat}}$ is equivalent to $L_A(h) < h$ for every $h > 0$.

We begin with the fixed-point and linearization facts for V_A that will be used throughout the paper.

Lemma 2.1. *The map V_A is strictly increasing on $[0,1]$ and strictly concave on $[0,1]$, with $V_A(0) = 0$, $V_A'(0+) = A^2$, and $V_A(1) < 1$. If $A \leqslant 1$, then 0 is its only fixed point. If $A > 1$, then it has a unique nonzero fixed point $q_* \in (0,1)$ satisfying*

$$q_* = V_A(q_*), \qquad \mu_A := V_A'(q_*) \in (0,1). \tag{2.31}$$

Every unrounded deterministic orbit started from $(0,1]$ converges to q_*. If $q_0 \in (0,1]\backslash\{q_*\}$, then there is a continuous function C in a neighborhood of q_0, with $C(q_0) \neq 0$, such that

$$V_A^t(q) - q_* = C(q)\mu_A^t + o(\mu_A^t) \tag{2.32}$$

locally uniformly for q near q_0.

Proof. Fix $a > 0$ and put $f_a(q) := \tanh^2(a\sqrt{q})$ for $q > 0$. With $x = a\sqrt{q}$,

$$f_a'(q) = a^2\frac{\tanh x}{x}\operatorname{sech}^2 x.$$

The functions $x \mapsto \tanh x/x$ and $x \mapsto \operatorname{sech}^2 x$ are strictly decreasing on $(0,\infty)$, hence f_a' is strictly decreasing and f_a is strictly concave. Since $\tanh^2(A\sqrt{q}\,G) = f_{A|G|}(q)$ and $\mathbb{P}(|G| > 0) = 1$, integration gives strict concavity of V_A. The same formula gives strict monotonicity, and the expansion in (2.12) gives $V_A'(0+) = A^2$. Finally $V_A(1) < 1$ since $\tanh^2(AG) < 1$ almost surely.

Concavity gives no positive fixed point when $A \leqslant 1$ and exactly one when $A > 1$. At the positive fixed point, concavity gives

$$0 < V_A'(q_*) < \frac{V_A(q_*)}{q_*} = 1.$$

If $A > 1$, then $V_A(q) > q$ for $0 < q < q_*$ and $V_A(q) < q$ for $q_* < q \leqslant 1$. Since V_A is increasing, the orbit is monotone and bounded on each side of q_*, and therefore converges to q_*.

The asymptotic (2.32) follows from the one-dimensional Koenigs linearization; see [CG93, Ch. II, Sec. 2, Theorem 2.1]. The Taylor expansion at the attracting fixed point is

$$V_A(q) - q_* = \mu_A(q - q_*) + O((q - q_*)^2)$$

near q_*. After a finite time the orbit is in a neighborhood on which $|V_A(q) - q_*| \leqslant \rho|q - q_*|$ for some $\rho < 1$. For all later times before the orbit reaches q_*, write

$$\frac{V_A^{t+1}(q) - q_*}{\mu_A^{t+1}} = \frac{V_A^t(q) - q_*}{\mu_A^t} \frac{V_A(V_A^t(q)) - q_*}{\mu_A(V_A^t(q) - q_*)}.$$

The last factor is $1 + O(|V_A^t(q) - q_*|)$, uniformly for q in a small neighborhood of any fixed $q_0 \neq q_*$. Since the orbit is eventually contained in a contraction neighborhood of q_*, the series

$$\sum_{t=0}^{\infty} |V_A^t(q) - q_*|$$

is locally uniformly summable. Hence the logarithms of the product factors are locally uniformly summable, the Koenigs product converges locally uniformly, and we may define

$$C(q) = (V_A^s(q) - q_*)\mu_A^{-s} \prod_{j=s}^{\infty} \frac{V_A(V_A^j(q)) - V_A(q_*)}{\mu_A(V_A^j(q) - q_*)}$$

for any entrance time s after which the orbit stays in the contraction neighborhood. This formula is independent of the chosen sufficiently large s, locally uniform in q, and continuous. It is nonzero at q_0 because every factor is positive and finite and the orbit approaches q_* monotonically from one side without crossing it. This proves (2.32). □

2.3. Common scalar estimates. The deterministic coefficients in the stopped recursion may amplify the one-step noise. We retain this amplification in the second-moment estimate.

Let $T_N \in \mathbb{N}$, let $(\mathcal{F}_t)_{0 \leqslant t \leqslant T_N}$ be a filtration, and let $(\mathcal{R}_{t,N})_{0 \leqslant t \leqslant T_N}$ be adapted. Let τ_N be a stopping time. Suppose that, on $\{t < \tau_N\}$ and for $0 \leqslant t < T_N$,

$$\mathcal{R}_{t+1,N} = A_{t,N}\mathcal{R}_{t,N} + \eta_{t+1,N}, \tag{2.33}$$

where $A_{t,N}$ is $\mathcal{F}_t$-measurable, $|A_{t,N}| \leqslant \alpha_{t,N}$ for deterministic $\alpha_{t,N} \geqslant 0$, and

$$\mathbb{E}[\eta_{t+1,N} \mid \mathcal{F}_t] = 0, \qquad \mathbb{E}[\eta_{t+1,N}^2 \mid \mathcal{F}_t] \leqslant \frac{C\sigma_{t,N}^2}{N} \tag{2.34}$$

for a constant $C < \infty$ and deterministic $\sigma_{t,N} \geqslant 0$. Define

$$K_N := 1 \vee \max_{0 \leqslant s < t \leqslant T_N} \left\{ \prod_{u=s}^{t-1}(1 \vee \alpha_{u,N}), \sigma_{s,N} \prod_{u=s+1}^{t-1}(1 \vee \alpha_{u,N}) \right\}. \tag{2.35}$$

To control the orbit before it reaches the terminal layer, we need a stopped second-moment estimate.

Lemma 2.2. *Under the preceding assumptions, set* $\overline{\mathcal{R}}_{t,N} := \mathcal{R}_{t\wedge\tau_N,N}$. *Then*

$$\mathbb{E}\overline{\mathcal{R}}^2_{T_N,N} \leqslant CK_N^2\left(\mathbb{E}\mathcal{R}^2_{0,N} + \frac{T_N}{N}\right). \tag{2.36}$$

In addition, for every $t \leqslant T_N$,

$$\mathbb{E}\big[\mathcal{R}^2_{t,N}\mathbb{1}_{\{\tau_N>t\}}\big] \leqslant \left(\prod_{u=0}^{t-1}\alpha^2_{u,N}\right)\mathbb{E}\mathcal{R}^2_{0,N} + \frac{C}{N}\sum_{s=0}^{t-1}\sigma^2_{s,N}\prod_{u=s+1}^{t-1}\alpha^2_{u,N}, \tag{2.37}$$

with empty products equal to one and empty sums equal to zero. In particular, if $\tau_N = \inf\{t : |\mathcal{R}_{t,N}| > \delta\}$ *for some* $\delta > 0$, *then*

$$\mathbb{P}(\tau_N \leqslant T_N) \leqslant \frac{CK_N^2}{\delta^2}\left(\mathbb{E}\mathcal{R}^2_{0,N} + \frac{T_N}{N}\right). \tag{2.38}$$

Affine recursions with negative mean log multiplier form the contractive case; see [BDM16]. The next lemma gives the quantitative logarithmic entrance estimate needed here.

Lemma 2.3. *Let* $(M_j)_{j\geqslant 1}$ *be independent and identically distributed positive random variables with* $\mathbb{E}\log M_1 < 0$, *and assume* $\log(M_1+\varepsilon)$ *has exponential moments in a neighborhood of the origin for each fixed* $\varepsilon > 0$. *For every* $b < \infty$, *there exist* $K_* < \infty$ *and* $c_1, c_2, c_3 > 0$, *depending only on* b *and the law of* M_1, *such that the affine recursion*

$$U_{t+1} = M_{t+1}U_t + b, \qquad U_0 = K,$$

satisfies, uniformly for $K \geqslant 2$,

$$\mathbb{P}(\inf\{t : U_t \leqslant K_*\} > c_1\log K + r) \leqslant c_2 e^{-c_3 r}, \qquad r \geqslant 0. \tag{2.39}$$

See Appendices A.1 and A.2 for the proofs.

2.4. Transition and reconstruction kernels. After one Gaussian step, the vector law depends on the previous state only through its normalized squared radius. This factorization transfers invariant laws and total variation estimates between the vector and squared-radius chains.

Let $r_N(x) := N^{-1}\|x\|_2^2$, and let the squared-radius-to-vector reconstruction kernel $J_{A,N}(q,\cdot)$ be the law of $(\tanh(A\sqrt{q}\,G_1),\ldots,\tanh(A\sqrt{q}\,G_N))$, where $G \sim \mathcal{N}(0, I_N)$. For a probability measure η on $[0,1]$, we write $\eta J_{A,N}$ for the probability measure on $[-1,1]^N$ defined by

$$\eta J_{A,N}(B) := \int_0^1 J_{A,N}(q,B)\,\eta(\mathrm{d}q).$$

Gaussian symmetry now reduces the vector-chain comparison to the squared-radius chain.

Proposition 2.4. *Let* $P_{A,N}$ *be the unrounded vector transition kernel in* (2.2). *Then, for every bounded measurable* $f : [-1,1]^N \to \mathbb{R}$,

$$P_{A,N}f(x) = \int_{[-1,1]^N} f(z)\,J_{A,N}(r_N(x),\mathrm{d}z). \tag{2.40}$$

Consequently, for every deterministic $x \in [-1,1]^N$, *with* $q = r_N(x)$, *and every* $t \geqslant 1$,

$$P^t_{A,N}(x,\cdot) = K^{t-1}_{A,N}(q,\cdot)J_{A,N}, \qquad (r_N)_\# P^t_{A,N}(x,\cdot) = K^t_{A,N}(q,\cdot). \tag{2.41}$$

If $\nu_{A,N}$ is invariant for $K_{A,N}$, then $\pi_{A,N} := \nu_{A,N} J_{A,N}$, that is,

$$\pi_{A,N} = \int_0^1 J_{A,N}(q,\cdot)\,\nu_{A,N}(\mathrm{d}q) \tag{2.42}$$

is an invariant law for $P_{A,N}$. Conversely, if $\pi_{A,N}$ is invariant for $P_{A,N}$ and $\nu_{A,N} := (r_N)_\# \pi_{A,N}$, then $\nu_{A,N}$ is invariant for $K_{A,N}$ and $\pi_{A,N}$ is given by (2.42).

For this pair of invariant laws and $t \geqslant 1$,

$$\|K_{A,N}^t(q,\cdot) - \nu_{A,N}\|_{\mathrm{TV}} \leqslant \|P_{A,N}^t(x,\cdot) - \pi_{A,N}\|_{\mathrm{TV}} \leqslant \|K_{A,N}^{t-1}(q,\cdot) - \nu_{A,N}\|_{\mathrm{TV}}. \tag{2.43}$$

Proof. For a fixed vector x, the random vector $\mathsf{W}_A^{(N)} x$ has law $A\sqrt{r_N(x)}G$. Applying tanh coordinatewise gives (2.40). Since $(r_N)_\# J_{A,N}(q,\cdot) = K_{A,N}(q,\cdot)$, induction gives (2.41). The same identities also give the statement about invariant laws. Indeed, if $\nu_{A,N} K_{A,N} = \nu_{A,N}$, then

$$(\nu_{A,N} J_{A,N}) P_{A,N} = (\nu_{A,N} K_{A,N}) J_{A,N} = \nu_{A,N} J_{A,N}.$$

Conversely, if $\pi_{A,N} P_{A,N} = \pi_{A,N}$ and $\nu_{A,N} = (r_N)_\# \pi_{A,N}$, then

$$\pi_{A,N} = \pi_{A,N} P_{A,N} = \nu_{A,N} J_{A,N}$$

and, after applying $(r_N)_\#$, we obtain $\nu_{A,N} K_{A,N} = \nu_{A,N}$.

The lower bound in (2.43) is the contraction of total variation under the measurable map r_N. For the upper bound, use (2.41) and (2.42); the reconstruction kernel $J_{A,N}$ contracts total variation, and hence

$$\|K_{A,N}^{t-1}(q,\cdot) J_{A,N} - \nu_{A,N} J_{A,N}\|_{\mathrm{TV}} \leqslant \|K_{A,N}^{t-1}(q,\cdot) - \nu_{A,N}\|_{\mathrm{TV}}.$$

$\square$

2.5. Invariant laws. If $\nu_{A,N}$ is invariant for $K_{A,N}$ on $(0,1]$, then

$$\int_0^1 \phi(q)\,\nu_{A,N}(\mathrm{d}q) = \int_0^1 \int_{\mathbb{R}^N} \phi(F_{A,N}(q,g))\mathcal{G}_N(\mathrm{d}g)\,\nu_{A,N}(\mathrm{d}q) \tag{2.44}$$

for every bounded measurable $\phi : [0,1] \to \mathbb{R}$. Equivalently,

$$\nu_{A,N} K_{A,N} = \nu_{A,N}. \tag{2.45}$$

Cesàro averages have invariant subsequential limits because $[0,1]$ is compact. A limit may equal δ_0; the final proposition rules this out in the supercritical fixed-width regime.

Fix $A > 0$, $N \geqslant 1$, and $q \in [0,1]$. For $T \geqslant 1$, define

$$\overline{\nu}_T^q := \frac{1}{T} \sum_{t=0}^{T-1} K_{A,N}^t(q,\cdot). \tag{2.46}$$

Proposition 2.5. *The family $(\overline{\nu}_T^q)_{T \geqslant 1}$ is relatively compact for weak convergence. Every subsequential limit $\overline{\nu}$ is invariant for $K_{A,N}$ and satisfies the stationary equation (2.44). We denote by $\mathcal{I}_{A,N}(q)$ the nonempty set of all such limits.*

If $\overline{\nu} \in \mathcal{I}_{A,N}(q)$ and $\overline{\nu} \neq \delta_0$, then

$$\nu^+ := \frac{\overline{\nu}|_{(0,1]}}{1 - \overline{\nu}(\{0\})}$$

is invariant for $K_{A,N}$ on $(0,1]$. We write $\mathcal{I}^+_{A,N}(q)$ for the collection of all such nonzero components. For every $\nu^+ \in \mathcal{I}^+_{A,N}(q)$, the measure

$$\pi^+ := \int_0^1 J_{A,N}(s,\cdot)\,\nu^+(\mathrm{d}s) \tag{2.47}$$

is an invariant law for the unrounded vector chain. Equivalently,

$$\int_{[-1,1]^N} \varphi(y)\,\pi^+(\mathrm{d}y) = \int_{[-1,1]^N}\int_{\mathbb{R}^N} \varphi\big(\tanh(A\sqrt{r_N(x)}\,g)\big)\,\mathcal{G}_N(\mathrm{d}g)\,\pi^+(\mathrm{d}x) \tag{2.48}$$

for every bounded measurable $\varphi : [-1,1]^N \to \mathbb{R}$.

For $A > A_c(N)$, a negative-moment Foster–Lyapunov estimate prevents Cesàro limits started from positive radii from charging the origin.

The preceding compactness argument produces a candidate invariant law. In the supercritical regime it does not collapse to zero, as we now show.

Proposition 2.6. *Let $N \geqslant 1$ and assume $A > A_c(N)$. Then, for every $q \in (0,1]$, $\mathcal{I}^+_{A,N}(q)$ is nonempty. In particular, the unrounded squared-radius transition kernel $K_{A,N}$ admits at least one invariant probability on $(0,1]$, and the unrounded vector chain admits a corresponding nonzero invariant law.*

The proofs of Propositions 2.5 and 2.6 are in Appendix A.3.

2.6. Notation guide. The table collects the notation used throughout the proofs.

Symbol	Meaning and first definition
$\mathsf{W}_{t,A}^{(N)}$	Gaussian weight matrix with entries $\mathcal{N}(0, A^2/N)$, (2.1).
$\mathcal{G}_N$	standard Gaussian measure on $\mathbb{R}^N$, (2.8).
$X_t^{(N)}$, $Y_t^{(N)}$	unrounded and rounded vector chains, (2.2) and (2.3).
Q_ϱ, Q_1	coordinatewise nearest-grid rounding at mesh ϱ and at unit mesh, defined before (2.3).
$\tau_\varrho^{(N)}$	absorption time of the rounded vector chain, (2.4).
$Z_t^{(N)}$, $R_t^{(N)}$	unrounded squared-radius and radius chains, introduced before (2.6).
$\mathscr{H}_t^{(N)}$, $U_{A,\varrho}$	rounded squared-radius chain and the associated one-coordinate observable, (2.17) and (2.18).
$F_{A,N}$, $K_{A,N}$	random squared-radius map and unrounded squared-radius transition kernel, (2.7) and (2.8).
$P_{A,N}$, $P_{\varrho,A,N}$	unrounded and rounded vector transition kernels, defined before (2.4) and (2.40).
$J_{A,N}$	squared-radius-to-vector reconstruction kernel with $\pi_{A,N} = \nu_{A,N} J_{A,N}$, introduced before Proposition 2.4.
V_A	unrounded squared-radius mean map, (2.10).
$A_c(N)$, A_c	fixed-width critical gain, defined by zero mean log-radius increment; we sometimes write A_c when N is fixed, (2.26).
$\gamma_{A,N}$	negative mean log-radius increment, (2.25) and (2.27).
χ_N	Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$, (2.25).
L_A	unit-grid lattice comparison map for the linearized rounded dynamics, (2.29).
$V_{A,\varrho}$	rounded squared-radius mean map, (2.21).
A_{lat}	threshold for global contraction of the lattice comparison map, (2.30).
q_*, μ_A, $C(q_0)$	positive fixed point, multiplier, and deterministic linearization coefficient $C(q_0) = \lim_{t\to\infty} \mu_A^{-t}(V_A^t(q_0) - q_*)$, (2.31) and (2.32).
$\nu_{A,N}$, $\pi_{A,N}$	nonzero invariant laws for the unrounded squared-radius and vector chains, (2.42) and (2.45).

Table 1: Frequently used notation.

3. Subcritical cutoff as the mesh vanishes

We prove total variation cutoff to the absorbing state for the rounded vector chain as the mesh vanishes at fixed width. The activation is tanh. Throughout the section $N \geqslant 1$ and $0 < A < A_c(N)$ are fixed, and $x_0 \in \mathbb{R}^N \backslash \{0\}$ is deterministic. We write

$$R_t := \|X_t^{(N)}\|_2, \qquad \widehat{R}_t := \|Y_t^{(N)}\|_2,$$

where $X^{(N)}$ starts from x_0 and the rounded vector chain $Y^{(N)}$ starts from $Q_\varrho x_0$. Thus R and $\widehat{R}$ are the unrounded and rounded radius chains. Generic constants c, C, c_N, C_N may change from line to line; a subscript N indicates dependence only on the fixed width unless further dependence is displayed.

The logarithm of the unrounded radius is a positive-drift random walk plus an almost surely finite correction. A synchronous coupling transfers its first-passage asymptotics to the rounded chain up to the grid scale, where a finite-grid estimate completes the absorption argument.

3.1. Unrounded first-passage asymptotics. Let $(\chi_{N,j})_{j\geqslant 1}$ be independent copies of χ_N. In the notation of (2.25), the fixed-width subcritical condition is $\gamma_{A,N} > 0$. Put

$$\xi_j := -\log\Big(\frac{A}{\sqrt{N}}\chi_{N,j}\Big), \qquad \sigma_N^2 := \mathrm{Var}(\xi_1). \tag{3.1}$$

By (2.27),

$$\mathbb{E}\xi_1 = \gamma_{A,N} = \log\frac{A_c(N)}{A}, \qquad \sigma_N^2 = \mathrm{Var}(\log\chi_N) = \frac{1}{4}\psi_1\Big(\frac{N}{2}\Big), \tag{3.2}$$

where ψ_1 is the trigamma function. For $0 < \varrho < R_0$, the cutoff scale is

$$L_\varrho := \log\frac{R_0}{\varrho}, \qquad t_\varrho(a) := \max\left\{0, \left\lfloor \frac{L_\varrho}{\gamma_{A,N}} + a\frac{\sigma_N}{\gamma_{A,N}^{3/2}}\sqrt{L_\varrho}\right\rfloor\right\}. \tag{3.3}$$

Here and below Φ_{G} denotes the distribution function of a standard Gaussian random variable. The nonlinear correction to the logarithmic radius is summable.

Lemma 3.1. *There are an increasing adapted sequence $(E_t)_{t\geqslant 0}$ and an event $\Omega_{\log}$ of probability one such that, on $\Omega_{\log}$, $R_t > 0$ for every $t \geqslant 0$,*

$$-\log R_t = -\log R_0 + \sum_{j=1}^{t}\xi_j + E_t \qquad \text{for every } t \geqslant 0 \tag{3.4}$$

and $E_t \uparrow E_\infty < \infty$. Consequently, $(E_t)_{t\geqslant 0}$ is tight.

Proof. For each fixed t, the unrounded vector chain satisfies $R_t > 0$ almost surely. Since the time set is countable, the event

$$\Omega_{\mathrm{nz}} := \bigcap_{t\geqslant 0}\{R_t > 0\}$$

has probability one. For each $j \geqslant 1$, define, on $\{R_{j-1} > 0\}$,

$$G^{(j)} := \frac{\sqrt{N}}{AR_{j-1}}\mathsf{W}_{j,A}^{(N)}X_{j-1}^{(N)}.$$

Extend $G^{(j)}$ arbitrarily across the null event $\{R_{j-1} = 0\}$. Conditionally on the past, $G^{(j)}$ is standard Gaussian and its law does not depend on the past. Thus $(G^{(j)})_{j\geqslant 1}$ are independent standard Gaussian vectors. Taking $\chi_{N,j} := \|G^{(j)}\|_2$, the radius update is

$$R_j = \Big\|\tanh\Big(\frac{A}{\sqrt{N}}G^{(j)}R_{j-1}\Big)\Big\|_2.$$

Set $M_j := (A/\sqrt{N})\chi_{N,j}$. Since $|\tanh z| \leqslant |z|$ coordinatewise,

$$R_j \leqslant M_j R_{j-1} \qquad \text{for every } j \geqslant 1. \tag{3.5}$$

Define

$$\eta_j := \log\frac{M_j R_{j-1}}{R_j} \geqslant 0, \qquad E_t := \sum_{j=1}^{t}\eta_j, \qquad \xi_j := -\log M_j,$$

which gives (3.4) simultaneously for every $t \geqslant 0$ on Ω_{nz}.

To prove finiteness of the error series, let $u \in \mathbb{R}$, with the value at zero defined by continuity. Then

$$0 \leqslant \log\frac{|u|}{|\tanh u|} \leqslant Cu^2.$$

Applying this coordinatewise, if $u = (u_1, \dots, u_N)$ and $v_i = \tanh(u_i)$, with ratios at $u_i = 0$ understood by continuity, then

$$\|u\|_2^2 = \sum_{i=1}^{N} v_i^2 \frac{u_i^2}{v_i^2} \leqslant \max_{1\leqslant i\leqslant N} \frac{u_i^2}{\tanh^2(u_i)} \|v\|_2^2,$$

and hence $\log(\|u\|_2/\|v\|_2) \leqslant C\|u\|_2^2$. With $u = (A/\sqrt{N})G^{(j)}R_{j-1}$ and $v = \tanh(u)$, this yields

$$0 \leqslant \eta_j \leqslant CM_j^2 R_{j-1}^2. \tag{3.6}$$

Since $A < A_c(N)$, the variables $\log M_j$ have negative mean. By the strong law of large numbers, for some $\kappa > 0$ and all sufficiently large j,

$$\prod_{i=1}^{j} M_i \leqslant e^{-\kappa j}$$

almost surely. Combining this with (3.5) gives $R_j \leqslant R_0 e^{-\kappa j}$ eventually almost surely. On the other hand, $\chi_{N,j}^2 \leqslant e^{\kappa j}$ eventually almost surely by the Gaussian tail bound and Borel–Cantelli. Therefore the tail of the error series is almost surely bounded by

$$C \sum_{j \text{ large}} M_j^2 R_{j-1}^2 \leqslant C\frac{A^2}{N} R_0^2 \sum_{j \text{ large}} e^{-2\kappa(j-1)} \chi_{N,j}^2 \leqslant C\frac{A^2}{N} R_0^2 \sum_{j \text{ large}} e^{-\kappa(j-2)} < \infty.$$

The finitely many remaining terms are finite almost surely. Intersecting Ω_{nz} with the probability-one events furnished by the strong law and Borel–Cantelli gives an event $\Omega_{\log}$ on which all the preceding identities hold simultaneously for every time and $E_t \uparrow E_\infty < \infty$. $\square$

For a positive threshold ε_ϱ, define

$$L_{\varepsilon_\varrho} := \log \frac{R_0}{\varepsilon_\varrho}, \qquad \theta_{\varepsilon_\varrho}^{\mathrm{rad}} := \inf\{t \geqslant 0 : R_t \leqslant \varepsilon_\varrho\}.$$

The unrounded entrance time has the same profile for thresholds whose logarithmic displacement from ϱ is negligible on the cutoff scale.

The remaining probabilistic input is a first-passage estimate for a moving threshold.

Proposition 3.2. *Let $\varepsilon_\varrho \downarrow 0$ satisfy*

$$L_{\varepsilon_\varrho} = L_\varrho + o(\sqrt{L_\varrho}).$$

Then, for every fixed $a \in \mathbb{R}$,

$$\mathbb{P}\Big(\theta_{\varepsilon_\varrho}^{\mathrm{rad}} > t_\varrho(a)\Big) \to \Phi_{\mathrm{G}}(-a), \qquad \varrho \downarrow 0. \tag{3.7}$$

The same limit holds if $t_\varrho(a)$ is replaced by $\max\{0, \lfloor t_\varrho(a) + u_\varrho \rfloor\}$ for any deterministic $u_\varrho = o(\sqrt{L_\varrho})$.

Proof. On the probability-one event supplied by Lemma 3.1, simultaneously for every $t \geqslant 0$,

$$-\log R_t = -\log R_0 + S_t + E_t, \qquad S_t := \sum_{j=1}^{t} \xi_j,$$

where $0 \leqslant E_t \leqslant E_\infty < \infty$. Taking logarithms in the definition of the entrance time is therefore legitimate on this event, and, pathwise,

$$\theta_{\varepsilon_\varrho}^{\mathrm{rad}} = \inf\{t \geqslant 0 : S_t + E_t \geqslant L_{\varepsilon_\varrho}\}.$$

Let $H_u = \inf\{t \geqslant 0 : S_t \geqslant u\}$, and let $\nu(u) = \inf\{t \geqslant 0 : S_t > u\}$ be the strict first-passage time used in [Gut09]. The law of ξ_1 is absolutely continuous, and hence so is the law of S_t for every $t \geqslant 1$. Thus, for every fixed $u > 0$,

$$\mathbb{P}\big(S_t = u \text{ for some } t \geqslant 0\big) = 0,$$

and therefore $H_u = \nu(u)$ almost surely. The increments have mean $\gamma_{A,N} > 0$ and variance $\sigma_N^2 < \infty$ by (3.2). Hence [Gut09, Theorem 3.5.1], with $\mu = \gamma_{A,N}$ and $\sigma = \sigma_N$, gives

$$\frac{H_u - u/\gamma_{A,N}}{\sigma_N \gamma_{A,N}^{-3/2} \sqrt{u}} \Rightarrow N(0,1) \qquad \text{as } u \to \infty.$$

The same limit holds with H_u replaced by H_{u+b_u} whenever $b_u = o(\sqrt{u})$, because the resulting changes in the centering and normalization are negligible.

Fix $B < \infty$. On $\{E_\infty \leqslant B\}$, the deterministic sandwich is

$$H_{L_{\varepsilon_\varrho} - B} \leqslant \theta^{\mathrm{rad}}_{\varepsilon_\varrho} \leqslant H_{L_{\varepsilon_\varrho}}. \tag{3.8}$$

Therefore, for $t = t_\varrho(a)$,

$$\mathbb{P}(H_{L_{\varepsilon_\varrho} - B} > t) - \mathbb{P}(E_\infty > B) \leqslant \mathbb{P}(\theta^{\mathrm{rad}}_{\varepsilon_\varrho} > t) \leqslant \mathbb{P}(H_{L_{\varepsilon_\varrho}} > t) + \mathbb{P}(E_\infty > B).$$

Applying the first-passage limit with $u = L_\varrho$, both hitting probabilities in the last display converge to $\Phi_{\mathrm{G}}(-a)$, since $L_{\varepsilon_\varrho} - L_\varrho = o(\sqrt{L_\varrho})$ and $B = o(\sqrt{L_\varrho})$. Thus

$$\Phi_{\mathrm{G}}(-a) - \mathbb{P}(E_\infty > B) \leqslant \liminf_{\varrho \downarrow 0} \mathbb{P}(\theta^{\mathrm{rad}}_{\varepsilon_\varrho} > t_\varrho(a))$$

and

$$\limsup_{\varrho \downarrow 0} \mathbb{P}(\theta^{\mathrm{rad}}_{\varepsilon_\varrho} > t_\varrho(a)) \leqslant \Phi_{\mathrm{G}}(-a) + \mathbb{P}(E_\infty > B).$$

Sending $B \to \infty$ proves the displayed limit. For the final assertion, fix $\delta > 0$. Since

$$\frac{u_\varrho}{\sigma_N \gamma_{A,N}^{-3/2} \sqrt{L_\varrho}} \longrightarrow 0,$$

monotonicity sandwiches the perturbed survival probability, for all sufficiently small ϱ, between those at $t_\varrho(a-\delta)$ and $t_\varrho(a+\delta)$, up to a uniformly bounded time shift. The first part and continuity of $a \mapsto \Phi_{\mathrm{G}}(-a)$ give the result after $\delta \downarrow 0$. $\square$

3.2. Rounding comparison and absorption. We couple the rounded and unrounded vector chains with the same Gaussian matrices. The following estimate quantifies the accumulated rounding error and shows that it is polylogarithmically larger than the grid size on the cutoff time scale.

Lemma 3.3. *Assume $0 < A < A_c(N)$, and let $x_0 \in \mathbb{R}^N$ be deterministic. Run the unrounded vector chain $X^{(N)}$ from $X_0^{(N)} = x_0$ and the rounded vector chain $Y^{(N)}$ from $Y_0^{(N)} = Q_\varrho(x_0)$ with the same Gaussian matrices. There is $C_N < \infty$, depending only on N, such that the following holds. For $0 < C_0 < \infty$, define*

$$T_\varrho := \lfloor C_0 \log(1/\varrho) \rfloor.$$

There exists $p = p(A, N, C_0)$ satisfying

$$0 \leqslant p \leqslant C_N(1 + C_0)\left(1 + \frac{1}{\gamma_{A,N}}\right), \tag{3.9}$$

and

$$\mathbb{P}\left(\max_{0 \leqslant t \leqslant T_\varrho} \|Y_t^{(N)} - X_t^{(N)}\|_2 > \varrho\,(\log(1/\varrho))^p\right) \to 0 \tag{3.10}$$

as $\varrho \downarrow 0$.

Proof. Step 1: Let $D_t := Y_t^{(N)} - X_t^{(N)}$ and $d_t := \|D_t\|_2$. For the common Gaussian matrix at time $t+1$, set

$$\mathcal{T}_{t+1}(z) := \tanh\Big(\mathsf{W}_{t+1,A}^{(N)} z\Big).$$

The update rules (2.2) and (2.3) give

$$d_{t+1} \leqslant \|\mathcal{T}_{t+1}(Y_t^{(N)}) - \mathcal{T}_{t+1}(X_t^{(N)})\|_2 + \|Q_\varrho(\mathcal{T}_{t+1}(Y_t^{(N)})) - \mathcal{T}_{t+1}(Y_t^{(N)})\|_2.$$

The second term is at most $c_N \varrho$. By the one-Lipschitz property of tanh, the first term is at most

$$\|\mathsf{W}_{t+1,A}^{(N)} Y_t^{(N)} - \mathsf{W}_{t+1,A}^{(N)} X_t^{(N)}\|_2 = \|\mathsf{W}_{t+1,A}^{(N)} D_t\|_2.$$

Thus

$$d_{t+1} \leqslant \|\mathsf{W}_{t+1,A}^{(N)} D_t\|_2 + c_N \varrho. \tag{3.11}$$

Let $\mathcal{F}_t := \sigma(\mathsf{W}_{1,A}^{(N)}, \ldots, \mathsf{W}_{t,A}^{(N)})$, so that D_t is $\mathcal{F}_t$-measurable. Fix a reference unit vector $e_1 \in \mathbb{R}^N$ and define the $\mathcal{F}_t$-measurable unit vector and the multiplier

$$\theta_t := \frac{D_t}{\|D_t\|_2} \mathbb{1}_{\{D_t \neq 0\}} + e_1 \mathbb{1}_{\{D_t = 0\}}, \qquad M_{t+1} := \|\mathsf{W}_{t+1,A}^{(N)} \theta_t\|_2.$$

Conditionally on $\mathcal{F}_t$, the vector θ_t is a fixed unit vector, so the Gaussian weights give

$$\mathsf{W}_{t+1,A}^{(N)} \theta_t \sim \mathcal{N}(0, (A^2/N) I_N),$$

independently of $\mathcal{F}_t$. Thus the multipliers $(M_j)_{j \geqslant 1}$ are independent copies of the linearized radius multiplier $\ell_{A,N}$ in (2.25); in particular, $\mathbb{E} \log M_j = -\gamma_{A,N} < 0$. Moreover $\|\mathsf{W}_{t+1,A}^{(N)} D_t\|_2 = M_{t+1}\, d_t$ holds identically, since both sides vanish on $\{D_t = 0\}$, so (3.11) gives the pathwise bound $d_{t+1} \leqslant M_{t+1} d_t + c_N \varrho$. Since $d_0 \leqslant c_N \varrho$ and each $M_{t+1} \geqslant 0$, induction on t yields $d_t \leqslant U_t$ for all t, where U_t is the scalar affine recursion driven by the same multipliers,

$$U_{t+1} = M_{t+1} U_t + c_N \varrho, \qquad U_0 = c_N \varrho. \tag{3.12}$$

Step 2: Since $M_1 = (A/\sqrt{N})\chi_N$ and χ_N^2 is chi-squared with N degrees of freedom, the Gamma moment formula $\mathbb{E}(\chi_N^2)^{\alpha/2} = 2^{\alpha/2}\Gamma((N+\alpha)/2)/\Gamma(N/2)$ gives, for every $\alpha \geqslant 0$,

$$\log \mathbb{E} M_1^\alpha = \alpha \log \frac{A}{\sqrt{N}} + \frac{\alpha}{2} \log 2 + \log \Gamma\Big(\frac{N+\alpha}{2}\Big) - \log \Gamma\Big(\frac{N}{2}\Big). \tag{3.13}$$

In particular all positive moments are finite, and differentiating (3.13) twice in α gives

$$\frac{\mathrm{d}^2}{\mathrm{d}\alpha^2} \log \mathbb{E} M_1^\alpha = \frac{1}{4} \psi_1\Big(\frac{N+\alpha}{2}\Big),$$

with ψ_1 the trigamma function. As ψ_1 is positive and decreasing on $(0,\infty)$, this second derivative is bounded on $[0,1]$ by $K := \frac{1}{4}\psi_1(N/2)$. Put

$$\Lambda(\alpha) := \log \mathbb{E} M_1^\alpha, \qquad \alpha_\gamma := \min\Big\{1, \frac{\gamma_{A,N}}{2K}\Big\}.$$

Since $\Lambda(0) = 0$ and $\Lambda'(0) = \mathbb{E} \log M_1 = -\gamma_{A,N}$, Taylor's theorem yields

$$\mathbb{E} M_1^{\alpha_\gamma} \leqslant \exp\Big(-\frac{\alpha_\gamma \gamma_{A,N}}{2}\Big) =: q < 1, \qquad \alpha_\gamma^{-1} \leqslant C_N\Big(1 + \gamma_{A,N}^{-1}\Big).$$

Let $L_\varrho := \log(1/\varrho)$. Markov's inequality and a union bound therefore give, for every $p_0 > 0$,

$$\mathbb{P}\Bigg(\max_{0 \leqslant s \leqslant t \leqslant T_\varrho} \prod_{j=s+1}^{t} M_j > L_\varrho^{p_0}\Bigg) \leqslant L_\varrho^{-\alpha_\gamma p_0} \sum_{0 \leqslant s \leqslant t \leqslant T_\varrho} q^{t-s} \leqslant \frac{T_\varrho + 1}{1-q} L_\varrho^{-\alpha_\gamma p_0}.$$

We may choose p_0 so that $\alpha_\gamma p_0 > 2$ and, after increasing C_N,

$$p_0 \leqslant C_N(1 + C_0)\Big(1 + \gamma_{A,N}^{-1}\Big).$$

Since $T_\varrho = O(L_\varrho)$, this probability tends to zero. Thus

$$\mathbb{P}\left(\max_{0\leqslant s\leqslant t\leqslant T_\varrho} \prod_{j=s+1}^{t} M_j > L_\varrho^{p_0}\right) \to 0. \tag{3.14}$$

On the complementary event, the explicit solution of (3.12) gives, for every $t \leqslant T_\varrho$,

$$U_t \leqslant c_N \varrho \sum_{s=0}^{t} \prod_{j=s+1}^{t} M_j \leqslant C_N \varrho\, T_\varrho (\log(1/\varrho))^{p_0} \leqslant \varrho(\log(1/\varrho))^{p_0 + C_N(1+C_0)}$$

for all sufficiently small ϱ. Set $p := p_0 + \lceil C_N(1 + C_0)\rceil$. After increasing C_N, this exponent satisfies (3.9). Since $d_t \leqslant U_t$, this proves (3.10). $\square$

After entering a ball of radius $K\varrho$, the rounded vector chain is absorbed on a logarithmic time scale. The constants below may depend on (A, N), but not on ϱ, K, or the initial grid point.

Lemma 3.4. *Let $0 < A < A_c(N)$. There are constants $c > 0$ and $C < \infty$ such that, uniformly in $\varrho \in (0, 1)$, $K \geqslant 2$, and deterministic initial grid points $y \in \varrho\mathbb{Z}^N$ satisfying $\|y\|_2 \leqslant K\varrho$,*

$$\mathbb{P}_y\Big(\tau_\varrho^{(N)} > C \log K + r\Big) \leqslant Ce^{-cr}, \qquad r \geqslant 0. \tag{3.15}$$

Here $\mathbb{P}_y$ is the law of the rounded vector chain started from y.

Proof. Step 1: Let

$$K_t := \frac{\widehat{R}_t}{\varrho}.$$

The coordinatewise nearest-grid map satisfies $\|Q_\varrho(z) - z\|_2 \leqslant c_N\varrho$, and tanh is one-Lipschitz coordinatewise, so $\widehat{R}_{t+1} \leqslant \|\mathsf{W}_{t+1,A}^{(N)} Y_t^{(N)}\|_2 + c_N\varrho$. Let $\mathcal{F}_t := \sigma(\mathsf{W}_{1,A}^{(N)}, \ldots, \mathsf{W}_{t,A}^{(N)})$ and fix a reference unit vector $e_1 \in \mathbb{R}^N$. Exactly as in the proof of Lemma 3.3, set

$$\theta_t := \frac{Y_t^{(N)}}{\|Y_t^{(N)}\|_2}\mathbb{1}_{\{Y_t^{(N)}\neq 0\}} + e_1\mathbb{1}_{\{Y_t^{(N)}=0\}}, \qquad M_{t+1} := \|\mathsf{W}_{t+1,A}^{(N)}\theta_t\|_2.$$

As before, $\|\mathsf{W}_{t+1,A}^{(N)} Y_t^{(N)}\|_2 = M_{t+1}\widehat{R}_t$ holds identically, with both sides zero when $Y_t^{(N)} = 0$. The law of M_{t+1} is independent of θ_t, and pathwise

$$K_{t+1} \leqslant M_{t+1}K_t + c_N. \tag{3.16}$$

Consider the affine recursion

$$U_{t+1} = M_{t+1}U_t + c_N, \qquad U_0 = K,$$

The multipliers satisfy the hypotheses of Lemma 2.3. Since $K_0 \leqslant K = U_0$, (3.16) gives $K_t \leqslant U_t$. Hence there are $K_* = K_*(A, N) < \infty$ and $c_1, c_2, c_3 > 0$ such that, with $\kappa := \inf\{t \geqslant 0 : K_t \leqslant K_*\}$,

$$\mathbb{P}(\kappa > c_1 \log K + r) \leqslant c_2 e^{-c_3 r}, \qquad r \geqslant 0. \tag{3.17}$$

Step 2: Once $K_t \leqslant K_*$, the rounded vector chain has a uniform positive probability of absorption in the next step. By the Gaussianity of the weight matrix $\mathsf{W}_{t+1,A}^{(N)}$, there is a positive probability $p_* = p_*(A,N,K_*) > 0$ such that

$$\mathbb{P}\Big(\|\mathsf{W}_{t+1,A}^{(N)}\|_{\ell^2\to\ell^2} \leqslant \frac{1}{2K_*}\Big) \geqslant p_*.$$

On this small operator norm event, if $\widehat{R}_t \leqslant K_*\varrho$, then

$$\|\tanh(\mathsf{W}_{t+1,A}^{(N)} Y_t^{(N)})\|_\infty \leqslant \|\mathsf{W}_{t+1,A}^{(N)} Y_t^{(N)}\|_2 \leqslant \frac{\varrho}{2},$$

and hence $Y_{t+1}^{(N)} = 0$. Thus each visit to $\{K_t \leqslant K_*\}$ is followed by absorption with conditional probability at least p_*. To control a failed attempt, start with $K_0 \leqslant K_*$ and define

$$\kappa^+ := \inf\{t \geqslant 1 : K_t \leqslant K_*\}.$$

After one step, the rounded vector chain satisfies $K_1 \leqslant M_1 K_* + c_N$, where M_1 has law $(A/\sqrt{N})\chi_N$. Hence (3.17), conditionally applied from time one with initial radius $\max\{2, K_1\}$, gives

$$\mathbb{P}\big(\kappa^+ > 1 + c_1 \log(\max\{2,K_1\}) + r \,\big|\, \mathcal{F}_1\big) \leqslant c_2 e^{-c_3 r}, \qquad r \geqslant 0.$$

Choose $s_0 \in (0, c_3)$. The layer-cake formula and $K_1 \leqslant M_1 K_* + c_N$ give

$$B := \sup_{\varrho\in(0,1)} \sup_{K_0\leqslant K_*} \mathbb{E} e^{s_0\kappa^+} \leqslant C\mathbb{E}(\max\{2, M_1K_* + c_N\})^{c_1 s_0} < \infty,$$

since χ_N has moments of every positive order.

The one-step absorption event above is independent of the starting point in the fixed ball, and therefore

$$\sup_{\varrho\in(0,1)} \sup_{K_0\leqslant K_*} \mathbb{P}(Y_1^{(N)} \neq 0) \leqslant 1 - p_*.$$

For $s \in (0, s_0)$, Hölder's inequality consequently gives

$$q_s := \sup_{\varrho\in(0,1)} \sup_{K_0\leqslant K_*} \mathbb{E}\Big[e^{s\kappa^+} \mathbb{1}_{\{Y_1^{(N)}\neq 0\}}\Big] \leqslant B^{s/s_0}(1-p_*)^{1-s/s_0}.$$

The right-hand side converges to $1 - p_*$ as $s \downarrow 0$. We may thus fix $s \in (0, s_0)$ sufficiently small that $q_s < 1$.

Step 3: For $n \geqslant 1$, set

$$F_{s,n} := \sup_{\varrho\in(0,1)} \sup_{K_0\leqslant K_*} \mathbb{E} e^{s(\min\{\tau_\varrho^{(N)}, n\})} < \infty.$$

On $\{Y_1^{(N)} = 0\}$ we have $\tau_\varrho^{(N)} = 1$. On its complement, the strong Markov property at κ^+, together with $K_{\kappa^+} \leqslant K_*$, gives

$$F_{s,n} \leqslant e^s + q_s F_{s,n}.$$

Consequently $F_{s,n} \leqslant e^s/(1-q_s)$ uniformly in n. Monotone convergence now shows that

$$\sup_{\varrho\in(0,1)} \sup_{K_0\leqslant K_*} \mathbb{E} e^{s\tau_\varrho^{(N)}} \leqslant \frac{e^s}{1-q_s} < \infty.$$

Markov's inequality gives $C_0 < \infty$ and $c_0 > 0$ such that, uniformly in ϱ and deterministic starts with $K_0 \leqslant K_*$,

$$\mathbb{P}\Big(\tau_\varrho^{(N)} > u\Big) \leqslant C_0 e^{-c_0 u}, \qquad u \geqslant 0. \tag{3.18}$$

Combining (3.17) with (3.18) and the Markov property at κ gives, for $r \geqslant 0$,

$$\mathbb{P}\Big(\tau_\varrho^{(N)} > c_1 \log K + r\Big) \leqslant c_2 e^{-c_3 r/2} + C_0 e^{-c_0 r/2} \leqslant (c_2 + C_0) e^{-\frac{1}{2}\min\{c_3,c_0\} r}.$$

After increasing the prefactor and the coefficient of $\log K$, this is (3.15). $\square$

3.3. Proof of the cutoff theorem.

Proof of Theorem 1.4. Fix $a \in \mathbb{R}$ and write $\ell_\varrho := \log(1/\varrho)$ and $t_\varrho := t_\varrho(a)$. By (2.5), it is enough to prove $\mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \to \Phi_{\mathrm{G}}(-a)$.

Step 1: Since $L_\varrho = \ell_\varrho + \log R_0$ and a is fixed, (3.3) gives $t_\varrho = O(\ell_\varrho)$. Choose $C_0 < \infty$ such that, for all sufficiently small ϱ, $0 \leqslant t_\varrho \leqslant T_\varrho := \lfloor C_0 \ell_\varrho \rfloor$. Let $p = p(A, N, C_0)$ be the exponent from Lemma 3.3, put $d_\varrho := \varrho \ell_\varrho^p$ and $M_\varrho := \ell_\varrho^{p+3}$. For the synchronous coupling, define

$$\mathcal{G}_\varrho := \left\{ \max_{0 \leqslant s \leqslant T_\varrho} \|Y_s^{(N)} - X_s^{(N)}\|_2 \leqslant d_\varrho \right\}, \qquad \mathbb{P}(\mathcal{G}_\varrho^{\mathrm{c}}) = o(1), \tag{3.19}$$

where the estimate follows from (3.10).

We compare absorption with the unrounded entrance times $\Theta_\varrho^- := \theta_{d_\varrho}^{\mathrm{rad}}$ and $\Theta_\varrho^+ := \theta_{M_\varrho \varrho}^{\mathrm{rad}}$. Since both $\log(R_0/d_\varrho)$ and $\log(R_0/(M_\varrho \varrho))$ satisfy the hypotheses of Proposition 3.2,

$$\mathbb{P}(\Theta_\varrho^- > t_\varrho) \longrightarrow \Phi_{\mathrm{G}}(-a), \qquad \mathbb{P}(\Theta_\varrho^+ > t_\varrho) \longrightarrow \Phi_{\mathrm{G}}(-a) \tag{3.20}$$

as $\varrho \downarrow 0$.

Step 2: The lower bound. On $\{\tau_\varrho^{(N)} \leqslant t_\varrho\} \cap \mathcal{G}_\varrho$, absorption gives $Y_{\tau_\varrho^{(N)}}^{(N)} = 0$, while $\tau_\varrho^{(N)} \leqslant t_\varrho \leqslant T_\varrho$; hence $R_{\tau_\varrho^{(N)}} = \|X_{\tau_\varrho^{(N)}}^{(N)} - Y_{\tau_\varrho^{(N)}}^{(N)}\|_2 \leqslant d_\varrho$ and thus $\Theta_\varrho^- \leqslant \tau_\varrho^{(N)}$. Consequently,

$$\{\Theta_\varrho^- > t_\varrho\} \cap \mathcal{G}_\varrho \subseteq \{\tau_\varrho^{(N)} > t_\varrho\}, \qquad \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \geqslant \mathbb{P}(\Theta_\varrho^- > t_\varrho) - o(1), \tag{3.21}$$

where we used (3.19) in the last estimate.

Step 3: The upper bound. Let $c_{\mathrm{abs}} > 0$ and $C_{\mathrm{abs}} < \infty$ be constants for which (3.15) holds. Choose $B > C_{\mathrm{abs}}(p+3)$ and set $h_\varrho := \lceil B \log \ell_\varrho \rceil$, $K_\varrho := 2M_\varrho$, and $r_\varrho := h_\varrho - C_{\mathrm{abs}} \log K_\varrho$. Since $\log K_\varrho = \log 2 + (p+3) \log \ell_\varrho$, we have $r_\varrho \to \infty$, while $h_\varrho = o(\sqrt{L_\varrho})$ and $t_\varrho - h_\varrho \geqslant 0$ for all sufficiently small ϱ.

Write $\mathcal{E}_\varrho := \{\Theta_\varrho^+ \leqslant t_\varrho - h_\varrho\} \cap \mathcal{G}_\varrho$. On $\mathcal{E}_\varrho$, we have $\Theta_\varrho^+ \leqslant T_\varrho$, and therefore

$$\widehat{R}_{\Theta_\varrho^+} = \|Y_{\Theta_\varrho^+}^{(N)}\|_2 \leqslant \|X_{\Theta_\varrho^+}^{(N)}\|_2 + \|Y_{\Theta_\varrho^+}^{(N)} - X_{\Theta_\varrho^+}^{(N)}\|_2 \leqslant M_\varrho \varrho + d_\varrho \leqslant K_\varrho \varrho$$

for all sufficiently small ϱ. Define the $\mathcal{F}_{\Theta_\varrho^+}$-measurable event $\mathcal{A}_\varrho := \{\Theta_\varrho^+ \leqslant t_\varrho - h_\varrho,\ \widehat{R}_{\Theta_\varrho^+} \leqslant K_\varrho \varrho\}$. The preceding bound gives $\mathcal{E}_\varrho \subseteq \mathcal{A}_\varrho$. Therefore

$$\begin{aligned}
\mathbb{P}(\tau_\varrho^{(N)} > t_\varrho, \mathcal{E}_\varrho) &\leqslant \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho, \mathcal{A}_\varrho) \\
&= \mathbb{E}\left[\mathbb{1}_{\mathcal{A}_\varrho} \mathbb{P}_{Y_{\Theta_\varrho^+}^{(N)}} \left(\tau_\varrho^{(N)} > t_\varrho - \Theta_\varrho^+ \right) \right] \\
&\leqslant \mathbb{E}\left[\mathbb{1}_{\mathcal{A}_\varrho} \mathbb{P}_{Y_{\Theta_\varrho^+}^{(N)}} \left(\tau_\varrho^{(N)} > h_\varrho \right) \right] \\
&\leqslant C_{\mathrm{abs}} e^{-c_{\mathrm{abs}} r_\varrho} = o(1).
\end{aligned}$$

Here $\mathbb{P}_y$ denotes the law of the rounded vector chain started from y. The equality is the strong Markov property at Θ_ϱ^+; the next inequality uses $t_\varrho - \Theta_\varrho^+ \geqslant h_\varrho$ on $\mathcal{A}_\varrho$; and the last one follows from Lemma 3.4, since the entrance radius is at most $K_\varrho \varrho$ and $h_\varrho = C_{\mathrm{abs}} \log K_\varrho + r_\varrho$.

The event inclusion and its union-bound consequence are

$$\begin{aligned}
\{\tau_\varrho^{(N)} > t_\varrho\} &\subseteq \{\Theta_\varrho^+ > t_\varrho - h_\varrho\} \cup \mathcal{G}_\varrho^{\mathrm{c}} \cup \big(\{\tau_\varrho^{(N)} > t_\varrho\} \cap \mathcal{E}_\varrho\big), \\
\mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) &\leqslant \mathbb{P}(\Theta_\varrho^+ > t_\varrho - h_\varrho) + \mathbb{P}(\mathcal{G}_\varrho^{\mathrm{c}}) + \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho, \mathcal{E}_\varrho) \\
&\leqslant \mathbb{P}(\Theta_\varrho^+ > t_\varrho - h_\varrho) + o(1).
\end{aligned} \tag{3.22}$$

Step 4: The profile. By the level calculations in Step 1, the conclusion of (3.20) also holds for Θ_ϱ^+ at $t_\varrho - h_\varrho$, because $h_\varrho = o(\sqrt{L_\varrho})$. Combining these limits with (3.21) and (3.22) yields

$$\Phi_{\mathrm{G}}(-a) \leqslant \liminf_{\varrho \downarrow 0} \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \leqslant \limsup_{\varrho \downarrow 0} \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \leqslant \Phi_{\mathrm{G}}(-a).$$

Thus the absorption tail converges to $\Phi_{\mathrm{G}}(-a)$, and (2.5) gives (1.19).

Since A and N are fixed, $\sigma_N \gamma_{A,N}^{-3/2}\sqrt{L_\varrho} = o(L_\varrho/\gamma_{A,N})$. The profile just proved satisfies $\Phi_{\mathrm{G}}(-a) \to 1$ as $a \to -\infty$ and $\Phi_{\mathrm{G}}(-a) \to 0$ as $a \to \infty$. Therefore (1.13) holds with center $L_\varrho/\gamma_{A,N}$ and window $\sigma_N \gamma_{A,N}^{-3/2}\sqrt{L_\varrho}$. □

Remark 3.5. The dependence on $A_c(N) - A$ is explicit in the leading cutoff scale. Since

$$\gamma_{A,N} = \log \frac{A_c(N)}{A} = \frac{A_c(N) - A}{A_c(N)} + O_N((A_c(N) - A)^2) \qquad \text{as } A \uparrow A_c(N),$$

the cutoff center and window behave as

$$\frac{L_\varrho}{\gamma_{A,N}} \asymp_N \frac{L_\varrho}{A_c(N) - A}, \qquad \frac{\sigma_N}{\gamma_{A,N}^{3/2}}\sqrt{L_\varrho} \asymp_N \frac{\sqrt{L_\varrho}}{(A_c(N) - A)^{3/2}}.$$

Here $\asymp_N$ denotes comparison up to constants depending only on N. For each fixed C_0, the exponent in Lemma 3.3 can be chosen so that $p = O_{N,C_0}(\gamma_{A,N}^{-1})$ as $A \uparrow A_c(N)$, by (3.9).

The constants in Lemma 3.4 depend on (A, N) and may deteriorate as $A \uparrow A_c(N)$. For fixed $A < A_c(N)$, they affect only the polylogarithmic comparison scale and the $O(\log\log(1/\varrho))$ absorption buffer, not the cutoff center or window.

4. Subcritical cutoff in the dimension

Fix $\varrho \in (0, 1)$ and let $N \to \infty$. We first use global contraction of the deterministic rounded mean map and a normalized stopped-error estimate to track the squared-radius chain until the orbit reaches the scale $(\log N)^{-1}$. The fixed-precision map then makes the radius polynomially small in N, and a Gaussian maximum estimate gives absorption in one further step, while the same tracking estimate rules out absorption one step earlier.

4.1. Deterministic rounded dynamics. Recall the rounded squared-radius transition and centered noise decomposition from (2.19) and (2.22). The deterministic orbit and terminal-scale entrance time are

$$\bar{h}_{0,N} := \frac{1}{N\varrho^2}\|Q_\varrho x_N\|_2^2, \qquad \bar{h}_{t+1,N} = V_{A,\varrho}(\bar{h}_{t,N}), \tag{4.1}$$

and

$$\bar{t}_N := \inf\left\{t \in \mathbb{N}_0 : \bar{h}_{t,N} \leqslant \frac{1}{\log N}\right\}. \tag{4.2}$$

With the tie convention fixed in Section 2, the one-dimensional nearest-grid map satisfies

$$|Q_1(u)| \leqslant |Q_1(v)| \quad \text{whenever } |u| \leqslant |v|. \tag{4.3}$$

The threshold A_{lat} is defined by (2.30); equivalently, $A < A_{\mathrm{lat}}$ means that $L_A(h) < h$ for every $h > 0$.

The relevant consequence of the lattice-subcritical condition is global contraction.

Lemma 4.1. *The threshold in* (2.30) *belongs to* $(0, 1)$. *If* $A < A_{\mathrm{lat}}$, *then there exists* $\theta_A \in (0, 1)$ *such that*

$$L_A(h) \leqslant \theta_A h \qquad \textit{for every } h > 0. \tag{4.4}$$

Consequently,

$$V_{A,\varrho}(h) \leqslant \theta_A h, \qquad h \geqslant 0. \tag{4.5}$$

Proof. For every $\alpha > 0$, $m(\alpha) > 0$, since $\mathbb{P}(|\alpha G| > 1/2) > 0$. The map $\alpha \mapsto m(\alpha)$ is continuous on $(0, \infty)$ by dominated convergence on compact α-intervals: if $\alpha \leqslant R$, then

$$Q_1(\alpha G)^2 \leqslant (|\alpha G| + \tfrac{1}{2})^2 \leqslant (R|G| + \tfrac{1}{2})^2.$$

Near the origin the zero bin gives a sharper estimate. Since $Q_1(u) = 0$ on $\{|u| \leqslant 1/2\}$ and $|Q_1(u)| \leqslant |u| + 1/2 \leqslant 2|u|$ on $\{|u| > 1/2\}$,

$$0 \leqslant \frac{m(\alpha)}{\alpha^2} \leqslant 4\mathbb{E}\big[G^2 \mathbb{1}_{\{|G|>1/(2\alpha)\}}\big] \xrightarrow[\alpha\downarrow 0]{} 0. \tag{4.6}$$

At infinity, nearest-integer rounding satisfies $|Q_1(u) - u| \leqslant 1/2$. Hence $Q_1(\alpha G)/\alpha \to G$ pointwise and

$$\left(\frac{|Q_1(\alpha G)|}{\alpha}\right)^2 \leqslant \left(|G| + \frac{1}{2\alpha}\right)^2 \leqslant (|G| + 1)^2 \qquad \text{for } \alpha \geqslant \frac{1}{2}.$$

Dominated convergence therefore gives

$$\frac{m(\alpha)}{\alpha^2} = \mathbb{E}\left[\left(\frac{Q_1(\alpha G)}{\alpha}\right)^2\right] \xrightarrow[\alpha\to\infty]{} \mathbb{E}[G^2] = 1. \tag{4.7}$$

It follows from (4.6) that $\alpha^2/m(\alpha) \to \infty$ as $\alpha \downarrow 0$, while (4.7) gives $\alpha^2/m(\alpha) \to 1$ as $\alpha \to \infty$. Since $\alpha^2/m(\alpha)$ is continuous and positive on $(0, \infty)$, these two endpoint limits imply that its infimum is positive: outside a compact subinterval of $(0, \infty)$ the ratio is bounded below, while on that compact subinterval it has a positive minimum. Hence the threshold A_{lat} belongs to $(0, 1]$. The upper bound is strict. Indeed, for $\alpha = 1/2$,

$$Q_1(\alpha G)^2 \geqslant 1 \quad \text{on } \{|G| > 1\},$$

and hence

$$m(1/2) \geqslant \mathbb{P}(|G| > 1) > \frac{1}{4}.$$

The final strict inequality can be checked by an elementary Gaussian tail bound. Therefore $A_{\text{lat}}^2 \leqslant (1/2)^2/m(1/2) < 1$.

By the change of variables $\alpha = A\sqrt{h}$,

$$\frac{L_A(h)}{h} = A^2 \frac{m(A\sqrt{h})}{(A\sqrt{h})^2}, \qquad h > 0.$$

Thus $A < A_{\text{lat}}$ implies $L_A(h) < h$ for every $h > 0$. The same endpoint estimates give

$$\frac{L_A(h)}{h} \to 0 \quad \text{as } h \downarrow 0, \qquad \frac{L_A(h)}{h} \to A^2 < 1 \quad \text{as } h \to \infty.$$

Since $h \mapsto L_A(h)/h$ is continuous on $(0, \infty)$, choose $0 < r < R < \infty$ so that the ratio is bounded away from one on $(0, r] \cup [R, \infty)$. Compactness gives the same bound on $[r, R]$, and hence (4.4).

The inequality $|\tanh z| \leqslant |z|$ and (4.3) give, pointwise in G,

$$\left|Q_1\Big(\varrho^{-1}\tanh(\varrho A\sqrt{h}\, G)\Big)\right| \leqslant \left|Q_1(A\sqrt{h}\, G)\right|.$$

Taking squares and expectations yields $V_{A,\varrho}(h) \leqslant L_A(h)$ for $h > 0$. The case $h = 0$ is immediate, since both sides vanish. Combining this with (4.4) proves (4.5). □

Let $\bar{\Phi}(x) := \mathbb{P}(G > x)$ and $\phi(x) := (2\pi)^{-1/2}\exp(-x^2/2)$. For $0 < \varrho < 1$ set

$$b_{\varrho,k} := \varrho^{-1}\operatorname{arctanh}\big(\varrho(k+\tfrac{1}{2})\big), \qquad \mathcal{K}_\varrho := \{k \geqslant 0 : \varrho(k+\tfrac{1}{2}) < 1\}. \tag{4.8}$$

Then

$$V_{A,\varrho}(h) = 2\sum_{k\in\mathcal{K}_\varrho}(2k+1)\bar{\Phi}\left(\frac{b_{\varrho,k}}{A\sqrt{h}}\right), \qquad V'_{A,\varrho}(h) = \sum_{k\in\mathcal{K}_\varrho}(2k+1)\frac{b_{\varrho,k}}{Ah^{3/2}}\phi\left(\frac{b_{\varrho,k}}{A\sqrt{h}}\right). \tag{4.9}$$

Indeed, $Q_1(u)^2 = \sum_{k\geqslant 0}(2k+1)\mathbb{1}_{\{|u|>k+\frac{1}{2}\}}$ outside Gaussian-null boundary events, and the preimage of the layer $k+\frac{1}{2}$ under $u \mapsto \varrho^{-1}\tanh(\varrho u)$ is $b_{\varrho,k}$. Since $\mathcal{K}_\varrho$ is finite, differentiating term by term is immediate. The small-radius estimate in (4.11) also gives $V'_{A,\varrho}(h) \to 0$ as $h \downarrow 0$. We use this one-sided extension at the origin.

On compact subsets of $(0,\infty)$, the layer formulas give positivity and a uniform derivative bound. Near the origin, the positive threshold $b_{\varrho,0}$ yields exponential decay in $1/h$.

We will use regularity of the fixed-precision mean map both away from and near the origin.

Lemma 4.2. *Fix $\varrho \in (0,1)$. For every $0 < r < R < \infty$ there are constants $m, L \in (0,\infty)$, depending on A, ϱ, r, R, such that*

$$m \leqslant V_{A,\varrho}(h), \qquad V'_{A,\varrho}(h) \leqslant L, \qquad r \leqslant h \leqslant R. \tag{4.10}$$

For every $r \in (0,1)$ there are constants $c, C \in (0,\infty)$, depending on A, ϱ, r, such that, for $0 < h \leqslant r$,

$$V_{A,\varrho}(h) \leqslant C\exp\Big(-\frac{c}{h}\Big), \qquad V'_{A,\varrho}(h) \leqslant Ch^{-2}\exp\Big(-\frac{c}{h}\Big). \tag{4.11}$$

Proof. Since $0 < \varrho < 1$, the set $\mathcal{K}_\varrho$ is finite and nonempty; in particular $0 \in \mathcal{K}_\varrho$ and

$$b_{\varrho,0} = \varrho^{-1}\operatorname{arctanh}(\varrho/2) > 0.$$

The representation (4.9) is a finite sum of continuous functions of $h > 0$. Hence $V_{A,\varrho}$ is continuous on $(0,\infty)$. Moreover, for every $h > 0$,

$$V_{A,\varrho}(h) \geqslant 2\bar{\Phi}\left(\frac{b_{\varrho,0}}{A\sqrt{h}}\right) > 0.$$

It follows that

$$m := \min_{h\in[r,R]} V_{A,\varrho}(h) > 0.$$

Similarly, (4.9) is a finite sum of continuous functions on $[r,R]$. Thus

$$L := \max_{h\in[r,R]} V'_{A,\varrho}(h) < \infty,$$

which proves (4.10).

For the small-radius bounds, $C < \infty$ may change from line to line and depends only on A, ϱ, r. Put $b_0 := b_{\varrho,0}$ and

$$c := \frac{b_0^2}{2A^2}.$$

Since $k \mapsto b_{\varrho,k}$ is increasing on $\mathcal{K}_\varrho$, we have $b_{\varrho,k} \geqslant b_0$ for every $k \in \mathcal{K}_\varrho$. Using $\bar{\Phi}(x) \leqslant \exp(-x^2/2)$ for $x \geqslant 0$ in (4.9), we obtain, for $h > 0$,

$$V_{A,\varrho}(h) \leqslant 2\sum_{k\in\mathcal{K}_\varrho}(2k+1)\exp\left(-\frac{b_{\varrho,k}^2}{2A^2h}\right) \leqslant C\exp\Big(-\frac{c}{h}\Big).$$

For the derivative, the same lower bound on the thresholds and (4.9) give

$$V'_{A,\varrho}(h) \leqslant \frac{1}{\sqrt{2\pi}}\sum_{k\in\mathcal{K}_\varrho}(2k+1)\frac{b_{\varrho,k}}{Ah^{3/2}}\exp\left(-\frac{b_{\varrho,k}^2}{2A^2h}\right) \leqslant Ch^{-3/2}\exp\Big(-\frac{c}{h}\Big).$$

If $0 < h \leqslant r < 1$, then $h^{-3/2} \leqslant h^{-2}$, and the desired derivative estimate follows after increasing C. This proves (4.11). $\square$

4.2. Normalized radius tracking. Assume $A < A_{\mathrm{lat}}$, put $a_N = (\log N)^{-1}$, and consider an orbit $h_{t+1,N} = V_{A,\varrho}(h_{t,N})$ with $h_{0,N} \leqslant C_0$. For $\delta \in (0, 1/4)$, set

$$L_{t,N} := \sup_{\substack{u \geqslant 0 \\ |u - h_{t,N}| \leqslant \delta(h_{t,N} + a_N)}} V'_{A,\varrho}(u), \qquad \beta_{t,N} := \frac{h_{t,N} + a_N}{h_{t+1,N} + a_N}, \qquad \alpha_{t,N} := L_{t,N}\beta_{t,N}.$$

The normalized error is the quantity that may be amplified along the rounded deterministic orbit; the estimate below controls this amplification.

Lemma 4.3. *There is a constant $C < \infty$, depending only on A, ϱ, δ, C_0, such that, for all $T_N \in \mathbb{N}_0$ and all sufficiently large N,*

$$\prod_{u=s}^{t-1}(1 \vee \alpha_{u,N}) + \beta_{s,N} \prod_{u=s+1}^{t-1}(1 \vee \alpha_{u,N}) \leqslant \exp\{C(\log\log N)^2\} \tag{4.12}$$

for all $0 \leqslant s < t \leqslant T_N$.

Proof. Let $\tau_r := \inf\{t : h_{t,N} \leqslant r\}$ and $\tau_a := \inf\{t : h_{t,N} \leqslant a_N\}$. Choose $r \in (0, 1/2)$ and apply the small-radius estimates in (4.11) with radius $2r$. Since $h_{0,N} \leqslant C_0$ and (4.5) gives $h_{t,N} \leqslant \theta_A^t C_0$, the number of times with $h_{t,N} \geqslant r$ is at most a constant depending on A, r, C_0. On this part of the orbit, $h_{t,N} \in [r, C_0]$. Since $a_N \leqslant r$ for large N and $\delta < 1/4$, the interval defining $L_{t,N}$ is contained in $[r/2, 2C_0]$, after increasing the upper endpoint if necessary. The lower bound in (4.10) applied to $V_{A,\varrho}(h_{t,N}) = h_{t+1,N}$, and the derivative bound in the same display, show that $\alpha_{t,N}$ and $\beta_{t,N}$ are uniformly bounded for all $t < \tau_r$.

For the tail $a_N < h_{t,N} \leqslant r$, (4.11) gives

$$V_{A,\varrho}(h) \leqslant C\exp(-c/h), \qquad V'_{A,\varrho}(h) \leqslant Ch^{-2}\exp(-c/h).$$

The contraction implies that the number of times with $a_N < h_{t,N} \leqslant r$ is at most $C\log\log N$. During these times the interval defining $L_{t,N}$ is contained in $[0, 2r]$ for all large N. More precisely, if u belongs to this interval, then

$$u \geqslant h_{t,N} - \delta(h_{t,N} + a_N) \geqslant (1 - 2\delta)h_{t,N}, \qquad u \leqslant h_{t,N} + \delta(h_{t,N} + a_N) \leqslant (1 + 2\delta)h_{t,N},$$

because $a_N < h_{t,N}$ and $\delta < 1/4$. Hence u is comparable to $h_{t,N}$ on the interval. The preceding display then gives $L_{t,N} \leqslant C$. Moreover $V_{A,\varrho}(h_{t,N}) + a_N \geqslant a_N$, so $\beta_{t,N} \leqslant C\log N$. Thus $\alpha_{t,N} \leqslant C\log N$ on the tail. In the terminal regime $h_{t,N} \leqslant a_N$. The interval defining $L_{t,N}$ is then contained in $[0, 2a_N]$ for all large N. Since $u \mapsto u^{-2}\exp(-c/u)$ is increasing in a sufficiently small right-neighborhood of the origin, the small-radius derivative estimate gives

$$L_{t,N} \leqslant Ca_N^{-2}\exp\left(-\frac{c}{2a_N}\right) = C(\log N)^2 N^{-c/2} \leqslant N^{-c/3}$$

for all sufficiently large N, after decreasing $c > 0$ if necessary. Moreover, $h_{t+1,N} + a_N \geqslant a_N$, and therefore

$$\beta_{t,N} = \frac{h_{t,N} + a_N}{h_{t+1,N} + a_N} \leqslant 2.$$

Consequently, after decreasing c once more, $\alpha_{t,N} = L_{t,N}\beta_{t,N} \leqslant N^{-c}$. The factors $1 \vee \alpha_{t,N}$ are therefore uniformly bounded for $O(1)$ times, bounded by $C\log N$ for at most $C\log\log N$ times, and equal to 1 after the orbit reaches a_N, for all sufficiently large N. Hence their products are bounded by $\exp\{C(\log\log N)^2\}$. For the second term in (4.12), the extra factor $\beta_{s,N}$ is uniformly bounded before τ_r, bounded by $C\log N$ on the tail, and bounded by 2 after τ_a, so it is absorbed into the same exponential bound. This proves (4.12). ☐

Conditional independence in the radius transition and (4.12) give a stopped second-moment estimate for the normalized error. This is the discrete-time counterpart of the stopped L^2 fluid-limit estimate of Darling and Norris [DN08]; compare also Warnke [War19] for a bounded-increment discrete-time formulation.

These estimates allow us to follow the stochastic radius around the rounded deterministic orbit.

Proposition 4.4. *Assume $A < A_{\mathrm{lat}}$, fix $\varrho \in (0,1)$, and let $x_N \in [-1,1]^N$. Let $\bar{T}_N \in \mathbb{N}_0$ be a sequence such that*

$$\frac{\bar{T}_N}{N} \exp\{C(\log\log N)^2\} \to 0 \quad \textit{for every fixed } C < \infty.$$

Then

$$\sup_{0 \leqslant t \leqslant \bar{T}_N} \frac{\left|\mathscr{H}_t^{(N)} - \bar{h}_{t,N}\right|}{\bar{h}_{t,N} + (\log N)^{-1}} \xrightarrow{\mathbb{P}} 0. \tag{4.13}$$

Proof. Put $a_N = (\log N)^{-1}$ and use the coordinate observable $U_{A,\varrho}$ from (2.18). If $\bar{T}_N = 0$, then the assertion is immediate because $\mathscr{H}_0^{(N)} = \bar{h}_{0,N}$. We may therefore restrict attention to those N for which $\bar{T}_N \geqslant 1$. Since $x_N \in [-1,1]^N$, nearest-grid rounding gives $|Q_\varrho x_{N,i}| \leqslant 1 + \varrho/2$ for every coordinate. Hence the deterministic initial radius satisfies

$$\bar{h}_{0,N} \leqslant C_\varrho := \frac{(1+\varrho/2)^2}{\varrho^2}. \tag{4.14}$$

Moreover, $U_{A,\varrho}(h,G)$ is bounded uniformly in h, because the argument of Q_1 belongs to $[-\varrho^{-1}, \varrho^{-1}]$. For $0 < h \leqslant 1$ the event $\{U_{A,\varrho}(h,G) \neq 0\}$ is contained in $\{A\sqrt{h}\,|G| > b_{\varrho,0}\}$, and hence has probability at most $C\exp(-c/h)$. Therefore, for every integer $p \geqslant 1$,

$$\mathbb{E}U_{A,\varrho}(h,G)^p \leqslant C_p h^p, \qquad h \geqslant 0. \tag{4.15}$$

Indeed, for $0 < h \leqslant 1$ we use $e^{-c/h} \leqslant C_p h^p$, for $h = 0$ the left-hand side vanishes, and for $h \geqslant 1$ the bound follows from the uniform boundedness of $U_{A,\varrho}$ and $h^p \geqslant 1$. Thus this coordinate moment estimate is intrinsic to the radius h and does not involve the terminal scale a_N.

Let $e_t := \mathscr{H}_t^{(N)} - \bar{h}_{t,N}$ and let $\mathcal{F}_t$ be the natural filtration. By (2.22) and (2.23), $\mathbb{E}[\widehat{\zeta}_{t+1}^{(N)} \mid \mathcal{F}_t] = 0$. Since $\mathscr{H}_{t+1}^{(N)}$ is an average of N conditionally independent copies of $U_{A,\varrho}(\mathscr{H}_t^{(N)}, G)$, the conditional variance is bounded by

$$\mathbb{E}[(\widehat{\zeta}_{t+1}^{(N)})^2 \mid \mathcal{F}_t] \leqslant \frac{1}{N}\mathbb{E}U_{A,\varrho}(\mathscr{H}_t^{(N)}, G)^2.$$

By (4.15), on the event $\left|\mathscr{H}_t^{(N)} - \bar{h}_{t,N}\right| \leqslant \delta(\bar{h}_{t,N} + a_N)$, we have

$$\mathscr{H}_t^{(N)} \leqslant (1+\delta)\bar{h}_{t,N} + \delta a_N \leqslant C(\bar{h}_{t,N} + a_N).$$

Therefore

$$\mathbb{E}[(\widehat{\zeta}_{t+1}^{(N)})^2 \mid \mathcal{F}_t] \leqslant \frac{C(\bar{h}_{t,N} + a_N)^2}{N}. \tag{4.16}$$

Fix $\delta \in (0, 1/4)$, and let τ_δ be the first time at which $|e_t| > \delta(\bar{h}_{t,N} + a_N)$, and set

$$\mathcal{R}_t := \frac{e_t}{\bar{h}_{t,N} + a_N}.$$

Define the $\mathcal{F}_t$-measurable difference quotient

$$D_{t,N} := \begin{cases} \dfrac{V_{A,\varrho}(\mathscr{H}_t^{(N)}) - V_{A,\varrho}(\bar{h}_{t,N})}{\mathscr{H}_t^{(N)} - \bar{h}_{t,N}}, & \mathscr{H}_t^{(N)} \neq \bar{h}_{t,N}, \\ V'_{A,\varrho}(\bar{h}_{t,N}), & \mathscr{H}_t^{(N)} = \bar{h}_{t,N}. \end{cases}$$

On $\{t < \tau_\delta\}$, the interval between $\mathscr{H}_t^{(N)}$ and $\bar{h}_{t,N}$ is contained in the interval used to define $L_{t,N}$ (see Lemma 4.3). If the two values are distinct, the mean-value theorem gives $|D_{t,N}| \leqslant L_{t,N}$; if they coincide, the same bound follows directly from the definition of $L_{t,N}$. Moreover, the definitions of e_t and $\widehat{\zeta}_{t+1}^{(N)}$, together with $\bar{h}_{t+1,N} = V_{A,\varrho}(\bar{h}_{t,N})$, give the error recursion

$$e_{t+1} = V_{A,\varrho}(\mathscr{H}_t^{(N)}) - V_{A,\varrho}(\bar{h}_{t,N}) + \widehat{\zeta}_{t+1}^{(N)} = D_{t,N} e_t + \widehat{\zeta}_{t+1}^{(N)}.$$

Hence, on $\{t < \tau_\delta\}$, division by $\bar{h}_{t+1,N} + a_N$ yields

$$\mathcal{R}_{t+1} = A_{t,N}\mathcal{R}_t + \beta_{t,N}\zeta_{t+1}, \qquad \zeta_{t+1} := \frac{\widehat{\zeta}_{t+1}^{(N)}}{\bar{h}_{t,N} + a_N}, \qquad A_{t,N} := \mathbb{1}_{\{t<\tau_\delta\}} \frac{D_{t,N}(\bar{h}_{t,N} + a_N)}{\bar{h}_{t+1,N} + a_N},$$

where $|A_{t,N}| \leqslant \alpha_{t,N}$, and where $\alpha_{t,N}$ and $\beta_{t,N}$ are defined as in Lemma 4.3 for the deterministic orbit $\bar{h}_{t,N}$. The hypotheses of that lemma are satisfied with $C_0 = C_\varrho$, since (4.14) gives $\bar{h}_{0,N} \leqslant C_\varrho$ and $\bar{h}_{t+1,N} = V_{A,\varrho}(\bar{h}_{t,N})$. Let

$$K_N := \sup\left\{1, \prod_{u=s}^{t-1}(1 \vee \alpha_{u,N}) + \beta_{s,N} \prod_{u=s+1}^{t-1} (1 \vee \alpha_{u,N}) : 0 \leqslant s < t \leqslant \bar{T}_N \right\}.$$

By Lemma 4.3, $K_N \leqslant \exp\{C(\log\log N)^2\}$.

For the later application of Lemma 2.2, set $\mathcal{R}_{t,N} = \mathcal{R}_t$, $\tau_N = \tau_\delta$, terminal time $T_N = \bar{T}_N$, and

$$\eta_{t+1,N} := \mathbb{1}_{\{t<\tau_\delta\}}\beta_{t,N}\zeta_{t+1}, \qquad \sigma_{t,N} := \beta_{t,N}.$$

On $\{t < \tau_\delta\}$, the indicators in $A_{t,N}$ and $\eta_{t+1,N}$ are equal to one, so the normalized error recursion above is precisely

$$\mathcal{R}_{t+1,N} = A_{t,N}\mathcal{R}_{t,N} + \eta_{t+1,N}.$$

The coefficient $A_{t,N}$ is $\mathcal{F}_t$-measurable and satisfies $|A_{t,N}| \leqslant \alpha_{t,N}$: this was proved on $\{t < \tau_\delta\}$, while $A_{t,N} = 0$ on its complement. Moreover, $\mathbb{1}_{\{t<\tau_\delta\}}$ is $\mathcal{F}_t$-measurable, and ζ_{t+1} is conditionally centered because its denominator is deterministic. Hence $\mathbb{E}[\eta_{t+1,N} \mid \mathcal{F}_t] = 0$. Moreover, (4.16) gives

$$\mathbb{E}[\eta_{t+1,N}^2 \mid \mathcal{F}_t] = \mathbb{1}_{\{t<\tau_\delta\}}\beta_{t,N}^2 \frac{\mathbb{E}[(\widehat{\zeta}_{t+1}^{(N)})^2 \mid \mathcal{F}_t]}{(\bar{h}_{t,N} + a_N)^2} \leqslant \frac{C\beta_{t,N}^2}{N}.$$

Thus all hypotheses of Lemma 2.2 hold, and the present K_N dominates the amplification constant in (2.35). Since $\mathscr{H}_0^{(N)} = \bar{h}_{0,N}$, we also have $\mathcal{R}_0 = 0$. Therefore (2.38) and the bound on K_N yield

$$\begin{aligned} \mathbb{P}\left(\sup_{0\leqslant t\leqslant \bar{T}_N} |\mathcal{R}_t| > \delta\right) &= \mathbb{P}(\tau_\delta \leqslant \bar{T}_N) \\ &\leqslant \frac{CK_N^2\bar{T}_N}{\delta^2 N} \leqslant \frac{C\bar{T}_N}{\delta^2 N} \exp\{C'(\log\log N)^2\} \longrightarrow 0. \end{aligned}$$

This holds for every fixed $\delta \in (0, 1/4)$, and hence $\sup_{0\leqslant t\leqslant \bar{T}_N}|\mathcal{R}_t| \to 0$ in probability. By the definition of $\mathcal{R}_t$, this is exactly (4.13). $\square$

4.3. Terminal layer and cutoff.

Theorem 4.5 (Fixed-precision subcritical dimension cutoff)**.** *Assume $A < A_{\mathrm{lat}}$, fix $\varrho \in (0,1)$, and let $x_N \in [-1,1]^N$ be deterministic. Suppose that $\bar{t}_N \to \infty$. Then*

$$\mathbb{P}\Big(\tau_\varrho^{(N)} > \bar{t}_N - 1\Big) \to 1, \qquad \mathbb{P}\Big(\tau_\varrho^{(N)} > \bar{t}_N + 2\Big) \to 0. \tag{4.17}$$

Equivalently, by the total-variation/absorption identity (2.5),

$$\|P_{\varrho,A,N}^{\bar{t}_N-1}(Q_\varrho x_N, \cdot) - \delta_0\|_{\mathrm{TV}} \to 1, \qquad \|P_{\varrho,A,N}^{\bar{t}_N+2}(Q_\varrho x_N, \cdot) - \delta_0\|_{\mathrm{TV}} \to 0. \tag{4.18}$$

Thus the rounded vector chain has total-variation cutoff to δ_0 at $\bar{t}_N$ with a window of order one. In particular, for every fixed $\varepsilon \in (0,1)$,

$$t_{\mathrm{mix}}^{(N)}(\varepsilon) = \bar{t}_N + O(1).$$

Proof. Step 1: Put $a_N := (\log N)^{-1}$ and $\bar{T}_N := \bar{t}_N + 1$. By Lemma 4.1 and (4.5),

$$\bar{h}_{t,N} \leqslant C_\varrho \theta_A^t, \qquad t \geqslant 0,$$

for some $\theta_A \in (0,1)$. The definition of $\bar{t}_N$ therefore gives $\bar{t}_N = O(\log\log N)$, and the same bound holds for $\bar{T}_N$. Since $(\log\log N)^2 = o(\log N)$, there is $C_0 < \infty$ such that, for every fixed $C < \infty$ and all sufficiently large N,

$$\frac{\bar{T}_N}{N} \exp\{C(\log\log N)^2\} \leqslant \frac{C_0 \log\log N}{N} \exp\{C(\log\log N)^2\} \longrightarrow 0.$$

The assumptions $A < A_{\mathrm{lat}}$, fixed $\varrho \in (0,1)$, and $x_N \in [-1,1]^N$ are precisely the remaining hypotheses of Proposition 4.4; hence (4.13) holds on $[0, \bar{T}_N]$.

Step 2: The lower bound. Since $\bar{t}_N \to \infty$, we have $\bar{t}_N \geqslant 1$ for all sufficiently large N. By the definition of $\bar{t}_N$,

$$\bar{h}_{\bar{t}_N - 1, N} > a_N.$$

On the event $\{\mathscr{H}_{\bar{t}_N-1}^{(N)} = 0\}$, it follows that

$$\frac{\Big|\mathscr{H}_{\bar{t}_N-1}^{(N)} - \bar{h}_{\bar{t}_N-1,N}\Big|}{\bar{h}_{\bar{t}_N-1,N} + a_N} = \frac{\bar{h}_{\bar{t}_N-1,N}}{\bar{h}_{\bar{t}_N-1,N} + a_N} > \frac{1}{2}.$$

Proposition 4.4 therefore gives $\mathbb{P}(\mathscr{H}_{\bar{t}_N-1}^{(N)} = 0) \to 0$. By (2.20) and the fact that 0 is absorbing, this is equivalent to $\mathbb{P}(\tau_\varrho^{(N)} > \bar{t}_N - 1) \to 1$.

Step 3: Define

$$b_\varrho := \varrho^{-1} \operatorname{arctanh}(\varrho/2). \tag{4.19}$$

By the zero-bin characterization of nearest-grid rounding in Section 2, conditionally on $\mathscr{H}_t^{(N)} = h$, the next rounded vector is zero precisely when

$$\Big|\varrho^{-1} \tanh(\varrho A \sqrt{h}\, G_{t+1,i})\Big| \leqslant \frac{1}{2}, \qquad 1 \leqslant i \leqslant N.$$

Since tanh is increasing, this event is equivalently

$$\max_{1 \leqslant i \leqslant N} \Big|A\sqrt{h}\, G_{t+1,i}\Big| \leqslant b_\varrho. \tag{4.20}$$

Step 4: Set

$$s_N := \bar{t}_N + 1 = \bar{T}_N.$$

By the definition of $\overline{t}_N$, we have $\overline{h}_{\overline{t}_N,N} \leqslant a_N$. The small-radius estimate in Lemma 4.2 therefore yields, for some constants $c, C > 0$ and all sufficiently large N,

$$\overline{h}_{s_N,N} = V_{A,\varrho}(\overline{h}_{\overline{t}_N,N}) \leqslant C \exp\left(-\frac{c}{a_N}\right) = CN^{-c}.$$

Here the inequality is immediate if $\overline{h}_{\overline{t}_N,N} = 0$. Consequently, $\overline{h}_{s_N,N} \log N \to 0$ and

$$(\overline{h}_{s_N,N} + a_N) \log N \to 1.$$

Since $s_N = \overline{T}_N$, the uniform concentration estimate (4.13) applies at time s_N. Writing the exact radius as its deterministic value plus the normalized tracking error, we obtain

$$\begin{aligned}\mathscr{H}_{s_N}^{(N)} \log N &= \overline{h}_{s_N,N} \log N \\ &\quad + \frac{\mathscr{H}_{s_N}^{(N)} - \overline{h}_{s_N,N}}{\overline{h}_{s_N,N} + a_N} (\overline{h}_{s_N,N} + a_N) \log N \xrightarrow{\mathbb{P}} 0.\end{aligned}$$

Step 5: Final absorption. Fix

$$0 < \eta < \frac{b_\varrho^2}{2A^2}$$

and define

$$M_{s_N+1,N} := \max_{1 \leqslant i \leqslant N} |G_{s_N+1,i}|,$$

where G_{s_N+1} is the standard Gaussian vector used in the update from s_N to $s_N + 1$. On the event $\{\mathscr{H}_{s_N}^{(N)} \leqslant \eta / \log N\}$, the survival criterion complementary to (4.20) implies

$$\{\tau_\varrho^{(N)} > s_N + 1\} \cap \left\{\mathscr{H}_{s_N}^{(N)} \leqslant \frac{\eta}{\log N}\right\} \subseteq \left\{M_{s_N+1,N} > \frac{b_\varrho}{A}\sqrt{\frac{\log N}{\eta}}\right\}.$$

Combining the convergence $\mathscr{H}_{s_N}^{(N)} \log N \to 0$ established in Step 4 with Gaussian tails and a union bound over the N coordinates gives

$$\begin{aligned}\mathbb{P}(\tau_\varrho^{(N)} > \overline{t}_N + 2) &\leqslant \mathbb{P}\left(\mathscr{H}_{s_N}^{(N)} \log N > \eta\right) + \mathbb{P}\left(M_{s_N+1,N} > \frac{b_\varrho}{A}\sqrt{\frac{\log N}{\eta}}\right) \\ &\leqslant \mathbb{P}\left(\mathscr{H}_{s_N}^{(N)} \log N > \eta\right) + 2N^{1-b_\varrho^2/(2A^2\eta)} \longrightarrow 0.\end{aligned}$$

Hence (4.17) holds. The total variation formulation (4.18) follows from the absorption identity (2.5). The mixing-time estimate follows from the monotonicity of total variation distance. □

5. Qualitative metastability at fixed precision

Fix $A > 0$ and $\varrho \in (0, 1)$. We construct a rounded metastable well, prove entrance into that well in bounded time, and use one-step concentration to obtain persistence for time $\exp(cN)$. At each fixed N, the finite-state rounded vector chain nevertheless absorbs almost surely.

5.1. Rounded metastable wells. Let $\mathscr{H}_t^{(N)}$ be the rounded squared-radius chain from (2.17). Its mean map is $V_{A,\varrho}$ from (2.21); throughout this section we write

$$V := V_{A,\varrho}.$$

Let

$$M_\varrho := \max_{|u| \leqslant \varrho^{-1}} Q_1(u)^2 < \infty. \tag{5.1}$$

For $\alpha > 0$ define the fixed-precision profile

$$m_\varrho(\alpha) := \mathbb{E}\Big[Q_1\big(\varrho^{-1}\tanh(\varrho\alpha G)\big)^2\Big]. \tag{5.2}$$

The rounding monotonicity in (4.3) implies that V is nondecreasing on $[0,\infty)$. In the layer notation of (4.9),

$$m_\varrho(\alpha) = 2\sum_{k\in\mathcal{K}_\varrho}(2k+1)\bar{\Phi}\left(\frac{b_{\varrho,k}}{\alpha}\right). \tag{5.3}$$

The fixed-precision existence threshold is

$$A_{\mathrm{ex}}(\varrho)^2 := \inf_{\alpha>0}\frac{\alpha^2}{m_\varrho(\alpha)}. \tag{5.4}$$

Define the positive-drift set

$$\mathcal{O}_{A,\varrho} := \{h \in (0, M_\varrho) : V_{A,\varrho}(h) > h\}. \tag{5.5}$$

We call the connected components of $\mathcal{O}_{A,\varrho}$ positive-drift components. The rightmost such component produces a rounded metastable well near its attracting right endpoint; in the vector state space, this well is a grid shell.

The metastable well appears precisely when the rounded mean map has a positive fixed-point component. We first identify these fixed points.

Proposition 5.1. *The threshold $A_{\mathrm{ex}}(\varrho)$ belongs to $(0,\infty)$. If $A > A_{\mathrm{ex}}(\varrho)$, then the rounded squared-radius mean map $V_{A,\varrho}$ has at least two positive fixed points in $(0, M_\varrho)$. There exist $0 < \alpha_- < \alpha_+$ such that*

$$A^2 m_\varrho(\alpha_\pm) = \alpha_\pm^2, \tag{5.6}$$

and $\widehat{h}_\pm := \alpha_\pm^2/A^2$ are fixed points of $V_{A,\varrho}$.

Moreover, $\mathcal{O}_{A,\varrho}$ is nonempty and has finitely many connected components. If (h_-, h_+) is its rightmost positive-drift component, then

$$0 < h_- < h_+ < M_\varrho, \qquad V(h_-) = h_-, \qquad V(h_+) = h_+, \tag{5.7}$$

and $V(h) > h$ on (h_-, h_+). The fixed point h_+ is isolated, and there exists $\eta_0 > 0$ such that

$$h_+ - \eta_0 > h_-, \qquad h_+ + \eta_0 < M_\varrho, \qquad V(h) < h \quad \text{for } h \in (h_+, h_+ + \eta_0]. \tag{5.8}$$

Proof. By the definitions in (4.8), the set $\mathcal{K}_\varrho$ is finite and nonempty, and each threshold $b_{\varrho,k}$ is strictly positive. Hence the finite sum (5.3) shows that m_ϱ is continuous on $(0,\infty)$. Moreover, the first layer gives $m_\varrho(\alpha) > 0$ for every $\alpha > 0$.

For the two endpoints in (5.4), put $b_{\varrho,0} = \varrho^{-1}\operatorname{arctanh}(\varrho/2)$. The coordinate is nonzero only on the event $\{|G| > b_{\varrho,0}/\alpha\}$. Since the coordinate square is bounded by M_ϱ, this gives

$$m_\varrho(\alpha) \leqslant 2M_\varrho\bar{\Phi}\left(\frac{b_{\varrho,0}}{\alpha}\right), \qquad \alpha > 0,$$

and the Gaussian tail bound implies $m_\varrho(\alpha)/\alpha^2 \to 0$ as $\alpha \downarrow 0$. At the other endpoint, the bound $m_\varrho(\alpha) \leqslant M_\varrho$ gives $m_\varrho(\alpha)/\alpha^2 \to 0$ as $\alpha \to \infty$. Therefore

$$\frac{\alpha^2}{m_\varrho(\alpha)} \longrightarrow \infty \quad \text{as } \alpha \downarrow 0 \quad \text{and as } \alpha \to \infty. \tag{5.9}$$

Since the quotient is continuous and strictly positive on $(0,\infty)$, (5.9) implies that the infimum in (5.4) is attained on a compact subinterval of $(0,\infty)$ and is positive and finite. This proves $A_{\mathrm{ex}}(\varrho) \in (0,\infty)$.

Assume now that $A > A_{\mathrm{ex}}(\varrho)$. Then there is $\alpha_0 > 0$ such that $A^2 m_\varrho(\alpha_0) - \alpha_0^2 > 0$. The preceding endpoint estimates imply that $A^2 m_\varrho(\alpha) - \alpha^2 < 0$ for all sufficiently small and all sufficiently large α. The intermediate value theorem therefore gives two roots $0 < \alpha_- < \alpha_+$ of $A^2 m_\varrho(\alpha) = \alpha^2$.

The change of variables $\alpha = A\sqrt{h}$ gives

$$V_{A,\varrho}(h) = m_\varrho(A\sqrt{h}).$$

Thus, if $\widehat{h}_\pm = \alpha_\pm^2/A^2$, then

$$V_{A,\varrho}(\widehat{h}_\pm) = m_\varrho(\alpha_\pm) = \frac{\alpha_\pm^2}{A^2} = \widehat{h}_\pm.$$

Hence $\widehat{h}_\pm$ are fixed points. They are positive because $\alpha_\pm > 0$. They are strictly below M_ϱ because, at a fixed point,

$$\widehat{h}_\pm = V_{A,\varrho}(\widehat{h}_\pm) = \mathbb{E}\Big[Q_1\big(\varrho^{-1}\tanh(\varrho\alpha_\pm G)\big)^2\Big],$$

the integrand is bounded by M_ϱ, and it is strictly smaller than M_ϱ with positive probability, for instance on the zero-bin event $\{|G| \leqslant b_{\varrho,0}/\alpha_\pm\}$.

The roots show that $\mathcal{O}_{A,\varrho}$ is nonempty, because $A^2 m_\varrho(\alpha_0) - \alpha_0^2 > 0$ is equivalent, with $h_0 = \alpha_0^2/A^2$, to $V(h_0) > h_0$. The endpoint estimates already proved also translate to the h variable. As $h \downarrow 0$,

$$\frac{V(h)}{h} = A^2 \frac{m_\varrho(A\sqrt{h})}{(A\sqrt{h})^2} \longrightarrow 0,$$

and hence $V(h) - h < 0$ for all sufficiently small $h > 0$. At the upper endpoint, $V(M_\varrho) < M_\varrho$ by the same zero-bin argument used above. By continuity, there is $\delta > 0$ such that

$$V(h) - h < 0 \quad \text{for } h \in (0, \delta] \cup [M_\varrho - \delta, M_\varrho].$$

The finite layer formula shows that V is real analytic on $(0, \infty)$. The analytic function $V(h) - h$ is not identically zero, because it is negative for all sufficiently small positive h. Its zeros are therefore isolated. Its zero set in the compact interval $[\delta, M_\varrho - \delta]$ is consequently finite. Since every component of $\mathcal{O}_{A,\varrho}$ is contained in this interval, there are only finitely many such components.

Let (h_-, h_+) be the rightmost component. Continuity of $V - h$ gives (5.7), and the defining inequality gives $V(h) > h$ on (h_-, h_+). The zero h_+ is isolated, so $V(h) - h$ has a constant nonzero sign immediately to its right. This sign cannot be positive, since that would produce a positive-drift component to the right of (h_-, h_+). It is therefore negative. We may choose $\eta_0 > 0$ so small that $h_+ - \eta_0 > h_-$, $h_+ + \eta_0 < M_\varrho$, and $V(u) < u$ for every $u \in (h_+, h_+ + \eta_0]$. This proves (5.8). □

5.2. Entrance and exponential persistence. Conditionally on $\mathscr{H}_t^{(N)} = h$, the random variables in (2.19) are independent and take values in $[0, M_\varrho]$. Hence Hoeffding's inequality gives, for every $u > 0$,

$$\mathbb{P}\Big(\Big|\mathscr{H}_{t+1}^{(N)} - V(\mathscr{H}_t^{(N)})\Big| > u \,\Big|\, \mathscr{H}_t^{(N)}\Big) \leqslant 2\exp\Big(-\frac{2Nu^2}{M_\varrho^2}\Big). \tag{5.10}$$

Theorem 5.2 (Metastability from the rightmost positive-drift component)**.** *Assume that $A > A_{\mathrm{ex}}(\varrho)$, and let (h_-, h_+) be the rightmost positive-drift component of $\mathcal{O}_{A,\varrho}$ in (5.5). Let $\eta_0 > 0$ be as in (5.8), and let $B \subset (h_-, h_+]$ be compact. For every $0 < \eta < \eta_0$, there are $T_\eta \in \mathbb{N}_0$ and* *constants $c_0, c_1, C \in (0,\infty)$, depending only on A, ϱ, B, η, with $c_1 < c_0$, such that the following holds. If $Y_0^{(N)}$ is deterministic and $\mathscr{H}_0^{(N)} \in B$, then*

$$\mathbb{P}\Big(\Big|\mathscr{H}_{T_\eta}^{(N)} - h_+\Big| > \frac{\eta}{2}\Big) \leqslant C\exp(-c_0 N), \tag{5.11}$$

and, for every $T \in \mathbb{N}_0$,

$$\mathbb{P}\bigg(\max_{0\leqslant s\leqslant T}\Big|\mathscr{H}_{T_\eta+s}^{(N)} - h_+\Big| > \eta\bigg) \leqslant C(1+T)\exp(-c_0 N). \tag{5.12}$$

In particular,

$$\mathbb{P}\Big(\tau_\varrho^{(N)} > T_\eta + \lfloor \exp(c_1 N)\rfloor\Big) \to 1. \tag{5.13}$$

For each fixed N and each deterministic rounded vector initial condition in the finite rounded state space, the absorption time $\tau_\varrho^{(N)}$ is nevertheless finite almost surely.

Proof. Step 1: Deterministic entrance. By (4.3), the map V is nondecreasing. Since (h_-, h_+) is a connected component of $\mathcal{O}_{A,\varrho}$, this monotonicity gives

$$h < V(h) \leqslant h_+ \quad \text{for } h \in (h_-, h_+),$$

while (5.8) gives, for $h \in (h_+, h_+ + \eta_0]$,

$$h_+ \leqslant V(h) < h.$$

Hence every rounded deterministic orbit starting from $(h_-, h_+]$ converges monotonically to h_+. Since B is compact and separated from h_-, this entrance is uniform on B: outside $[h_+ - \eta/4, h_+ + \eta/4]$ the signed displacement toward h_+ is bounded from below by a positive constant. Thus there is a deterministic time T_η such that

$$V^{T_\eta}(B) \subset [h_+ - \eta/4, h_+ + \eta/4]. \tag{5.14}$$

To pass from the rounded deterministic orbit to the rounded squared-radius chain, let

$$\omega(r) := \sup\big\{|V(u) - V(v)| : u, v \in [0, M_\varrho],\ |u - v| \leqslant r\big\}.$$

Uniform continuity gives $\omega(r) \to 0$ as $r \downarrow 0$. If $q_s := V^s(\mathscr{H}_0^{(N)})$ and $e_s := \Big|\mathscr{H}_s^{(N)} - q_s\Big|$, then

$$e_{s+1} \leqslant \Big|\mathscr{H}_{s+1}^{(N)} - V(\mathscr{H}_s^{(N)})\Big| + \omega(e_s).$$

For $\delta > 0$, define $E_0(\delta) := 0$ and $E_{s+1}(\delta) := \delta + \omega(E_s(\delta))$. Since T_η is fixed, induction gives $E_s(\delta) \to 0$ as $\delta \downarrow 0$ for every $s \leqslant T_\eta$. We may therefore choose $\delta_0 > 0$ such that $E_{T_\eta}(\delta_0) \leqslant \eta/4$.

Suppose now that $\mathscr{H}_0^{(N)} \in B$ and

$$\Big|\mathscr{H}_{s+1}^{(N)} - V(\mathscr{H}_s^{(N)})\Big| \leqslant \delta_0 \qquad \text{for } 0 \leqslant s < T_\eta,$$

Then $e_s \leqslant E_s(\delta_0)$ for $s \leqslant T_\eta$. Combining this bound with (5.14) yields

$$\Big|\mathscr{H}_{T_\eta}^{(N)} - h_+\Big| \leqslant e_{T_\eta} + \big|q_{T_\eta} - h_+\big| \leqslant \frac{\eta}{4} + \frac{\eta}{4} = \frac{\eta}{2}.$$

A union bound and (5.10) prove (5.11). If $T_\eta = 0$, the estimate is trivial after increasing C.

Step 2: Persistence. Put

$$J_\eta := [h_+ - \eta, h_+ + \eta].$$

The inequalities on the two sides of h_+ imply $V(J_\eta) \subset J_\eta^\circ$. Since J_η is compact, there exists $d_\eta > 0$ such that
$$\operatorname{dist}(V(J_\eta), J_\eta^c) \geqslant d_\eta.$$
Therefore, on the event $\mathscr{H}_t^{(N)} \in J_\eta$, the additional event
$$\left|\mathscr{H}_{t+1}^{(N)} - V(\mathscr{H}_t^{(N)})\right| \leqslant d_\eta/2$$
implies $\mathscr{H}_{t+1}^{(N)} \in J_\eta$. Applying (5.10) and taking a union bound over the times $T_\eta, \ldots, T_\eta + T - 1$, together with (5.11), gives (5.12) with an exponential rate $c_0 = c_0(A, \varrho, B, \eta) > 0$.

Since $J_\eta \subset (0, M_\varrho]$, the event in (5.12) prevents absorption. Decrease the persistence exponent if necessary so that $0 < c_1 < c_0$. Taking $T = \lfloor \exp(c_1 N) \rfloor$, the right-hand side of (5.12) is bounded by
$$2C \exp\bigl(-(c_0 - c_1)N\bigr) \longrightarrow 0.$$
This is (5.13).

Step 3: Eventual absorption. For fixed N, the rounded vector chain is supported on the finite set
$$(\varrho\mathbb{Z})^N \cap [-1 - \varrho/2, 1 + \varrho/2]^N.$$
If $y \neq 0$ belongs to this set, then conditionally on $Y_t^{(N)} = y$ the coordinates of $\mathsf{W}_{t+1,A}^{(N)} y$ are independent nondegenerate Gaussians. Thus each coordinate has a strictly positive probability to fall in the zero rounding bin, and consequently
$$\mathbb{P}\Bigl(Y_{t+1}^{(N)} = 0 \,\Big|\, Y_t^{(N)} = y\Bigr) > 0.$$
Taking the minimum over the finite set of nonzero states gives a uniform positive one-step absorption probability before absorption. Hence $\mathbb{P}(\tau_\varrho^{(N)} > m) \leqslant (1 - p_N)^m$ for some $p_N > 0$, and $\tau_\varrho^{(N)} < \infty$ almost surely. □

Remark 5.3. The metastable object for the rounded vector chain is a shell, not a point. If $J = [j_-, j_+]$, then on the event $\mathscr{H}_t^{(N)} \in J$ the vector satisfies
$$\varrho\sqrt{N j_-} \leqslant \|Y_t^{(N)}\|_2 \leqslant \varrho\sqrt{N j_+}.$$
Thus exponential persistence of the radius is exponential persistence in a nonzero grid shell. The theorem does not determine the sharp escape exponent, the quasi-stationary law in the well, or the limiting absorption-time law.

6. Supercritical cutoff in the dimension

Fix $A > 1$ and let $N \to \infty$. We use the unrounded squared-radius chain $Z_t^{(N)}$, its transition kernel $K_{A,N}$, and its mean map V_A from Section 2.1. Negative moments localize the invariant law, while synchronous contraction brings the evolving and stationary chains within the range of the one-step total-variation estimate. Generic constants depend only on the fixed parameters in the surrounding statement.

6.1. The unrounded squared-radius chain. This subsection records the basic structure of the unrounded radial chain. We first compute the one-coordinate density and its score. Positivity of the resulting radial transition kernel yields uniqueness of the nonzero invariant law, which the reconstruction kernel then transfers to the vector chain.

Let $Y_q := \tanh^2(A\sqrt{q}\,G)$ for $q > 0$. For $y \in (0,1)$, the change of variables $u = \operatorname{arctanh}(\sqrt{y})$ gives the strictly positive density

$$f_q(y) = \frac{1}{A\sqrt{2\pi q}} \frac{1}{\sqrt{y}(1-y)} \exp\left(-\frac{\operatorname{arctanh}(\sqrt{y})^2}{2A^2 q}\right). \tag{6.1}$$

Differentiating the same density with respect to q gives the score used in the one-step total-variation estimate. Its positivity also implies positivity of the N-coordinate transition density, which is the key input for uniqueness.

Proposition 6.1. *Fix $A > 0$ and $N \geqslant 1$. If $K_{A,N}$ admits a nonzero invariant probability $\nu_{A,N}$ on $(0,1]$, then this probability is unique and has a strictly positive density on $(0,1)$.*

Proof. The density of $\sum_{i=1}^N Y_{i,q}$ is the convolution f_q^{*N}. It is strictly positive on $(0,N)$: for $s \in (0,N)$, choose $y_1, \dots, y_N \in (0,1)$ with $\sum_i y_i = s$ and integrate over a small neighborhood of $(y_1, \dots, y_{N-1})$ on which all coordinates, including the remaining coordinate, stay in $(0,1)$. After scaling by N^{-1}, the transition kernel $K_{A,N}(q,\cdot)$ therefore has a density $p_{A,N}(q,r)$ that is strictly positive for every $r \in (0,1)$. Consequently, for every $q > 0$ and every Borel set $B \subset (0,1)$ with positive Lebesgue measure,

$$K_{A,N}(q,B) > 0.$$

In particular, $K_{A,N}(q,O) > 0$ for every $q > 0$ and every nonempty open set $O \subset (0,1)$. Thus the squared-radius chain restricted to $(0,1]$ is Lebesgue-irreducible. We use the standard uniqueness theorem for ψ-irreducible Markov chains: such a chain admits at most one invariant probability measure; see, for instance, Nummelin [Num84]. Equivalently, two distinct invariant probabilities would have ergodic decompositions containing disjoint invariant ergodic components, but one-step positivity forces every component with positive invariant mass to communicate with every nonempty open subset of $(0,1)$. This contradiction proves uniqueness.

If $\nu K_{A,N} = \nu$ on $(0,1]$, then invariance and Fubini's theorem give

$$\nu(\mathrm{d}r) = \left(\int_{(0,1]} p_{A,N}(q,r)\,\nu(\mathrm{d}q)\right)\mathrm{d}r.$$

The density in parentheses is strictly positive for every $r \in (0,1)$. □

Remark 6.2. On $[0,1]$, the point 0 is absorbing. Hence, whenever the nonzero invariant law exists, the invariant probabilities on $[0,1]$ are exactly

$$\alpha\delta_0 + (1-\alpha)\nu_{A,N}, \qquad 0 \leqslant \alpha \leqslant 1. \tag{6.2}$$

Indeed, any invariant law has a fixed mass α at $\{0\}$, and its normalized restriction to $(0,1]$, if nonzero, is invariant for the restricted chain.

Corollary 6.3 (Uniqueness of the nonzero vector invariant law)**.** *Fix $A > 0$ and $N \geqslant 1$ for which $K_{A,N}$ admits a nonzero invariant law $\nu_{A,N}$ on $(0,1]$. Then $P_{A,N}$ has a unique invariant probability $\pi_{A,N}$ on $[-1,1]^N$ with $\pi_{A,N}(\{0\}) = 0$, namely*

$$\pi_{A,N} = \nu_{A,N} J_{A,N}. \tag{6.3}$$

Proof. For existence, Proposition 2.4 shows that $\nu_{A,N}J_{A,N}$ is invariant for $P_{A,N}$; it has no mass at 0 because $J_{A,N}(q,\cdot)$ is absolutely continuous, and hence charges no point, for every $q > 0$, while $\nu_{A,N}$ is carried by $(0,1]$. For uniqueness, let π be invariant for $P_{A,N}$ with $\pi(\{0\}) = 0$. By Proposition 2.4, $\nu := (r_N)_\#\pi$ is invariant for $K_{A,N}$, and since $r_N(x) = 0$ if and only if $x = 0$, we have $\nu(\{0\}) = \pi(\{0\}) = 0$; thus ν is a nonzero invariant law of $K_{A,N}$. By Proposition 6.1, $\nu = \nu_{A,N}$. Although $(r_N)_\#$ is not injective on arbitrary probability measures, invariance and (2.40) give the reconstruction identity

$$\pi = \pi P_{A,N} = \big((r_N)_\#\pi\big) J_{A,N} = \nu_{A,N} J_{A,N}.$$

Hence the squared-radius law determines the invariant vector law uniquely. □

6.2. Cutoff scale. For $A > 1$, Lemma 2.1 gives the unique nonzero fixed point q_* and the multiplier $\mu_A = V_A'(q_*) \in (0,1)$.

Let $q_0 \in (0,1]$ with $q_0 \neq q_*$, and let $C(q_0)$ be the nonzero constant in (2.32). By (2.32), the deterministic bias decays like $C(q_0)\mu_A^t$. The cutoff scale is therefore the time at which this bias reaches the $N^{-1/2}$ fluctuation scale, and the resulting center and integer times are given by (1.20) and (1.21). The proof of Theorem 1.12 is given in Section 6.5.

6.3. Localization in the stable interval. This subsection establishes localization of the invariant law in a stable interval around q_*. Negative moments make the invariant mass near zero exponentially small, while deterministic entrance and concentration control the remaining radii. The estimates are proved in Appendix B.

Assumption 6.4 (Macroscopic initial radius)**.** *We fix $A > 1$ and let q_* and μ_A be the unique nonzero fixed point and multiplier from Lemma 2.1. We also fix $q_0 \in (0,1]$ with $q_0 \neq q_*$; every deterministic initial-radius sequence used below is assumed to converge to this q_0.*

Throughout the remainder of this subsection, Assumption 6.4 is in force.

For a deterministic initial-radius sequence $q_N \to q_0$, put

$$q_{0,N} := q_N, \qquad q_{t+1,N} := V_A(q_{t,N}). \tag{6.4}$$

The deterministic orbit must first enter a region where the mean map is contractive. We record that entrance estimate here.

Lemma 6.5. *Let Assumption 6.4 hold, i.e. $q_N \to q_0$. There are an integer s_*, a compact interval $I_* \Subset (0,1)$, constants $\delta_* > 0$ and $\kappa < 1$, and $N_0 < \infty$ such that, for all $N \geqslant N_0$,*

$$q_{t,N} \in I_* \quad \textit{for every } t \geqslant s_*, \qquad \inf_{t \geqslant s_*} \operatorname{dist}(q_{t,N}, \partial I_*) \geqslant \delta_*, \tag{6.5}$$

and

$$\sup_{q \in I_*} \left| V_A'(q) \right| \leqslant \kappa. \tag{6.6}$$

Proof. Since q_* is stable, choose a compact interval $I_* \Subset (0,1)$ containing q_* in its interior such that $\sup_{I_*} |V_A'| < 1$. Since q_0 lies in the basin of q_*, the unrounded deterministic orbit $V_A^t(q_0)$ enters the interior of I_* and remains in a smaller compact subinterval of I_* after a finite time. By continuity of V_A^t for each fixed t, the same is true uniformly for all q_N sufficiently close to q_0, after increasing the entrance time if necessary. This proves (6.5), while the derivative bound follows from the choice of I_*. □

Negative moments provide the required small-radius control. The linearized calculation determines the dimension-dependent exponent, and the truncated estimate makes it uniform near the origin.

Remark 6.6. For every positive random variable X, Markov's inequality gives $\mathbb{P}(X \leqslant r) \leqslant r^p \mathbb{E} X^{-p}$, so negative moments provide small-ball bounds. Near $q = 0$,

$$\frac{F_{A,N}(q,G)}{q} \approx A^2 \frac{\chi_N^2}{N}.$$

Here χ_N is the Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$, as in the linearized multiplier (2.25). The cutoff localization uses $p = \gamma N$. If $Z_N \sim \chi_N^2$ and $\gamma \in (0, 1/2)$, direct integration gives

$$\mathbb{E}\left[\left(A^2 \frac{Z_N}{N} \right)^{-\gamma N} \right] = A^{-2\gamma N} \left(\frac{N}{2} \right)^{\gamma N} \frac{\Gamma(N/2 - \gamma N)}{\Gamma(N/2)}.$$

The moment is finite precisely when $\gamma < 1/2$, and Stirling's formula yields

$$\frac{1}{N}\log \mathbb{E}\left[\left(A^2\frac{Z_N}{N}\right)^{-\gamma N}\right] \to \Lambda_A(\gamma) := -2\gamma\log A + \gamma + \left(\frac{1}{2}-\gamma\right)\log(1-2\gamma).$$

Since $\Lambda_A(0) = 0$ and $\Lambda_A'(0) = -2\log A < 0$ for $A > 1$, this exponent is negative for all sufficiently small $\gamma > 0$. The truncated estimate below makes this linearized calculation uniform over a neighborhood of the origin.

Near zero, the linearized calculation must be made uniform. This is the role of the truncation estimate below.

Lemma 6.7. *Let $A > 1$. There are constants $\gamma \in (0, 1/2)$, $\kappa > 0$, $L \geqslant 1$, and $\varepsilon \in (0, 1)$ such that, for all sufficiently large N,*

$$\mathbb{E}\left[\left((1-\varepsilon)A^2\frac{1}{N}\sum_{i=1}^{N}\min(G_i^2, L^2)\right)^{-\gamma N}\right] \leqslant e^{-\kappa N}. \tag{6.7}$$

For sufficiently small R_0, the ratio $F_{A,N}(q, G)/q$ dominates the truncated quadratic average in Lemma 6.7, uniformly for $0 < q \leqslant R_0$. Since $x \mapsto x^{-\gamma N}$ is decreasing, the lemma then gives the required negative-moment contraction near zero.

Proposition 6.8. *Let Assumption 6.4 hold, and let I_* be the compact interval selected in Lemma 6.5. There are $R_0 \in (0, \inf I_*)$, $\gamma \in (0, 1/2)$, and $\kappa > 0$ such that, for all sufficiently large N,*

$$\sup_{0<q\leqslant R_0} \mathbb{E}\left[\left(\frac{F_{A,N}(q, G)}{q}\right)^{-\gamma N}\right] \leqslant e^{-\kappa N}. \tag{6.8}$$

Away from zero, there is $c_{A,R_0} > 0$ such that $\tanh^2(A\sqrt{q}\,g) \geqslant c_{A,R_0}\min(g^2, 1)$ for $q \in [R_0, 1]$. After averaging over the coordinates, this reduces the estimate to a negative-moment bound for a truncated Gaussian square average.

Lemma 6.9. *Let $R_0 \in (0, 1)$ and $\gamma \in (0, 1/2)$. There is $b < \infty$ such that, for every $N \geqslant 1$,*

$$\sup_{q\in[R_0,1]} \mathbb{E}\big[F_{A,N}(q, G)^{-\gamma N}\big] \leqslant e^{bN}. \tag{6.9}$$

See Appendix B.1 for the proofs of Lemmas 6.7 and 6.9 and Proposition 6.8.

By Propositions 2.6 and 6.1, and the fact that $A_c(N) \downarrow 1$, the squared-radius transition kernel has a unique nonzero invariant law for all sufficiently large N. We denote it by $\nu_{A,N}$.

Remark 6.10. For the mere non-emptiness of the positive invariant class, the preliminary Proposition 2.6 uses a simpler negative moment Lyapunov function. In the special case $p = 1$, the linearized estimate gives, for $N > 2$,

$$\mathbb{E}\left[\left(A^2\frac{\chi_N^2}{N}\right)^{-1}\right] = A^{-2}\frac{N}{N-2} < 1 \tag{6.10}$$

for fixed $A > 1$ and all large N. However, this only prevents collapse to δ_0. The cutoff proof needs the stronger fixed-threshold localization

$$\nu_{A,N}((0, r]) \leqslant e^{-cN} \tag{6.11}$$

for some fixed $r > 0$. This is why we use

$$V_N(q) = q^{-\gamma N}. \tag{6.12}$$

Then Markov's inequality gives

$$\nu_{A,N}((0,r]) \leqslant r^{\gamma N} \int q^{-\gamma N}\, \nu_{A,N}(\mathrm{d}q), \tag{6.13}$$

and an e^{bN} bound on the negative moment becomes

$$\nu_{A,N}((0,r]) \leqslant \exp\{N(b + \gamma \log r)\}. \tag{6.14}$$

Choosing $r > 0$ sufficiently small yields exponential smallness.

Combining the near-zero contraction estimate with the bound on $[R_0, 1]$ yields the invariant-law estimate below.

Proposition 6.11. *Let Assumption 6.4 hold. Then, for all sufficiently large N, the unique nonzero invariant law $\nu_{A,N}$ of $K_{A,N}$ satisfies*

$$\int_0^1 q^{-\gamma N}\, \nu_{A,N}(\mathrm{d}q) \leqslant \frac{e^{bN}}{1 - e^{-\kappa N}}, \tag{6.15}$$

where R_0, γ, κ are as in Proposition 6.8, and b is the constant in Lemma 6.9 for this choice of R_0, γ.

To prove the next proposition, choose $r < R_0$ so that the preceding negative-moment estimate gives $\nu_{A,N}((0,r]) \leqslant Ce^{-cN}$. Uniform deterministic entrance from $[r,1]$ into I_*, together with a fixed-time concentration estimate, then gives, for some fixed m, $\sup_{q\in[r,1]} K^m_{A,N}(q, I^c_*) \leqslant Ce^{-cN}$. Invariance combines these two bounds.

Proposition 6.12. *Let Assumption 6.4 hold. Let $\nu_{A,N}$ be the unique nonzero invariant law of $K_{A,N}$. Then there are constants $C, c > 0$ such that*

$$\nu_{A,N}(I^c_*) \leqslant Ce^{-cN} \tag{6.16}$$

for all sufficiently large N. In particular,

$$\nu_{A,N}(I^c_*) \leqslant \frac{C}{N} \tag{6.17}$$

after enlarging C.

The proofs of Propositions 6.11 and 6.12 are in Appendices B.1 and B.2.

6.4. Dynamic tracking and total variation. With the invariant law localized in I_*, we next compare the evolving chain with a stationary copy. The argument uses one-step total-variation smoothing inside I_*, concentration estimates for the stationary and evolving chains, and a synchronous coupling on the cutoff time scale.

We first record the one-step total-variation estimate inside the stable interval.

Lemma 6.13. *Let $I \Subset (0,1)$. There is a constant $C_I < \infty$ such that, for every $q, q' \in I$,*

$$\|K_{A,N}(q,\cdot) - K_{A,N}(q',\cdot)\|_{\mathrm{TV}} \leqslant C_I \sqrt{N}\, |q - q'|. \tag{6.18}$$

The stationary chain satisfies the analogous concentration estimate.

Proposition 6.14. *Let Assumption 6.4 hold, and let $\nu_{A,N}$ be the unique nonzero invariant law of $K_{A,N}$. Then*

$$\int_0^1 (q - q_*)^2\, \nu_{A,N}(\mathrm{d}q) \leqslant \frac{C}{N}. \tag{6.19}$$

The deterministic orbit has the matching concentration estimate.

Lemma 6.15. *Let Assumption 6.4 hold. Then, for every $0 < C_0 < \infty$ there are constants $C, c_0 > 0$ such that, with $T_N := \lfloor C_0 \log N \rfloor$ and*

$$\tau_*^{(N)} := \inf\{t \geqslant s_* : Z_t^{(N)} \notin I_*\},$$

one has, for all sufficiently large N,

$$\mathbb{P}\Big(\tau_*^{(N)} \leqslant T_N\Big) \leqslant C T_N \exp(-c_0 N), \tag{6.20}$$

and

$$\sup_{0 \leqslant t \leqslant T_N} \mathbb{E}\bigg[\Big(Z_t^{(N)} - q_{t,N}\Big)^2\bigg] \leqslant \frac{C}{N}. \tag{6.21}$$

See Appendix B.2 for the proofs.

We use this lemma with a horizon constant chosen before the cutoff offset. Fix

$$C_0 > \frac{1}{2|\log \mu_A|}. \tag{6.22}$$

For every fixed $c \in \mathbb{R}$, the definitions of $t_N(q_0)$ and $n_N(c)$ give

$$\frac{n_N(c)}{\log N} \to \frac{1}{2|\log \mu_A|}.$$

Hence, by (6.22),

$$0 \leqslant n_N(c) \leqslant C_0 \log N \tag{6.23}$$

for all sufficiently large N. Thus the constants supplied by Lemma 6.15 for this choice of C_0 are independent of c; only the lower bound on N may depend on the fixed value of c.

The next lemma uses a synchronous coupling on the last $O(\log\log N)$ steps before the cutoff. It shows that the difference between the evolving and stationary squared-radius chains is of order $\mu_A^c N^{-1/2}$ at the cutoff time, which is precisely the scale needed for the one-step total-variation estimate.

Lemma 6.16. *Let Assumption 6.4 hold. Let $q_N \to q_0$ with $q_N > 0$, let $\nu_{A,N}$ be the unique nonzero invariant law of $K_{A,N}$, and let $(Z_t^{(N)}, \widetilde{Z}_t^{(N)})$ be the synchronous coupling driven by the same Gaussian vectors, with $Z_0^{(N)} = q_N$ and $\widetilde{Z}_0^{(N)} \sim \nu_{A,N}$. Then there is a constant $C < \infty$ such that, for every $c \geqslant 0$,*

$$\limsup_{N\to\infty} \sqrt{N}\, \mathbb{E}\Big|Z^{(N)}_{(n_N(c)-1)_+} - \widetilde{Z}^{(N)}_{(n_N(c)-1)_+}\Big| \leqslant C\mu_A^c. \tag{6.24}$$

See Appendix B.3 for the proof.

6.5. Proof of the cutoff theorem. The preceding localization and coupling estimates reduce the theorem to separating the evolving and stationary laws before and after the deterministic cutoff scale. We now carry out these two bounds.

Proof of Theorem 1.12. Step 1: Setup. The theorem assumptions give Assumption 6.4; moreover, $q_N > 0$ for all sufficiently large N because $q_N \to q_0 > 0$. Since $A > 1$ and $A_c(N) \downarrow 1$, the existence of a nonzero invariant law for all sufficiently large N follows from Proposition 2.6. Uniqueness follows from Proposition 6.1. We denote this law by $\nu_{A,N}$.

Step 2: The lower bound. Fix $c > 0$ and put $t_N^- := n_N(-c)$, with n_N defined in (1.21). By (6.23), this time is positive and bounded by $C_0 \log N$ for all large N, with the same C_0 for every fixed c. Let $d_N := |q_{t_N^-,N} - q_*|$. Since $q_0 \neq q_*$, the unrounded deterministic orbit never crosses q_*; hence $d_N > 0$ for all large N. We separate the dynamic law and the invariant law by the half-line

$$H_N := \left\{ q \in [0,1] : \operatorname{sgn}(q_{t_N^-,N} - q_*)(q - q_*) \geqslant \frac{1}{2} d_N \right\}. \tag{6.25}$$

Write $\varepsilon_N := \operatorname{sgn}(q_{t_N^-,N} - q_*) \in \{-1, 1\}$. Then $\varepsilon_N(q_{t_N^-,N} - q_*) = d_N$. If $Z^{(N)}_{t_N^-} \notin H_N$, then, by the definition of H_N,

$$\varepsilon_N(Z^{(N)}_{t_N^-} - q_*) < \frac{1}{2} d_N.$$

Subtracting this inequality from the preceding identity gives

$$\varepsilon_N(q_{t_N^-,N} - Z^{(N)}_{t_N^-}) = \varepsilon_N(q_{t_N^-,N} - q_*) - \varepsilon_N(Z^{(N)}_{t_N^-} - q_*) > \frac{1}{2} d_N.$$

In particular,

$$\{Z^{(N)}_{t_N^-} \notin H_N\} \subseteq \left\{ \left| Z^{(N)}_{t_N^-} - q_{t_N^-,N} \right| \geqslant \frac{1}{2} d_N \right\}.$$

Chebyshev's inequality and Lemma 6.15 give

$$\begin{aligned} K^{t_N^-}_{A,N}(q_N, H_N^c) &= \mathbb{P}_{q_N}(Z^{(N)}_{t_N^-} \notin H_N) \\ &\leqslant \mathbb{P}_{q_N}\left(\left| Z^{(N)}_{t_N^-} - q_{t_N^-,N} \right| \geqslant \frac{1}{2} d_N \right) \\ &\leqslant \frac{4}{d_N^2} \mathbb{E}\left[\left(Z^{(N)}_{t_N^-} - q_{t_N^-,N} \right)^2 \right] \leqslant \frac{C}{N d_N^2}. \end{aligned} \tag{6.26}$$

In the last step we used (6.21), since $t_N^- \leqslant C_0 \log N$. Similarly, if $q \in H_N$, then $|q - q_*| \geqslant d_N/2$. Chebyshev's inequality and Proposition 6.14 therefore give

$$\nu_{A,N}(H_N) \leqslant \frac{4}{d_N^2} \int_0^1 (q - q_*)^2 \, \nu_{A,N}(\mathrm{d}q) \leqslant \frac{C}{N d_N^2}. \tag{6.27}$$

Since total variation is the supremum over measurable sets,

$$\begin{aligned} \|K^{t_N^-}_{A,N}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}} &\geqslant K^{t_N^-}_{A,N}(q_N, H_N) - \nu_{A,N}(H_N) \\ &\geqslant 1 - K^{t_N^-}_{A,N}(q_N, H_N^c) - \nu_{A,N}(H_N). \end{aligned}$$

The locally uniform asymptotic (2.32), applied at $t_N^- = n_N(-c)$, gives

$$d_N = |C(q_N)| \mu_A^{t_N^-} (1 + o(1)).$$

Moreover, $C(q_N) \to C(q_0) \neq 0$ and the definition of the cutoff center gives $\sqrt{N}\, |C(q_0)| \mu_A^{t_N(q_0)} = 1$. Since $t_N^- - (t_N(q_0) - c) \in [-1, 0]$, the integer-time rounding is uniformly bounded. Hence there is $a_0 > 0$, independent of c, such that

$$\liminf_{N \to \infty} \sqrt{N}\, d_N \geqslant a_0 \mu_A^{-c}.$$

Combining this bound with (6.26) and (6.27) yields

$$\liminf_{N \to \infty} \|K^{t_N^-}_{A,N}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}} \geqslant 1 - C \mu_A^{2c}.$$

Here C is independent of c. Sending $c \to \infty$ proves (1.22).

Step 3: The upper bound. Fix $c \geqslant 0$ and put $t_N^+ := (n_N(c) - 1)_+$. Let $(Z_t^{(N)}, \widetilde{Z}_t^{(N)})$ be the synchronous coupling from Lemma 6.16, with $Z_0^{(N)} = q_N$ and $\widetilde{Z}_0^{(N)} \sim \nu_{A,N}$. By invariance, $\widetilde{Z}_{t_N^+}^{(N)} \sim \nu_{A,N}$. The Markov property and convexity of total variation imply the standard coupling inequality

$$
\begin{aligned}
\|K_{A,N}^{t_N^+ + 1}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}} &= \|\mathbb{E}\Big[K_{A,N}(Z_{t_N^+}^{(N)}, \cdot) - K_{A,N}(\widetilde{Z}_{t_N^+}^{(N)}, \cdot)\Big]\|_{\mathrm{TV}} \\
&\leqslant \mathbb{E}\, \|K_{A,N}(Z_{t_N^+}^{(N)}, \cdot) - K_{A,N}(\widetilde{Z}_{t_N^+}^{(N)}, \cdot)\|_{\mathrm{TV}}.
\end{aligned}
\tag{6.28}
$$

Set

$$
E_N := \Big\{ Z_{t_N^+}^{(N)} \in I_*, \ \widetilde{Z}_{t_N^+}^{(N)} \in I_* \Big\}.
$$

We claim that $\mathbb{P}(E_N^c) = o_N(1)$. Indeed, (6.23) gives $t_N^+ \leqslant C_0 \log N$ for all large N, with C_0 independent of c, and $t_N^+ \geqslant s_*$. Thus Lemma 6.15 gives

$$
\mathbb{P}(Z_{t_N^+}^{(N)} \notin I_*) \leqslant C(\log N) e^{-c_0 N} = o_N(1),
$$

while stationarity of $\widetilde{Z}^{(N)}$ and Proposition 6.12 give

$$
\mathbb{P}(\widetilde{Z}_{t_N^+}^{(N)} \notin I_*) = \nu_{A,N}(I_*^c) \leqslant \frac{C}{N}.
$$

These two bounds give $\mathbb{P}(E_N^c) = o_N(1)$.

On E_N, the one-step total variation estimate Lemma 6.13, applied with $I = I_*$, gives

$$
\|K_{A,N}(Z_{t_N^+}^{(N)}, \cdot) - K_{A,N}(\widetilde{Z}_{t_N^+}^{(N)}, \cdot)\|_{\mathrm{TV}} \leqslant C\sqrt{N} \Big| Z_{t_N^+}^{(N)} - \widetilde{Z}_{t_N^+}^{(N)} \Big|.
$$

On E_N^c we only use the trivial bound by one. Hence (6.28) yields

$$
\|K_{A,N}^{t_N^+ + 1}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}} \leqslant C\sqrt{N}\, \mathbb{E} \Big| Z_{t_N^+}^{(N)} - \widetilde{Z}_{t_N^+}^{(N)} \Big| + \mathbb{P}(E_N^c)
$$

For all sufficiently large N, one has $n_N(c) \geqslant 1$ and hence $t_N^+ + 1 = n_N(c)$. Since $t_N^+ = (n_N(c) - 1)_+$, Lemma 6.16 gives

$$
\limsup_{N \to \infty} \|K_{A,N}^{n_N(c)}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}} \leqslant C \mu_A^c.
$$

Here C is independent of c. Sending $c \to \infty$ proves (1.23). □

Proof of Corollary 1.13. The lower bound follows from Proposition 2.4, the left inequality in (2.43), and Theorem 1.12. The right inequality in (2.43) gives

$$
\|P_{A,N}^{n_N(c)}(x_N, \cdot) - \pi_{A,N}\|_{\mathrm{TV}} \leqslant \|K_{A,N}^{(n_N(c)-1)_+}(q_N, \cdot) - \nu_{A,N}\|_{\mathrm{TV}}.
$$

For all sufficiently large N, one has $(n_N(c) - 1)_+ = n_N(c - 1)$. Thus (1.23) proves the upper bound after sending $c \to \infty$. Finally, Corollary 6.3 shows that $\pi_{A,N}$ is the unique nonzero vector invariant law. □

7. The N-dimensional supercritical stationary law

Fix N and assume $A > A_c(N)$. We study the nonzero invariant law of the unrounded vector chain near the repelling origin. The normalized squared radius determines the radial marginal, but the vector law also contains an angular variable. In log-polar coordinates, the stationary equation is a nonlinear perturbation of a Markov additive renewal equation. Exponential tilting at the Cramér root gives the power-law exponent, as in the implicit-renewal method for stationary random recursions [Kes73], [Gol91], and [BDM16].

We use the Gaussian matrix $\mathsf{W}_A^{(N)}$ from (2.1), without a time index. The unrounded vector chain and its transition kernel $P_{A,N}$ are given in (2.2) and Proposition 2.4. A standalone matrix W below denotes an independent copy of $\mathsf{W}_A^{(N)}$. We use the linearized radius multiplier $\ell_{A,N}$ from (2.25). Set

$$\mathcal{K}_N := [-1,1]^N.$$

The origin is absorbing, so δ_0 is always invariant. We write $\pi_{A,N}$ for the nonzero invariant law, which exists under the present assumption by Proposition 2.6; its uniqueness follows from Corollary 6.3.

7.1. The stationary density equation. For $x \neq 0$, the coordinatewise change of variables $z_i = \operatorname{arctanh}(y_i)$ shows that one Gaussian step has a product density. We record the resulting stationary equation.

Proposition 7.1. *Let $\pi_{A,N}$ be an invariant probability for $P_{A,N}$ on $\mathcal{K}_N$ with $\pi_{A,N}(\{0\}) = 0$. Then, for every bounded measurable $\varphi : \mathcal{K}_N \to \mathbb{R}$,*

$$\int_{\mathcal{K}_N} \varphi(y)\, \pi_{A,N}(\mathrm{d}y) = \int_{\mathcal{K}_N} \int_{\mathbb{R}^N} \varphi\big(\tanh(A\|x\|_2 g/\sqrt{N})\big)\, \mathcal{G}_N(\mathrm{d}g)\, \pi_{A,N}(\mathrm{d}x). \tag{7.1}$$

Moreover, $\pi_{A,N}(\partial[-1,1]^N) = 0$ and $\pi_{A,N}$ is absolutely continuous on $(-1,1)^N$. If $f_{A,N}$ denotes its density, then, for $x \neq 0$ and $y \in (-1,1)^N$,

$$\kappa_{A,N}(x,y) = \prod_{i=1}^N \frac{\sqrt{N}}{A\|x\|_2\sqrt{2\pi}}\frac{1}{1-y_i^2}\exp\left(-\frac{N\operatorname{arctanh}(y_i)^2}{2A^2\|x\|_2^2}\right) \tag{7.2}$$

is the transition density from x to y, and

$$f_{A,N}(y) = \int_{\mathcal{K}_N\setminus\{0\}} \kappa_{A,N}(x,y)\, \pi_{A,N}(\mathrm{d}x). \tag{7.3}$$

If $\pi_{A,N}(\mathrm{d}x) = f_{A,N}(x)\,\mathrm{d}x$, this becomes

$$f_{A,N}(y) = \int_{\mathcal{K}_N\setminus\{0\}} \kappa_{A,N}(x,y) f_{A,N}(x)\,\mathrm{d}x. \tag{7.4}$$

Proof. The weak equation (7.1) is the invariance relation $\pi_{A,N} = \pi_{A,N}P_{A,N}$. To identify the transition density, fix $x \neq 0$. By (2.1), the rows of $\mathsf{W}_A^{(N)}$ are independent, and the coordinates of $\mathsf{W}_A^{(N)}x$ are independent centered Gaussians with variance $A^2\|x\|_2^2/N$. Since $\tanh z \in (-1,1)$ for every finite z, the one-step law from $x \neq 0$ gives no mass to $\partial[-1,1]^N$. Since $\pi_{A,N}(\{0\}) = 0$, invariance therefore gives $\pi_{A,N}(\partial[-1,1]^N) = 0$. Thus, if $z_i = \operatorname{arctanh}(y_i)$, the change of variables $y_i = \tanh z_i$ gives the product density (7.2). Since $\pi_{A,N}(\{0\}) = 0$, integrating this transition density against the invariant law gives (7.3). This also proves absolute continuity on $(-1,1)^N$. The last display is the same identity when the invariant law is represented by its density. $\square$

7.2. Log-polar coordinates. To analyze the invariant law near 0, we use log-polar coordinates. Let $\mathbb{S}^{N-1}$ be the Euclidean unit sphere and write $\widehat{z} = z/\|z\|_2$ for $z \neq 0$. For $x \neq 0$, set

$$r = \|x\|_2, \qquad \theta = \frac{x}{\|x\|_2}, \qquad y = -\log r.$$

The origin corresponds to $y = +\infty$, so a power singularity at zero becomes an exponential tail for the logarithmic variable Y.

Let $\widetilde{\pi}_{A,N}$ be the image of $\pi_{A,N}$ under $x \mapsto (Y, \Theta) = (-\log\|x\|_2, x/\|x\|_2)$. Since $0 < \|x\|_2 \leqslant \sqrt{N}$ on $\mathcal{K}_N \backslash \{0\}$, the variable Y takes values in $[-\frac{1}{2}\log N, \infty)$. For $r > 0$, $\theta \in \mathbb{S}^{N-1}$, and $W\theta \neq 0$, define

$$\eta(r, \theta, W) := \log \frac{r\|W\theta\|_2}{\|\tanh(rW\theta)\|_2}, \tag{7.5}$$

$$\Theta_+(r, \theta, W) := \frac{\tanh(rW\theta)}{\|\tanh(rW\theta)\|_2}, \qquad \Theta_0(\theta, W) := \widehat{W\theta}. \tag{7.6}$$

Here η is the correction to the logarithmic radius caused by the nonlinearity, Θ_+ is the direction after the nonlinear update, and Θ_0 is the direction after the corresponding linear update. Since $W\theta \neq 0$ almost surely, the stationary equation (7.1) becomes in these coordinates

$$\int \varphi(y, \theta)\, \widetilde{\pi}_{A,N}(\mathrm{d}y, \mathrm{d}\theta) = \int \mathbb{E}\Big[\varphi\Big(y - \log\|W\theta\|_2 + \eta(e^{-y}, \theta, W),\, \Theta_+(e^{-y}, \theta, W)\Big)\Big]\, \widetilde{\pi}_{A,N}(\mathrm{d}y, \mathrm{d}\theta) \tag{7.7}$$

for every bounded measurable φ.

The linear part of (7.7) has increment $-\log\|W\theta\|_2$ and angular update Θ_0. Since $\tanh u = u + O(u^3)$ componentwise, the logarithmic and angular errors are quadratic in the radius near the origin. Their integrated total-variation bound is given in Lemma 7.7.

7.3. The linear renewal problem. For large y, equivalently for small radius, the nonlinear correction in (7.7) is lower order. We first consider the linearized Markov additive chain

$$(Y, \Theta) \mapsto (Y + \Delta, \Theta_0), \qquad \Delta = -\log\|\mathsf{W}_A^{(N)}\Theta\|_2. \tag{7.8}$$

This is the multidimensional analogue of the one-dimensional logarithmic increments in (3.1). For a general matrix ensemble this would be a genuine Markov additive chain on $\mathbb{R} \times \mathbb{S}^{N-1}$: the log norm is an additive functional of the direction chain; see [BL85]. Renewal theory for this product chain is developed in [Kes73]; see also [BDM16]. In the present Gaussian ensemble, isotropy collapses the angular part: conditionally on any direction, $W\Theta$ has the same centered Gaussian law.

Let $\bar{\sigma}_N$ denote normalized surface measure on $\mathbb{S}^{N-1}$. We equip $C(\mathbb{S}^{N-1})$ with the supremum norm. Define the linear log-polar transition kernel by

$$P^{\mathrm{lin}}_{A,N}(\theta, \cdot, \cdot) := \mathrm{Law}\Big(-\log\|\mathsf{W}_A^{(N)}\theta\|_2,\, \widehat{\mathsf{W}_A^{(N)}\theta}\Big).$$

For $0 \leqslant \beta < N$, define the transfer operator on $C(\mathbb{S}^{N-1})$ by

$$\mathcal{L}_{A,N,\beta}\varphi(\theta) := \int e^{\beta z}\varphi(\theta')\, P^{\mathrm{lin}}_{A,N}(\theta, \mathrm{d}z, \mathrm{d}\theta') = \mathbb{E}\Big[\|\mathsf{W}_A^{(N)}\theta\|_2^{-\beta}\varphi\big(\widehat{\mathsf{W}_A^{(N)}\theta}\big)\Big]. \tag{7.9}$$

The Gaussian transfer operator can be computed explicitly.

Lemma 7.2. *For every $0 \leqslant \beta < N$, the operator $\mathcal{L}_{A,N,\beta}$ is bounded on $C(\mathbb{S}^{N-1})$ and satisfies*

$$\mathcal{L}_{A,N,\beta}\varphi(\theta) = \mathcal{M}_{A,N}(\beta)\int_{\mathbb{S}^{N-1}} \varphi\, \mathrm{d}\bar{\sigma}_N, \qquad \theta \in \mathbb{S}^{N-1}, \tag{7.10}$$

where

$$\mathcal{M}_{A,N}(\beta) := \mathbb{E}\|\mathsf{W}_A^{(N)}\theta\|_2^{-\beta} = \left(\frac{\sqrt{N}}{A}\right)^{\beta} 2^{-\beta/2}\frac{\Gamma((N-\beta)/2)}{\Gamma(N/2)}. \tag{7.11}$$

Consequently, its spectral radius $r_{A,N}(\beta)$ is given by

$$r_{A,N}(\beta) = \mathcal{M}_{A,N}(\beta) = A^{-\beta}\mathcal{M}_{1,N}(\beta). \tag{7.12}$$

Proof. Fix $\theta \in \mathbb{S}^{N-1}$. By (2.1), the vector $\mathsf{W}_A^{(N)}\theta$ has law $(A/\sqrt{N})G$, where $G \sim \mathcal{N}(0, I_N)$. Hence

$$\mathbb{E}\|\mathsf{W}_A^{(N)}\theta\|_2^{-\beta} = \left(\frac{\sqrt{N}}{A}\right)^{\beta} \mathbb{E}\|G\|_2^{-\beta}.$$

The last expectation is finite exactly for $\beta < N$, since the standard Gaussian density is bounded and the singularity of $\|z\|_2^{-\beta}$ at the origin is integrable in dimension N precisely in this range. The displayed Gamma-function formula is the standard moment formula for $\chi_N = \|G\|_2$.

The same isotropy also shows that $\widehat{\mathsf{W}_A^{(N)}\theta}$ has law $\bar{\sigma}_N$ and is independent of $\|\mathsf{W}_A^{(N)}\theta\|_2$. Therefore (7.10) holds. A rank-one positive operator of the form $\mathcal{M}_{A,N}(\beta)\bar{\sigma}_N(\varphi)\mathbf{1}$ has spectral radius $\mathcal{M}_{A,N}(\beta)$. □

The Cramér exponent is the unique positive solution of the criticality equation for this operator.

Lemma 7.3. *Assume $A > A_c(N)$, with $A_c(N)$ defined in (2.26). Then there is a unique $\beta_{A,N} \in (0, N)$ satisfying*

$$r_{A,N}(\beta_{A,N}) = 1. \tag{7.13}$$

Equivalently, $\beta_{A,N}$ is the unique positive solution of

$$A^{-\beta}N^{\beta/2}2^{-\beta/2}\frac{\Gamma((N-\beta)/2)}{\Gamma(N/2)} = 1, \qquad 0 < \beta < N. \tag{7.14}$$

Proof. Recall that $\ell_{A,N}$ is the linearized radius multiplier from (2.25).

$$F(\beta) := \log \mathbb{E}\ell_{A,N}^{-\beta}.$$

By (2.26), the condition $A > A_c(N)$ is equivalent to $\mathbb{E}\log \ell_{A,N} > 0$. Thus $F(0) = 0$ and $F'(0) = -\mathbb{E}\log \ell_{A,N} < 0$. Moreover F is strictly convex by Hölder's inequality and the nondegeneracy of $\ell_{A,N}$, and $F(\beta) \to \infty$ as $\beta \uparrow N$, because $\mathbb{E}\chi_N^{-\beta} \uparrow \infty$. Therefore $F(\beta) = 0$ has exactly one solution in $(0, N)$. The explicit equation is just (7.11). □

Under the exponential tilt, the angular coordinate is refreshed from the uniform law at each step, so the renewal problem is scalar. Feller's whole-line renewal theorem gives the translated-interval limits [Fel71]. For another formulation of those interval limits, see also [Gut09]. Applying Feller's direct-Riemann-sum argument [Fel71] gives the directly Riemann integrable version used below.

We now state the renewal limit at the Cramér root.

Lemma 7.4. *Assume $A > A_c(N)$ and let $\beta_{A,N}$ be the exponent from Lemma 7.3. Using the multiplier $\ell_{A,N}$ from (2.25), define*

$$\int g(z)\,\widehat{\mu}_{A,N}(\mathrm{d}z) := \mathbb{E}\Big[\ell_{A,N}^{-\beta_{A,N}} g(-\log \ell_{A,N})\Big]. \tag{7.15}$$

Then $\widehat{\mu}_{A,N}$ is a nonlattice probability measure. Its drift satisfies

$$\widehat{m}_{A,N} := \int_{\mathbb{R}} z\,\widehat{\mu}_{A,N}(\mathrm{d}z) \in (0, \infty).$$

Set $\widehat{P}_{A,N}(\theta, \mathrm{d}z, \mathrm{d}\theta') := \bar{\sigma}_N(\mathrm{d}\theta')\,\widehat{\mu}_{A,N}(\mathrm{d}z)$.

Let $(\mathcal{H}_y)_{y\in\mathbb{R}}$ be a locally bounded family of finite signed measures on $\mathbb{S}^{N-1}$ satisfying the boundary conditions

$$\lim_{y\to-\infty}\|\mathcal{H}_y\|_{\mathrm{TV}} = 0, \qquad \lim_{y\to\infty} e^{-\beta_{A,N}y}\|\mathcal{H}_y\|_{\mathrm{TV}} = 0. \tag{7.16}$$

Assume also that the forcing $(\Psi_y)_{y\in\mathbb{R}}$ satisfies

$$\sum_{k\in\mathbb{Z}} \sup_{y\in[k,k+1]} \|\Psi_y\|_{\mathrm{TV}} < \infty. \tag{7.17}$$

For every continuous φ, assume that $y \mapsto \Psi_y(\varphi)$ is continuous and that

$$\mathcal{H}_y(\varphi) = \int_{\mathbb{S}^{N-1}} \int \varphi(\theta')\, \widehat{P}_{A,N}(\theta, \mathrm{d}z, \mathrm{d}\theta')\, \mathcal{H}_{y-z}(\mathrm{d}\theta) + \Psi_y(\varphi). \tag{7.18}$$

Then

$$\mathcal{H}_y(\varphi) \to \frac{1}{\widehat{m}_{A,N}} \left(\int_{\mathbb{R}} \Psi_t(\mathbf{1})\,\mathrm{d}t \right) \int_{\mathbb{S}^{N-1}} \varphi\, \mathrm{d}\bar{\sigma}_N, \qquad y \to \infty, \tag{7.19}$$

where $\mathbf{1}$ denotes the constant function on $\mathbb{S}^{N-1}$.

Proof. Step 1: Tilted law. The identity $r_{A,N}(\beta_{A,N}) = 1$ says precisely that $\widehat{\mu}_{A,N}$ has total mass one. Its law has a density, since χ_N has a density on $(0,\infty)$, and hence it is nonlattice. The drift is finite because $\mathbb{E}\chi_N^{-\beta}|\log \chi_N| < \infty$ for $\beta < N$. It is positive since

$$\widehat{m}_{A,N} = \frac{\mathrm{d}}{\mathrm{d}\beta} \log \mathcal{M}_{A,N}(\beta) \bigg|_{\beta=\beta_{A,N}},$$

and this derivative is positive at the unique positive zero $\beta_{A,N}$ of the strictly convex function $\log \mathcal{M}_{A,N}(\beta)$.

Step 2: Scalar renewal representation. Put $h_y := \mathcal{H}_y(\mathbf{1})$. Taking $\varphi = \mathbf{1}$ in (7.18) gives the scalar renewal equation

$$h_y = \int_{\mathbb{R}} h_{y-z}\, \widehat{\mu}_{A,N}(\mathrm{d}z) + \Psi_y(\mathbf{1}).$$

Let $S_n = Z_1 + \cdots + Z_n$, where the variables Z_i are independent with law $\widehat{\mu}_{A,N}$. Iterating the last display gives

$$h_y = \mathbb{E}h_{y-S_n} + \sum_{k=0}^{n-1} \mathbb{E}\Psi_{y-S_k}(\mathbf{1}).$$

We claim that, for this fixed y, the terminal term tends to zero as $n \to \infty$. Since $\widehat{m}_{A,N} > 0$, we have $S_n \to \infty$ almost surely. Fix $L < \infty$. On the half-line $(-\infty, L]$, local boundedness together with the first condition in (7.16) gives $\sup_{u\leqslant L} \|\mathcal{H}_u\|_{\mathrm{TV}} < \infty$. Since $y - S_n \to -\infty$ almost surely, dominated convergence gives

$$\mathbb{E}\big[\|\mathcal{H}_{y-S_n}\|_{\mathrm{TV}} \mathbb{1}_{\{y-S_n\leqslant L\}}\big] \to 0.$$

On the complementary event, the second condition in (7.16) gives, for L sufficiently large,

$$\mathbb{E}\big[\|\mathcal{H}_{y-S_n}\|_{\mathrm{TV}} \mathbb{1}_{\{y-S_n>L\}}\big] \leqslant \varepsilon e^{\beta_{A,N} y} \mathbb{E} e^{-\beta_{A,N} S_n} = \varepsilon e^{\beta_{A,N} y}.$$

The equality uses the defining tilt (7.15). Adding the two contributions and then letting $\varepsilon \downarrow 0$ gives

$$\mathbb{E}\|\mathcal{H}_{y-S_n}\|_{\mathrm{TV}} \to 0.$$

Since $\|\mathbf{1}\|_\infty = 1$, the signed terminal term is controlled by this total variation bound,

$$|\mathbb{E}h_{y-S_n}| = |\mathbb{E}\, \mathcal{H}_{y-S_n}(\mathbf{1})| \leqslant \mathbb{E}\|\mathcal{H}_{y-S_n}\|_{\mathrm{TV}} \to 0.$$

Therefore, after first fixing y and then letting $n \to \infty$,

$$h_y = \sum_{k=0}^{\infty} \mathbb{E}\Psi_{y-S_k}(\mathbf{1}).$$

Step 3: Scalar renewal limit. Let $y \to \infty$ and decompose the signed forcing as $\Psi_y(\mathbf{1}) = \Psi_y(\mathbf{1})^+ - \Psi_y(\mathbf{1})^-$; both parts are nonnegative and directly Riemann integrable by (7.17). The increment law $\widehat{\mu}_{A,N}$ is nonarithmetic with mean $\widehat{m}_{A,N} \in (0,\infty)$ and charges both half-lines, so the renewal measure $U := \sum_{k\geqslant 0} \mathcal{L}(S_k)$ obeys the general renewal theorem on the whole line: $U\{I+y\} \to |I|/\widehat{m}_{A,N}$ as $y \to +\infty$ and $U\{I+y\} \to 0$ as $y \to -\infty$ for every finite interval I [Fel71, Sec. XI.9, (9.2)–(9.3)]. Applying the upper- and lower-step-function argument of [Fel71, Sec. XI.1, (1.10)–(1.17)] to these whole-line interval limits extends the conclusion to any directly Riemann integrable integrand in the sense of (7.17). Apply it separately to $\Psi_\cdot(\mathbf{1})^+$ and $\Psi_\cdot(\mathbf{1})^-$. The identity $\int_{\mathbb{R}} z(y-s)\, U(\mathrm{d}s) = \sum_{k\geqslant 0} \mathbb{E}\, z(y-S_k)$ then gives

$$\sum_{k=0}^{\infty} \mathbb{E}\, \Psi_{y-S_k}(\mathbf{1})^{\pm} \to \frac{1}{\widehat{m}_{A,N}} \int_{\mathbb{R}} \Psi_t(\mathbf{1})^{\pm}\, \mathrm{d}t, \qquad y \to \infty,$$

and subtracting the two limits yields

$$h_y \to \frac{1}{\widehat{m}_{A,N}} \int_{\mathbb{R}} \Psi_t(\mathbf{1})\, \mathrm{d}t.$$

Step 4: Angular reconstruction. For a general continuous φ, the product form of $\widehat{P}_{A,N}$ gives

$$\mathcal{H}_y(\varphi) = \left(\int_{\mathbb{S}^{N-1}} \varphi\, \mathrm{d}\bar{\sigma}_N \right) \int_{\mathbb{R}} h_{y-z}\, \widehat{\mu}_{A,N}(\mathrm{d}z) + \Psi_y(\varphi).$$

Using the scalar renewal equation to replace the convolution by $h_y - \Psi_y(\mathbf{1})$, and using direct Riemann integrability to get $\Psi_y(\varphi) \to 0$ and $\Psi_y(\mathbf{1}) \to 0$, yields (7.19). $\square$

Remark 7.5. The simplification in Lemma 7.2 is special to the Gaussian ensemble in (2.1). For a non-isotropic matrix ensemble the direction would not be refreshed from $\bar{\sigma}_N$ in one step, and the analogue of Lemma 7.4 would have to be replaced by a genuine Markov renewal theorem for products of random matrices [Kes73] and [BDM16].

7.4. The nonlinear renewal equation. We next restore the tanh term. After the exponential tilt defining $\widehat{P}_{A,N}$, the forcing is the difference between the true log-polar update in (7.7) and the linearized update in (7.8). The perturbative estimate uses exponential moments below the Cramér exponent.

Lemma 7.6. *Assume $A > A_c(N)$ and let $\pi_{A,N}$ be a nonzero invariant law as in Proposition 7.1. Let (Y,Θ) have law $\widetilde{\pi}_{A,N}$. Then, for every $0 \leqslant \alpha < \beta_{A,N}$,*

$$\mathbb{E} e^{\alpha Y} < \infty. \tag{7.20}$$

Proof. The case $\alpha = 0$ is trivial, so assume $0 < \alpha < \beta_{A,N}$. Let ν be the image of $\pi_{A,N}$ under the squared-radius map $r_N(x) = N^{-1}\|x\|_2^2$, defined before Proposition 2.4. By Proposition 2.4, ν is a nonzero invariant law for the squared-radius transition kernel $K_{A,N}$.

Since $\alpha < \beta_{A,N}$ and $\beta_{A,N}$ is the first positive zero of $\log \mathcal{M}_{A,N}$, we have

$$\mathbb{E}\left[\left(\frac{A}{\sqrt{N}} \chi_N \right)^{-\alpha} \right] < 1.$$

Put $p = \alpha/2$ and $V(q) = q^{-p}$. Since

$$\frac{F_{A,N}(q,G)}{q} = \frac{1}{N} \sum_{i=1}^{N} A^2 G_i^2 \left(\frac{\tanh(A\sqrt{q}\, G_i)}{A\sqrt{q}\, G_i} \right)^2 \xrightarrow[q\downarrow 0]{} \frac{A^2}{N} \chi_N^2 \qquad \text{a.s.},$$

the fixed-N version of the truncation argument in Proposition 6.8 gives

$$\mathbb{E}\left[\left(\frac{F_{A,N}(q,G)}{q}\right)^{-p}\right] \longrightarrow \mathbb{E}\left[\left(\frac{A}{\sqrt{N}}\chi_N\right)^{-\alpha}\right] < 1.$$

Hence we may choose $R_0 > 0$ and $a < 1$ such that, for $0 < q \leqslant R_0$,

$$K_{A,N}V(q) = q^{-p}\mathbb{E}\left[\left(\frac{F_{A,N}(q,G)}{q}\right)^{-p}\right] \leqslant aV(q).$$

On the compact interval $[R_0, 1]$, the same small-ball estimate gives

$$B := \sup_{R_0 \leqslant q \leqslant 1} K_{A,N}V(q) < \infty.$$

Here finiteness uses $\alpha < N$, which follows from the finiteness of $\mathbb{E}[(A\chi_N/\sqrt{N})^{-\alpha}]$. Therefore

$$K_{A,N}V(q) \leqslant aV(q) + B, \qquad q \in (0,1].$$

The Cesàro construction in the proof of Proposition 2.6 therefore produces a nonzero invariant law $\overline{\nu}$ with $\int q^{-p}\,\overline{\nu}(\mathrm{d}q) < \infty$. By the uniqueness of the nonzero invariant law in Proposition 6.1, $\overline{\nu} = \nu$. Consequently,

$$e^{\alpha Y} = \|x\|_2^{-\alpha} = N^{-\alpha/2} r_N(x)^{-\alpha/2},$$

and (7.20) follows. □

The nonlinear correction is controlled by the following polar perturbation estimate.

Lemma 7.7. *Assume $A > A_c(N)$ and let $\beta_{A,N} \in (0,N)$ be the exponent from Lemma 7.3. Fix $\alpha \in (0,2)$ and set $h(t) := \min\{1, e^{-Nt}\}$. Then there exists $C < \infty$ such that*

$$\sum_{k\in\mathbb{Z}} \sup_{t\in[k,k+1]} e^{\beta_{A,N}t} h(t) < \infty \tag{7.21}$$

and, for every $0 < r \leqslant \sqrt{N}$ and every $t \in \mathbb{R}$,

$$\left\|\mathbb{E}\left[\mathbb{1}_{\{\|\tanh(rU)\|_2 < re^{-t}\}}\delta_{\widehat{\tanh(rU)}}\right] - \mathbb{E}\left[\mathbb{1}_{\{\|U\|_2 < e^{-t}\}}\delta_{\widehat{U}}\right]\right\|_{\mathrm{TV}} \leqslant Cr^{\alpha}h(t), \tag{7.22}$$

where δ_z denotes the Dirac mass at z, $U = (A/\sqrt{N})G$, and $G \sim \mathcal{N}(0, I_N)$.

Proof. Put

$$T_r u := r^{-1}\tanh(ru), \qquad u \in \mathbb{R}^N.$$

Then $\widehat{\tanh(rU)} = \widehat{T_rU}$ and $\{\|\tanh(rU)\|_2 < re^{-t}\} = \{\|T_rU\|_2 < e^{-t}\}$. Let p_0 be the density of U. The map T_r is a diffeomorphism from $\mathbb{R}^N$ onto $(-r^{-1}, r^{-1})^N$, and the density of T_rU is

$$p_r(v) = p_0\big(r^{-1}\operatorname{arctanh}(rv)\big)\prod_{i=1}^{N}(1 - r^2v_i^2)^{-1}\mathbb{1}_{\{|rv_i|<1,\ 1\leqslant i\leqslant N\}}. \tag{7.23}$$

Since angular projection and restriction to the ball $\{\|v\|_2 < e^{-t}\}$ cannot increase total variation, the left side of (7.22) is at most

$$\int_{\{\|v\|_2 < e^{-t}\}} |p_r(v) - p_0(v)|\,\mathrm{d}v. \tag{7.24}$$

We claim that, for every $\rho > 0$ and $0 < r \leqslant \sqrt{N}$,

$$\int_{\{\|v\|_2<\rho\}} |p_r(v) - p_0(v)| \,\mathrm{d}v \leqslant Cr^\alpha \min\{1, \rho^N\}. \tag{7.25}$$

For $1/4 \leqslant r \leqslant \sqrt{N}$, this follows from the uniform boundedness of the densities p_r on this compact range, together with the trivial total-mass bound; here r^α is bounded below by a positive constant. The uniform boundedness follows from (7.23): in one dimension the function

$$s \mapsto \exp\{-c \operatorname{arctanh}(s)^2\}(1 - s^2)^{-1}, \qquad |s| < 1,$$

is bounded for every $c > 0$, and here the constants range over a compact subset of $(0, \infty)$. It remains to consider $0 < r \leqslant 1/4$. On the set $\|v\|_2 \leqslant (2r)^{-1}$, Taylor's theorem applied to (7.23) gives

$$|p_r(v) - p_0(v)| \leqslant Cr^2(1 + \|v\|_2^4)e^{-c\|v\|_2^2}.$$

We spell out the estimate. If $\|v\|_2 \leqslant r^{-1/2}$, then $|rv_i| \leqslant \sqrt{r} \leqslant 1/2$, and hence

$$r^{-1} \operatorname{arctanh}(rv_i) = v_i + O(r^2|v_i|^3), \qquad -\log(1 - r^2v_i^2) = O(r^2v_i^2),$$

uniformly in i. Expanding the logarithm of the density ratio $p_r(v)/p_0(v)$ gives

$$|\log(p_r(v)/p_0(v))| \leqslant Cr^2(1 + \|v\|_2^4),$$

and the displayed bound follows after decreasing c. On the remaining annulus $r^{-1/2} < \|v\|_2 \leqslant (2r)^{-1}$, both $p_0(v)$ and $p_r(v)$ are bounded by $Ce^{-c\|v\|_2^2}$, while $r^2(1 + \|v\|_2^4) \geqslant 1$, so the same estimate follows, again after decreasing c. Integrating this bound over $\{\|v\|_2 < \rho\}$, and using that $\int_{\{\|v\|_2<\rho\}}(1 + \|v\|_2^4)e^{-c\|v\|_2^2} \,\mathrm{d}v \leqslant C \min\{1, \rho^N\}$, since the integrand is bounded near the origin and integrable over all of $\mathbb{R}^N$, this contributes at most $Cr^2 \min\{1, \rho^N\} \leqslant Cr^\alpha \min\{1, \rho^N\}$. On the complement, both the p_0 and p_r terms can contribute only through radii at least $(2r)^{-1}$; for the p_r term this is because $\|T_rU\|_2 \leqslant \|U\|_2$. The Gaussian tail is then bounded by $Ce^{-c/r^2} \leqslant Cr^\alpha$. This is enough when $\rho \geqslant 1$. If $\rho < 1$, then $\rho < (2r)^{-1}$ because $r \leqslant 1/4$, so this complement does not intersect $\{\|v\|_2 < \rho\}$. This proves (7.25).

Taking $\rho = e^{-t}$ in (7.25) proves (7.22). Since $\beta_{A,N} < N$, the function h in the statement satisfies (7.21). ☐

Definition 7.8 (Nonlinear renewal forcing)**.** Fix an invariant law $\pi_{A,N}$ as in Proposition 7.1, let (Y, Θ) have the associated log-polar law $\widetilde{\pi}_{A,N}$ from (7.7), and let W be an independent copy of $\mathsf{W}_A^{(N)}$. Put

$$\Delta = -\log\|W\Theta\|_2, \qquad \Theta_0 = \widehat{W\Theta}, \qquad \Theta_+ = \Theta_+(e^{-Y}, \Theta, W), \qquad \eta = \eta(e^{-Y}, \Theta, W).$$

For each continuous test function φ on $\mathbb{S}^{N-1}$, define

$$\begin{aligned}\Psi_y^\pi(\varphi) := e^{\beta_{A,N}y}\mathbb{E}\Big[&\varphi(\Theta_+)\mathbb{1}_{\{Y+\Delta+\eta>y\}} \\ &- \varphi(\Theta_0)\mathbb{1}_{\{Y+\Delta>y\}}\Big].\end{aligned} \tag{7.26}$$

This bounded linear functional on $C(\mathbb{S}^{N-1})$ defines a finite signed measure, also denoted by Ψ_y^π. Set

$$c_{\Psi^\pi} := \frac{1}{\widehat{m}_{A,N}} \int_{\mathbb{R}} \Psi_t^\pi(\mathbf{1}) \,\mathrm{d}t, \tag{7.27}$$

where $\widehat{m}_{A,N} := \int_{\mathbb{R}} z \,\widehat{\mu}_{A,N}(\mathrm{d}z)$ is the tilted drift. We say that the nonlinear renewal forcing is admissible if $(\Psi_y^\pi)_{y\in\mathbb{R}}$ satisfies (7.17) and $c_{\Psi^\pi} > 0$.

We next verify that this forcing is admissible.

Proposition 7.9. *Assume $A > A_c(N)$ and let $\pi_{A,N}$ be a nonzero invariant law as in Proposition 7.1. Then the forcing defined in Definition 7.8 is admissible.*

Proof. Choose $\alpha \in (0, \min\{2, \beta_{A,N}\})$. By Gaussian isotropy, after conditioning on (Y, Θ) the vector $W\Theta$ has the same law as $U = (A/\sqrt{N})G$, independently of (Y, Θ). With $r = e^{-Y}$ and $t = y - Y$, the signed measure inside (7.26) is exactly the polar perturbation error in Lemma 7.7. Therefore

$$\|\Psi_y^\pi\|_{\mathrm{TV}} \leqslant Ce^{\beta_{A,N}y}\mathbb{E}\big[e^{-\alpha Y}h(y-Y)\big] = C\mathbb{E}\Big[e^{(\beta_{A,N}-\alpha)Y}H(y-Y)\Big], \tag{7.28}$$

where $h(t) = \min\{1, e^{-Nt}\}$ is the function in Lemma 7.7, and set $H(t) := e^{\beta_{A,N}t}h(t)$. Then

$$H(t) = \begin{cases} e^{\beta_{A,N}t}, & t < 0, \\ e^{-(N-\beta_{A,N})t}, & t \geqslant 0. \end{cases}$$

In particular,

$$C_H := \sup_{a\in\mathbb{R}} \sum_{k\in\mathbb{Z}} \sup_{y\in[k,k+1]} H(y-a) < \infty.$$

Combining this uniform shifted summability with Lemma 7.6, with moment exponent $\beta_{A,N} - \alpha$, gives

$$\begin{aligned}\sum_{k\in\mathbb{Z}} \sup_{y\in[k,k+1]} \|\Psi_y^\pi\|_{\mathrm{TV}} &\leqslant C\mathbb{E}\left[e^{(\beta_{A,N}-\alpha)Y} \sum_{k\in\mathbb{Z}} \sup_{y\in[k,k+1]} H(y-Y)\right] \\ &\leqslant CC_H\mathbb{E}e^{(\beta_{A,N}-\alpha)Y} < \infty.\end{aligned}$$

For every continuous φ, the map $y \mapsto \Psi_y^\pi(\varphi)$ is continuous. To see this, condition on (Y, Θ). The vector $W\Theta$ has the nondegenerate Gaussian law $\mathcal{N}(0, (A^2/N)I_N)$ and therefore has a density. Since every Euclidean sphere has Lebesgue measure zero, $\|W\Theta\|_2$ has no atoms, and the same is true of

$$Y + \Delta = Y - \log\|W\Theta\|_2.$$

For each finite Y, the coordinatewise map

$$v \longmapsto \tanh(e^{-Y}v)$$

is a diffeomorphism from $\mathbb{R}^N$ onto $(-1,1)^N$. Its image of $W\Theta$ consequently has a density, so its Euclidean norm also has no atoms. By (7.5),

$$Y + \Delta + \eta = -\log\|\tanh(e^{-Y}W\Theta)\|_2,$$

and this second threshold variable is atomless as well. Averaging over (Y, Θ) preserves these zero atom probabilities. Hence, whenever $y_n \to y$, both indicators in (7.26) converge almost surely. Since φ is bounded, dominated convergence on compact y-intervals, followed by multiplication by $e^{\beta_{A,N}y}$, proves the claimed continuity.

For $\varphi = \mathbf{1}$, the angular variables disappear and $\eta \geqslant 0$ gives

$$\Psi_y^\pi(\mathbf{1}) = e^{\beta_{A,N}y}\mathbb{P}(Y + \Delta < y < Y + \Delta + \eta) \geqslant 0.$$

The direct Riemann integrability just proved implies that this function is integrable. By Tonelli's theorem,

$$\int_{\mathbb{R}} \Psi_y^\pi(\mathbf{1})\,\mathrm{d}y = \frac{1}{\beta_{A,N}}\mathbb{E}\Big[e^{\beta_{A,N}(Y+\Delta)}\Big(e^{\beta_{A,N}\eta} - 1\Big)\Big].$$

The right-hand side is positive. Indeed, $Y < \infty$ and $W\Theta \neq 0$ almost surely, while

$$\|\tanh(e^{-Y}W\Theta)\|_2 < e^{-Y}\|W\Theta\|_2 \qquad \text{almost surely.}$$

Since $\widehat{m}_{A,N} > 0$ by Lemma 7.4, we obtain $c_{\Psi^\pi} > 0$. ∎

It remains to transfer the scalar renewal limit to the nonlinear stationary law. We now prove the theorem stated in the introduction.

Proof of Theorem 1.15. We apply Lemma 7.4 to the weighted log-polar tail measures. The angular variable is retained throughout the argument.

Step 1: Weighted tail measure. Let (Y, Θ) have law $\widetilde{\pi}_{A,N}$ and define, for $y \in \mathbb{R}$,

$$\mathcal{H}_y(B) := e^{\beta_{A,N} y} \mathbb{E}\big[\mathbb{1}_{\{Y>y\}} \mathbb{1}_{\{\Theta \in B\}}\big]. \tag{7.29}$$

Each $\mathcal{H}_y$ is a nonnegative finite measure on $\mathbb{S}^{N-1}$. Since $\widetilde{\pi}_{A,N}$ is the image of $\pi_{A,N}$ under $x \mapsto (-\log\|x\|_2, \widehat{x})$, the event $\{Y > y,\ \Theta \in B\}$ is the image of $\{0 < \|x\|_2 < e^{-y},\ \widehat{x} \in B\}$, whence

$$\mathcal{H}_y(B) = e^{\beta_{A,N} y}\, \pi_{A,N}\big(0 < \|x\|_2 < e^{-y},\ \widehat{x} \in B\big). \tag{7.30}$$

By Proposition 7.1, $\pi_{A,N}$ is absolutely continuous on $(-1,1)^N$ and therefore charges no sphere $\{\|x\|_2 = s\}$; consequently the strict inequality in (7.30) may be replaced by $\|x\|_2 \leqslant e^{-y}$ without changing its value. Since $Y \geqslant -\frac{1}{2}\log N$ by the definition of $\widetilde{\pi}_{A,N}$ before (7.5), we have $\|\mathcal{H}_y\|_{\mathrm{TV}} \leqslant e^{\beta_{A,N} y}$ for all $y < -\frac{1}{2}\log N$. Hence the first condition in (7.16) holds. Moreover

$$e^{-\beta_{A,N} y}\|\mathcal{H}_y\|_{\mathrm{TV}} = \mathbb{P}(Y > y) \to 0, \qquad y \to \infty,$$

so the second condition in (7.16) holds as well. The same definition gives local boundedness of $(\mathcal{H}_y)_{y\in\mathbb{R}}$.

Step 2: Renewal equation. Let W be an independent copy of $\mathsf{W}_A^{(N)}$, independent of (Y, Θ), and use the notation of Definition 7.8. For a continuous test function φ on $\mathbb{S}^{N-1}$, stationarity and (7.7) give

$$\mathcal{H}_y(\varphi) = e^{\beta_{A,N} y} \mathbb{E}\big[\varphi(\Theta_+) \mathbb{1}_{\{Y+\Delta+\eta>y\}}\big]. \tag{7.31}$$

Add and subtract the linearized term with $(\Theta_0, Y + \Delta)$. The difference between the two terms is precisely $\Psi_y^\pi(\varphi)$ from (7.26). The remaining linearized term is the tilted convolution. Indeed, since $\widehat{P}_{A,N}$ is the $e^{\beta_{A,N} z}$-tilt of the linear log-polar transition kernel $P^{\mathrm{lin}}_{A,N}$ from (7.9),

$$\begin{aligned}\int_{\mathbb{S}^{N-1}} \int \varphi(\theta')\, \widehat{P}_{A,N}(\theta, \mathrm{d}z, \mathrm{d}\theta')\, \mathcal{H}_{y-z}(\mathrm{d}\theta) \\ = e^{\beta_{A,N} y} \mathbb{E}\big[\varphi(\Theta_0) \mathbb{1}_{\{Y+\Delta>y\}}\big].\end{aligned}$$

Therefore

$$\mathcal{H}_y(\varphi) = \int_{\mathbb{S}^{N-1}} \int \varphi(\theta')\, \widehat{P}_{A,N}(\theta, \mathrm{d}z, \mathrm{d}\theta')\, \mathcal{H}_{y-z}(\mathrm{d}\theta) + \Psi_y^\pi(\varphi). \tag{7.32}$$

This is (7.18).

By Lemma 7.4 and Proposition 7.9, there is a constant $c_{\Psi^\pi} \in (0, \infty)$. By the uniqueness in Corollary 6.3, it depends only on (A, N). We write $c_{A,N} := c_{\Psi^\pi}$. Then

$$\mathcal{H}_y(\varphi) \to c_{A,N} \int_{\mathbb{S}^{N-1}} \varphi(\theta)\, \overline{\sigma}_N(\mathrm{d}\theta) \qquad \text{as } y \to \infty. \tag{7.33}$$

Step 3: Angular reconstruction. By (7.33), the nonnegative finite measures $\mathcal{H}_y$ on the compact sphere $\mathbb{S}^{N-1}$ satisfy

$$\mathcal{H}_y(\varphi) \to c_{A,N} \int_{\mathbb{S}^{N-1}} \varphi \,\mathrm{d}\overline{\sigma}_N \qquad \varphi \in C(\mathbb{S}^{N-1}).$$

Thus $\mathcal{H}_y$ converges weakly to $c_{A,N}\overline{\sigma}_N$ as $y \to \infty$. By the Portmanteau theorem, weak convergence gives $\mathcal{H}_y(B) \to c_{A,N}\overline{\sigma}_N(B)$ for every Borel set B with $\overline{\sigma}_N(\partial B) = 0$. Substituting $s = e^{-y}$ into the identity (7.30), and using the vanishing of sphere mass,

$$\pi_{A,N}(0 < \|x\|_2 \leqslant s,\ \widehat{x} \in B) = s^{\beta_{A,N}} \mathcal{H}_y(B),$$

and since $\mathcal{H}_y(B) \to c_{A,N}\overline{\sigma}_N(B)$ as $s \downarrow 0$, this proves the directional asymptotic in the theorem. Taking $B = \mathbb{S}^{N-1}$, which has empty boundary, gives the radial asymptotic. □

8. Numerical illustrations

The figures compare the deterministic and stochastic predictions with computations. We evaluate the unrounded and rounded squared-radius mean maps by Gauss–Hermite quadrature or the available Poisson-summation formulas; these curves contain no sampling error. The unrounded radius chain and the unrounded and rounded squared-radius chains are simulated with fixed random seeds. Dashed local Gaussian curves in the supercritical cutoff plot are finite-N approximations and are not asserted by the cutoff theorem. The scripts are available from the author upon request.

Squared-radius mean dynamics.
The unrounded squared-radius mean map $V_A(q) = \mathbb{E}\tanh^2(A\sqrt{q}\,G)$ governs the deterministic squared-radius evolution. Figure 8.1 shows the transition at $A = 1$ and, in the supercritical regime, convergence to the unique positive fixed point q_*. Its multiplier $\mu_A = V_A'(q_*) \in (0, 1)$ determines the dimension-cutoff scale.

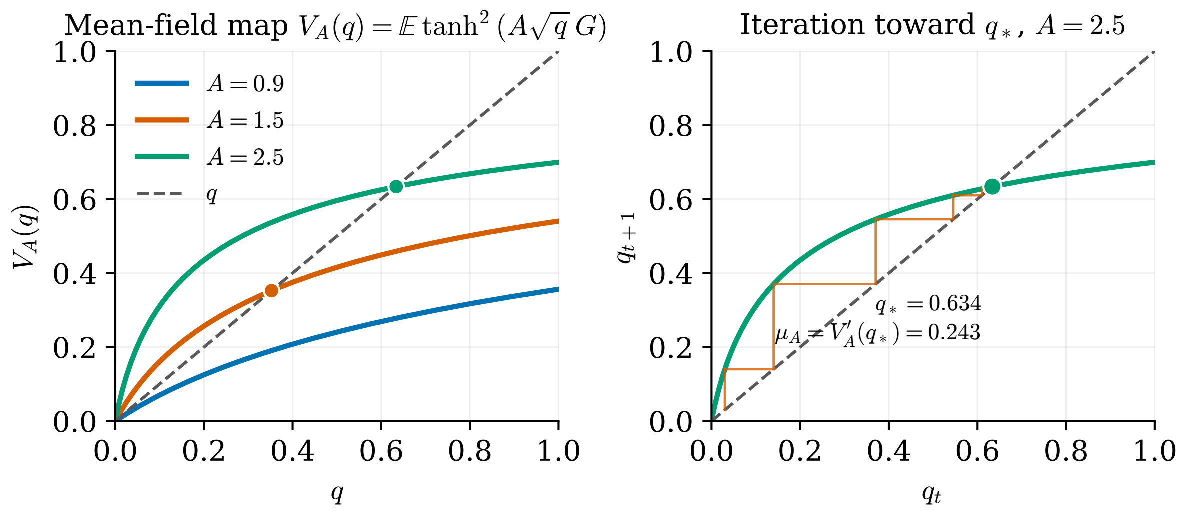

Figure 8.1: Unrounded squared-radius mean map. Left: V_A and the diagonal for three gains; a positive fixed point appears for $A > 1$. Right: iteration of V_A toward q_* for $A = 2.5$.

Supercritical dimension cutoff. Figure 8.2 shows the empirical total variation distance $\|K^t_{A,N}(q_0, \cdot) - \nu_{A,N}\|_{\mathrm{TV}}$ of the unrounded squared-radius chain to its nonzero invariant law, plotted against the centered time $t - t_N(q_0)$ for a range of widths N. The curves stay near one before the cutoff time and drop toward zero afterwards, consistently with Theorem 1.12. The dashed curves are heuristic local Gaussian approximations to the finite-N transition.

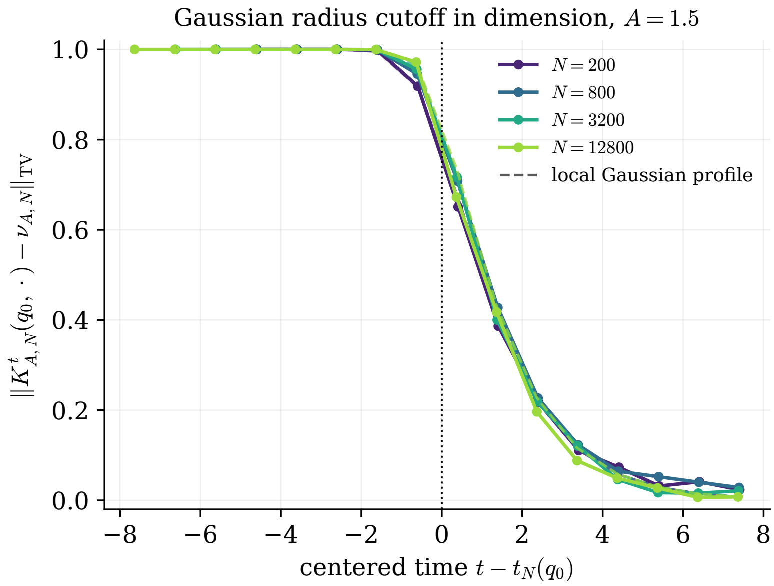

Figure 8.2: Numerical supercritical dimension cutoff for the unrounded squared-radius chain, with $A = 1.5$ and $q_0 = 0.05$. The solid curves are histogram estimates of the total variation distance to $\nu_{A,N}$ from 4,000 simulated paths. The dashed curves are heuristic local Gaussian approximations.

Invariant law and power-law singularity. In the supercritical regime, the unrounded vector chain has a nonzero invariant law characterized by Proposition 7.1. Figure 8.3 compares a long run of the unrounded radius chain with a numerical solution of a discretized invariant equation. For $N = 1$, the transition probabilities are computed from the exact conditional distribution on a mesh refined near zero. The logarithmic lower-tail plot also shows a reference line with the Cramér slope $\beta_{A,N}$ from Theorem 1.15.

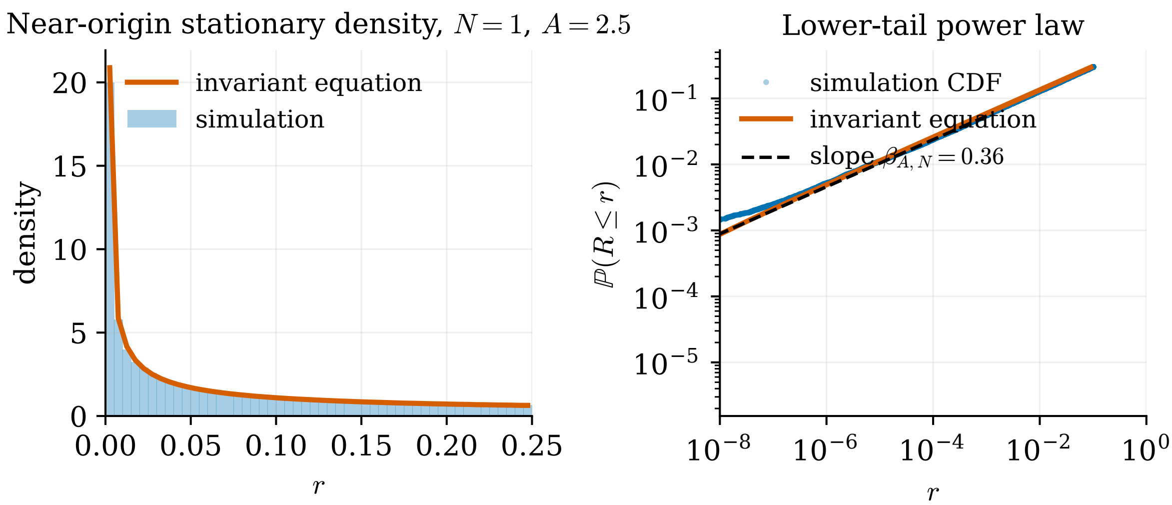

Figure 8.3: Supercritical stationary Euclidean radius for $N = 1$ and $A = 2.5$. Left: the near-origin density from simulation and a numerical solution of the invariant equation; the density increases toward zero because $\beta_{A,1} < 1$. Right: the empirical and numerical lower tails and a reference line of slope $\beta_{A,1}$. The observed slope is consistent with Theorem 1.15.

Vanishing-mesh cutoff at fixed width. At fixed width and subcritical gain, (2.5) identifies the survival probability with the total variation distance to δ_0. Figure 8.4 plots this probability on the theorem scale, with center $L_\varrho/\gamma_{A,N}$ and window $\sigma_N\gamma_{A,N}^{-3/2}\sqrt{L_\varrho}$. As ϱ decreases, the curves approach the Gaussian first-passage profile $\Phi_G(-a)$ of Theorem 1.4 and Proposition 3.2. For the smallest displayed ϱ, the figure also includes the finite-window log-random-walk reference, which displays the lattice correction.

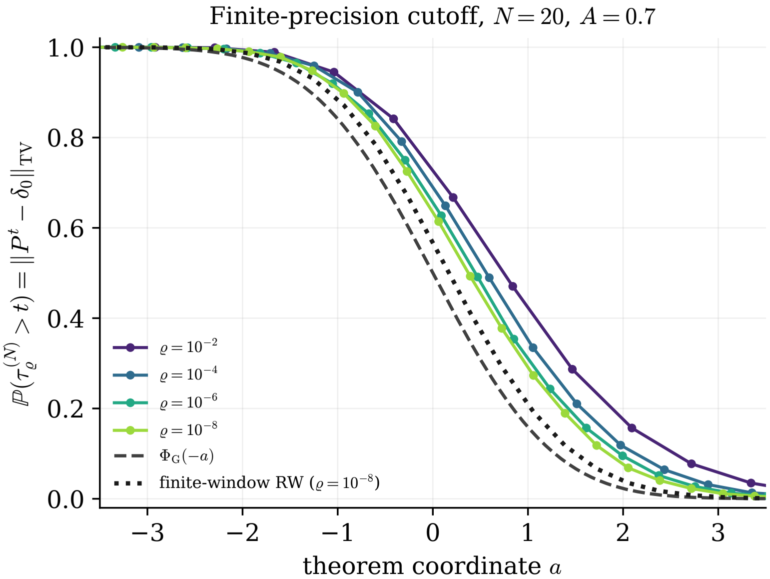

Figure 8.4: Vanishing-mesh cutoff at fixed width, with $N = 20$ and $A = 0.7$. The survival probability, equal to the total variation distance to δ_0, is plotted using the center and window in Theorem 1.4. The dashed curve is $\Phi_G(-a)$; the dotted curve is the finite-window log-random-walk comparison at $\varrho = 10^{-8}$.

Fixed-precision cutoff in dimension. Fix the mesh size and let the dimension grow in the lattice-subcritical regime. For the displayed parameters, $A = 0.7 < A_{\mathrm{lat}} \approx 0.8663$ and $\varrho = 0.5$. By Theorem 4.5, the survival probability remains near one at time $\bar{t}_N - 1$ and is near zero by time $\bar{t}_N + 2$, where $\bar{t}_N$ is the first time at which the rounded deterministic orbit enters the scale $(\log N)^{-1}$. Figure 8.5 plots the empirical survival probability against the centered integer time $t - \bar{t}_N$. The rounded squared-radius transition is sampled through its exact finite-layer count distribution, which permits the large dimensions displayed in the figure without changing the law of the rounded squared-radius chain.

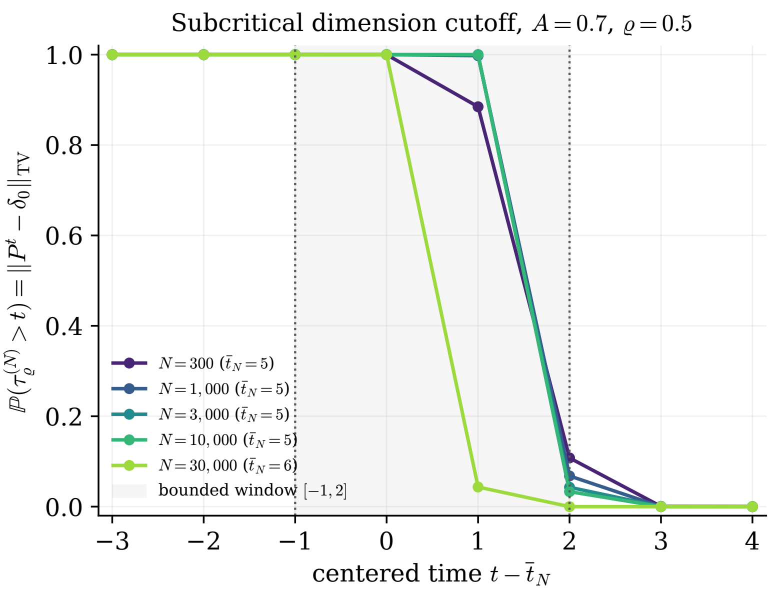

Figure 8.5: Fixed-precision subcritical dimension cutoff ($A = 0.7$, $\varrho = 0.5$, and $x_N = (1, \ldots, 1)$; 40,000 paths per dimension). The empirical survival probability $\mathbb{P}(\tau_\varrho^{(N)} > t) = \|P^t_{\varrho,A,N}(Q_\varrho x_N, \cdot) - \delta_0\|_{\mathrm{TV}}$ is plotted against $t - \bar{t}_N$. The dotted lines mark the lower and upper time bounds in Theorem 4.5, namely $\bar{t}_N - 1$ and $\bar{t}_N + 2$.

Metastability in the bistable rounded regime. When the rounded squared-radius mean map has a positive stable fixed point separated from the absorbing state by an unstable threshold (Proposition 5.1), a large finite-width rounded vector chain relaxes quickly toward the well and remains there for a long time before a rare fluctuation triggers absorption. Figure 8.6 shows the mean map, sample paths exhibiting the two time scales, and the growth of the absorption time with width. The numerical behavior is consistent with Theorem 5.2.

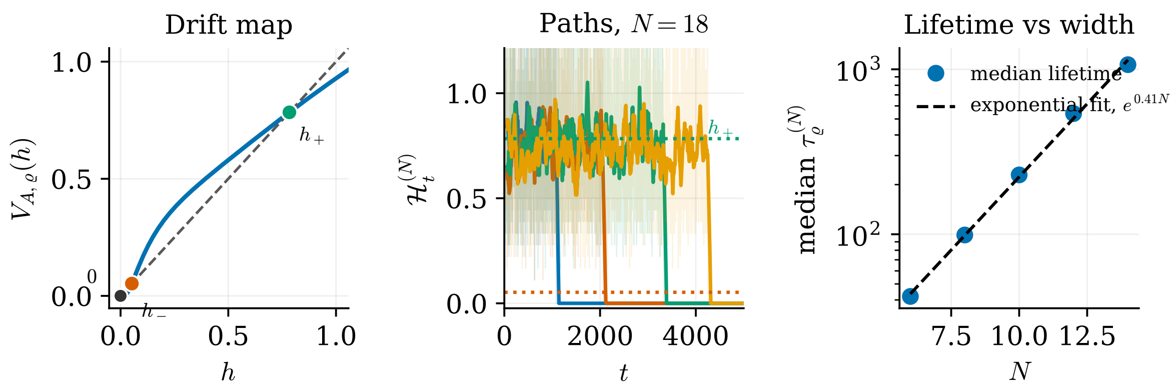

Figure 8.6: Metastability for $A = 1.15$ and $\varrho = 0.5$. Left: the squared-radius mean map $V_{A,\varrho}$ with unstable endpoint h_- and stable endpoint h_+. Middle: representative paths and moving averages with window 60. Right: median absorption times and an exponential fit; the observed growth is consistent with Theorem 5.2.

Critical curves. Figure 8.7 compares the fixed-width threshold $A_c(N)$ from (2.26), which decreases to 1, with the fixed-precision thresholds A_{lat} and $A_{\mathrm{ex}}(\varrho)$.

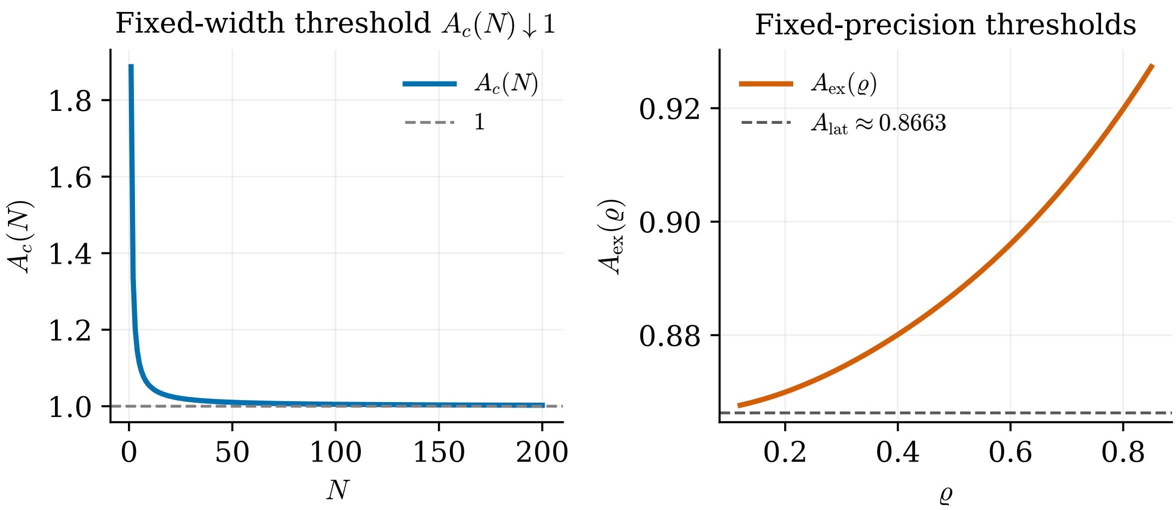

Figure 8.7: Threshold comparison. Left: the fixed-width critical gain $A_c(N)$ decreases to 1. Right: the positive-fixed-point existence threshold $A_{\mathrm{ex}}(\varrho)$, evaluated numerically from its variational definition, and the lattice-contraction threshold A_{lat}.

A. Common probabilistic estimates

This appendix proves Lemmas 2.2 and 2.3 and Propositions 2.5 and 2.6.

A.1. Stopped orbit tracking.

Proof of Lemma 2.2. Put $T := T_N$ and suppress the index N on $\mathcal{R}_t$, A_t, η_t, α_t, and σ_t throughout the proof.

Step 1: The killed moment. Fix $0 \leqslant t < T$. Since τ_N is a stopping time, $\{\tau_N > t\} \in \mathcal{F}_t$. Moreover, $\{\tau_N > t+1\} \subseteq \{\tau_N > t\}$, and on the latter event the recursion (2.33) applies. Consequently,

$$\mathcal{R}_{t+1}^2 \mathbb{1}_{\{\tau_N > t+1\}} \leqslant \mathbb{1}_{\{\tau_N > t\}} (A_t \mathcal{R}_t + \eta_{t+1})^2.$$

The variables $\mathbb{1}_{\{\tau_N > t\}}$, A_t, and $\mathcal{R}_t$ are $\mathcal{F}_t$-measurable. Thus the conditional cross term vanishes by (2.34), and

$$\mathbb{E}\big[\mathcal{R}_{t+1}^2 \mathbb{1}_{\{\tau_N > t+1\}} \,\big|\, \mathcal{F}_t\big] \leqslant \mathbb{1}_{\{\tau_N > t\}} \big(A_t^2 \mathcal{R}_t^2 + \mathbb{E}[\eta_{t+1}^2 \mid \mathcal{F}_t]\big) \leqslant \mathbb{1}_{\{\tau_N > t\}} \left(\alpha_t^2 \mathcal{R}_t^2 + \frac{C\sigma_t^2}{N}\right).$$

Set $m_t := \mathbb{E}[\mathcal{R}_t^2 \mathbb{1}_{\{\tau_N > t\}}]$. Taking expectations in the preceding display and using $\mathbb{P}(\tau_N > t) \leqslant 1$ gives

$$m_{t+1} \leqslant \alpha_t^2 m_t + \frac{C\sigma_t^2}{N}.$$

Iterating this deterministic recursion, and using $m_0 \leqslant \mathbb{E}\mathcal{R}_0^2$, yields

$$m_t \leqslant \left(\prod_{u=0}^{t-1} \alpha_u^2\right) \mathbb{E}\mathcal{R}_0^2 + \frac{C}{N} \sum_{s=0}^{t-1} \sigma_s^2 \prod_{u=s+1}^{t-1} \alpha_u^2.$$

This is (2.37).

Step 2: The stopped moment. Define the deterministic backward weights

$$W_t := \prod_{u=t}^{T-1} (1 \vee \alpha_u^2), \qquad 0 \leqslant t \leqslant T.$$

These weights satisfy

$$W_T = 1, \qquad W_{t+1} \leqslant W_t, \qquad \alpha_t^2 W_{t+1} \leqslant W_t.$$

In addition, the definition of K_N gives

$$W_0^{1/2} = \prod_{u=0}^{T-1} (1 \vee \alpha_u) \leqslant K_N, \qquad \sigma_t W_{t+1}^{1/2} = \sigma_t \prod_{u=t+1}^{T-1} (1 \vee \alpha_u) \leqslant K_N.$$

For the one-step weighted estimate, note that on $\{\tau_N \leqslant t\}$ the stopped process is constant, whereas on $\{\tau_N > t\}$ the recursion applies. Hence

$$\overline{\mathcal{R}}_{t+1,N} = \mathbb{1}_{\{\tau_N \leqslant t\}} \overline{\mathcal{R}}_{t,N} + \mathbb{1}_{\{\tau_N > t\}} (A_t \mathcal{R}_t + \eta_{t+1}).$$

The two events in this decomposition are disjoint. Taking conditional expectations, using (2.34), and applying the preceding bounds therefore gives

$$\begin{aligned}
&\mathbb{E}\Big[W_{t+1} \overline{\mathcal{R}}_{t+1,N}^2 \,\Big|\, \mathcal{F}_t\Big] \\
&\quad = \mathbb{1}_{\{\tau_N \leqslant t\}} W_{t+1} \overline{\mathcal{R}}_{t,N}^2 + \mathbb{1}_{\{\tau_N > t\}} W_{t+1} \big(A_t^2 \mathcal{R}_t^2 + \mathbb{E}[\eta_{t+1}^2 \mid \mathcal{F}_t]\big) \\
&\quad \leqslant \mathbb{1}_{\{\tau_N \leqslant t\}} W_t \overline{\mathcal{R}}_{t,N}^2 + \mathbb{1}_{\{\tau_N > t\}} W_t \mathcal{R}_t^2 + \frac{CK_N^2}{N} \\
&\quad = W_t \overline{\mathcal{R}}_{t,N}^2 + \frac{CK_N^2}{N}.
\end{aligned}$$

Taking expectations and using the tower property gives, for every $0 \leqslant t < T$,

$$\mathbb{E}\Big[W_{t+1}\overline{\mathcal{R}}_{t+1,N}^2\Big] - \mathbb{E}\Big[W_t\overline{\mathcal{R}}_{t,N}^2\Big] \leqslant \frac{CK_N^2}{N}.$$

Summing from $t = 0$ to $T - 1$, the left-hand side telescopes, and we obtain

$$\mathbb{E}\Big[W_T\overline{\mathcal{R}}_{T,N}^2\Big] - \mathbb{E}\Big[W_0\overline{\mathcal{R}}_{0,N}^2\Big] \leqslant \frac{CK_N^2 T}{N}.$$

Since $W_T = 1$, W_0 is deterministic, and $\overline{\mathcal{R}}_{0,N} = \mathcal{R}_{0,N}$, this is equivalent to

$$\mathbb{E}\overline{\mathcal{R}}_{T,N}^2 \leqslant W_0\mathbb{E}\mathcal{R}_{0,N}^2 + \frac{CK_N^2 T}{N}.$$

Since $W_0 \leqslant K_N^2$, this proves (2.36).

Step 3: Exit probability. Suppose that τ_N is the first time at which $|\mathcal{R}_{t,N}| > \delta$. On $\{\tau_N \leqslant T_N\}$ we have $T_N \wedge \tau_N = \tau_N$. Therefore, by the definition of the stopped process and the strict inequality in the definition of τ_N,

$$\overline{\mathcal{R}}_{T_N,N} = \mathcal{R}_{\tau_N,N}, \qquad |\mathcal{R}_{\tau_N,N}| > \delta.$$

Hence

$$\delta^2 \mathbb{1}_{\{\tau_N \leqslant T_N\}} \leqslant \overline{\mathcal{R}}_{T_N,N}^2.$$

Taking expectations and applying (2.36), we obtain

$$\begin{aligned}
\delta^2\mathbb{P}(\tau_N \leqslant T_N) &= \delta^2\mathbb{E}\mathbb{1}_{\{\tau_N \leqslant T_N\}} \\
&\leqslant \mathbb{E}\overline{\mathcal{R}}_{T_N,N}^2 \\
&\leqslant CK_N^2\left(\mathbb{E}\mathcal{R}_{0,N}^2 + \frac{T_N}{N}\right).
\end{aligned}$$

Dividing by δ^2 proves (2.38) and completes the proof. □

A.2. Negative-drift affine entrance.

Proof of Lemma 2.3. By our assumptions we can choose $\varepsilon > 0$ so small that $\mu_\varepsilon := \mathbb{E}\log(M_1+\varepsilon) < 0$. Take $K_* \geqslant b/\varepsilon$. Whenever $U_t \geqslant K_*$,

$$U_{t+1} = M_{t+1}U_t + b \leqslant M_{t+1}U_t + \varepsilon K_* \leqslant (M_{t+1} + \varepsilon)U_t.$$

Let $T := \inf\{t : U_t \leqslant K_*\}$. On the event $\{T > n\}$ we have $U_t > K_*$ for every $t \leqslant n$, so the previous bound applies at each step and

$$\log U_n \leqslant \log K + \widetilde{S}_n, \qquad \widetilde{S}_n := \sum_{j=1}^n \log(M_j + \varepsilon),$$

a random walk with increment mean $\mu_\varepsilon < 0$. Since $U_n > K_*$ on $\{T > n\}$, this forces $\widetilde{S}_n > -\log(K/K_*)$.

By hypothesis $\log(M_1+\varepsilon)$ has exponential moments near the origin, so $\Lambda(s) := \log\mathbb{E}(M_1+\varepsilon)^s$ is finite for s in a neighborhood of 0, with $\Lambda(0) = 0$ and $\Lambda'(0) = \mu_\varepsilon < 0$. Fix $s > 0$ small enough that $\Lambda(s) < 0$ and put $c_3 := -\Lambda(s) > 0$. Chernoff's inequality gives, for every integer $n \geqslant 1$,

$$\mathbb{P}(T > n) \leqslant \mathbb{P}\Big(\widetilde{S}_n > -\log(K/K_*)\Big) \leqslant \left(\frac{K}{K_*}\right)^s (\mathbb{E}(M_1+\varepsilon)^s)^n = \left(\frac{K}{K_*}\right)^s e^{-c_3 n}.$$

Since T is integer valued, $\mathbb{P}(T > c_1\log K + r) = \mathbb{P}(T > \lfloor c_1\log K + r\rfloor)$, and using $\lfloor c_1 \log K + r\rfloor > c_1\log K + r - 1$ with the choice $c_1 := s/c_3$,

$$\mathbb{P}(T > c_1\log K + r) \leqslant \left(\frac{K}{K_*}\right)^s e^{c_3}e^{-c_3(c_1\log K + r)} = K_*^{-s}e^{c_3}\,K^{s-c_3c_1}e^{-c_3 r} = K_*^{-s}e^{c_3}e^{-c_3 r},$$

because $s - c_3c_1 = 0$. This is (2.39) with $c_2 := \max\{K_*^{-s}e^{c_3}, 1\}$. □

A.3. Compactness and invariant-law selection.

Proof of Proposition 2.5. This is the Markov-kernel analogue of the classical Krylov–Bogolyubov averaging construction; compare [KB37]. The space $[0,1]$ is compact, and hence the set of probability measures on $[0,1]$ is weakly compact. The transition kernel is Feller: if ϕ is continuous and bounded on $[0,1]$, then

$$K_{A,N}\phi(q) = \int_{\mathbb{R}^N} \phi(F_{A,N}(q,g))\,\mathcal{G}_N(\mathrm{d}g)$$

is continuous in q, by the continuity of $F_{A,N}$ and dominated convergence.

Let $\overline{\nu}^q_{T_j} \Rightarrow \overline{\nu}$. For every continuous ϕ,

$$\overline{\nu}^q_{T_j} K_{A,N}\phi - \overline{\nu}^q_{T_j}\phi = \frac{1}{T_j}\Big(K^{T_j}_{A,N}\phi(q) - \phi(q)\Big).$$

The right-hand side tends to zero. Passing to the limit, using the Feller property for $K_{A,N}\phi$, gives $\overline{\nu}K_{A,N}\phi = \overline{\nu}\phi$ for every continuous ϕ. Thus $\overline{\nu}$ is invariant. The weak stationary equation (2.44) follows by identification with continuous functions since $[0,1]$ is compact.

Since $K_{A,N}(0,\cdot) = \delta_0$ and $K_{A,N}(s,\{0\}) = 0$ for $s > 0$ (all Gaussian coordinates would have to vanish), an invariant law $\overline{\nu} \neq \delta_0$ decomposes as

$$\overline{\nu} = \overline{\nu}(\{0\})\delta_0 + (1 - \overline{\nu}(\{0\}))\nu^+,$$

with ν^+ invariant on $(0,1]$. The assertion for the vector law is then exactly the invariant-law part of Proposition 2.4; the weak equation (2.48) is the corresponding stationary equation for the unrounded vector chain. $\square$

A.4. Nonzero invariant-law existence. Let $G_1,\dots,G_N$ be independent standard Gaussians and put

$$S_N := \frac{1}{N}\sum_{i=1}^N \min(G_i^2, 1).$$

If $0 < u < 1/N$, then $S_N \leqslant u$ implies $|G_i| \leqslant \sqrt{Nu}$ for every i. Hence

$$\mathbb{P}(S_N \leqslant u) \leqslant \prod_{i=1}^N \mathbb{P}(|G_i| \leqslant \sqrt{Nu}) \leqslant C_N u^{N/2}.$$

Consequently, $\mathbb{E}S_N^{-p} < \infty$ for every $p < N/2$.

Proof of Proposition 2.6. Since $A > A_c(N)$,

$$\frac{d}{dp}\mathbb{E}\left[\left(A^2\frac{\chi_N^2}{N}\right)^{-p}\right]_{p=0} = -\mathbb{E}\log\left(A^2\frac{\chi_N^2}{N}\right) = 2\gamma_{A,N} < 0,$$

where $\ell_{A,N}$ is as in (2.25), and $\gamma_{A,N} = -\mathbb{E}\log\ell_{A,N} < 0$ in the present supercritical regime. Therefore there exists $p \in (0, N/2)$ such that

$$\mathbb{E}\left[\left(A^2\frac{\chi_N^2}{N}\right)^{-p}\right] < 1. \tag{A.1}$$

We claim that, for some $R_0 \in (0,1)$ and $a < 1$,

$$\sup_{0<r\leqslant R_0} \mathbb{E}\left[\left(\frac{F_{A,N}(r,G)}{r}\right)^{-p}\right] \leqslant a. \tag{A.2}$$

Indeed, for fixed g the quotient $q^{-1}\tanh^2(A\sqrt{q}\,g)$ decreases in q and converges to A^2g^2 as $q \downarrow 0$. Therefore $F_{A,N}(q,G)/q$ converges almost surely to $A^2\chi_N^2/N$. Moreover, for each fixed $R > 0$ and $0 < q \leqslant R$, the same monotonicity gives

$$\frac{F_{A,N}(q,G)}{q} \geqslant \frac{F_{A,N}(R,G)}{R}.$$

The random variable on the right is bounded below by $c_{A,R,N}S_N$. The elementary fixed-N small-ball estimate above gives $\mathbb{E}S_N^{-p} < \infty$. Dominated convergence therefore gives the convergence of the expectations, and (A.2) follows from (A.1) by taking R_0 small.

On the complementary region there is $B < \infty$ such that

$$\sup_{r\in[R_0,1]} \mathbb{E}F_{A,N}(r,G)^{-p} \leqslant B. \tag{A.3}$$

This follows from the pointwise lower bound

$$\tanh^2(A\sqrt{r}\,g) \geqslant c_{A,R_0}\min(g^2,1), \qquad r \in [R_0,1],$$

together with the same elementary fixed-N small-ball estimate for S_N.

Let $V(r) := r^{-p}$ for $r > 0$. Combining the two estimates gives

$$K_{A,N}V(r) \leqslant aV(r) + B, \qquad r > 0. \tag{A.4}$$

Let the unrounded squared-radius chain start from $q \in (0,1]$. Iterating (A.4) gives

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}V(Z_t^{(N)}) \leqslant \frac{V(q)}{T(1-a)} + \frac{B}{1-a}.$$

Choose a weakly convergent Cesàro subsequence $\overline{\nu}_{T_j}^q \Rightarrow \overline{\nu}$. For $M < \infty$, the function $V^M(r) := \min\{r^{-p}, M\}$ for $r > 0$ and $V^M(0) := M$ is continuous on $[0,1]$. Passing to the limit gives a bound independent of M. Since $M\overline{\nu}(\{0\}) \leqslant \int V^M\,\mathrm{d}\overline{\nu}$, letting $M \to \infty$ first gives $\overline{\nu}(\{0\}) = 0$. Monotone convergence on $(0,1]$ then gives

$$\int_{(0,1]} r^{-p}\,\overline{\nu}(\mathrm{d}r) < \infty, \qquad \overline{\nu}(\{0\}) = 0.$$

Thus $\overline{\nu} \neq \delta_0$, and Proposition 2.5 shows that $\overline{\nu} \in \mathcal{I}_{A,N}^+(q)$. The vector invariant law follows from (2.47). $\square$

B. Technical estimates for the supercritical dimension cutoff

This appendix contains the estimates used in Section 6.3, with the notation introduced there for the unrounded squared-radius chain and its deterministic orbit.

B.1. Negative moments and Foster estimates.

Proof of Lemma 6.7. Let

$$U_i := \min(G_i^2, 1), \qquad S_N := \frac{1}{N}\sum_{i=1}^{N} U_i.$$

For $\lambda \geqslant 1$,

$$\mathbb{E}e^{-\lambda U_1} \leqslant \int_{-1}^{1} e^{-\lambda x^2}\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x + e^{-\lambda} \leqslant C\lambda^{-1/2}.$$

Chernoff's bound with $\lambda = u^{-1}$ gives

$$\mathbb{P}(S_N \leqslant u) \leqslant e^N \left(\mathbb{E}e^{-U_1/u}\right)^N \leqslant C^N u^{N/2},$$

for $0 < u \leqslant 1$.

For $x, p > 0$, the Gamma integral and the change of variables $v = ux$ give

$$x^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty u^{p-1} e^{-ux} \, \mathrm{d}u.$$

Let $p = \alpha N$ with $\alpha \in (0, 1/2)$. By Tonelli's theorem and independence,

$$\mathbb{E}S_N^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty u^{p-1} \left(\mathbb{E}e^{-(u/N)U_1}\right)^N \mathrm{d}u.$$

We split the integral at $u = N$. On $[0, N]$ we use the trivial bound $\mathbb{E}e^{-(u/N)U_1} \leqslant 1$. On $[N, \infty)$ we use the preceding one-coordinate Laplace bound with $\lambda = u/N \geqslant 1$. Since $p = \alpha N < N/2$, this gives

$$\mathbb{E}S_N^{-p} \leqslant \frac{N^p}{\Gamma(p+1)} + \frac{C^N N^{N/2}}{\Gamma(p)} \int_N^\infty u^{p-1-N/2} \, \mathrm{d}u$$
$$\leqslant C_\alpha^N \frac{N^p}{\Gamma(p+1)}.$$

Stirling's formula, after increasing the constant to handle bounded N, shows that the last expression is bounded by $e^{C_\alpha N}$. If $W_{L,i} := \min(G_i^2, L^2)$ and $L \geqslant 1$, then $W_{L,i} \geqslant U_i$. Consequently,

$$\sup_{L \geqslant 1} \mathbb{E}\left[\left(\frac{1}{N}\sum_{i=1}^N W_{L,i}\right)^{-\alpha N}\right] \leqslant e^{C_\alpha N}. \tag{B.1}$$

Put $\mu_L := \mathbb{E}W_{L,1}$. Choose first L large, then $\varepsilon > 0$ small enough that $(1-\varepsilon)A^2\mu_L > 1$, and finally $\delta > 0$ such that

$$a := (1-\varepsilon)A^2(\mu_L - \delta) > 1.$$

If $\overline{W}_{L,N} := N^{-1}\sum_i W_{L,i}$, Hoeffding's inequality gives

$$\mathbb{P}(\overline{W}_{L,N} < \mu_L - \delta) \leqslant e^{-c_0 N}$$

with $c_0 > 0$ depending on L and δ.

Fix $\alpha \in (0, 1/2)$. By (B.1), after increasing C_α to absorb the fixed factor $((1-\varepsilon)A^2)^{-\alpha N}$,

$$\mathbb{E}\left[\left((1-\varepsilon)A^2\overline{W}_{L,N}\right)^{-\alpha N}\right] \leqslant e^{C_\alpha N}.$$

On the event $\{\overline{W}_{L,N} \geqslant \mu_L - \delta\}$, the integrand in (6.7) is at most $a^{-\gamma N}$. On the complementary event, Holder's inequality gives, for $0 < \gamma < \alpha$,

$$\mathbb{E}\left[\left((1-\varepsilon)A^2\overline{W}_{L,N}\right)^{-\gamma N} \mathbb{1}_{\{\overline{W}_{L,N} < \mu_L - \delta\}}\right] \leqslant e^{-c_0 N(1-\gamma/\alpha)} e^{C_\alpha N\gamma/\alpha}.$$

Choose $\gamma > 0$ so small that $\gamma < \alpha c_0/(c_0 + C_\alpha)$. Then choose $\kappa > 0$ such that

$$\kappa < \min\{\gamma \log a, c_0(1-\gamma/\alpha) - C_\alpha\gamma/\alpha\}.$$

The good- and bad-event estimates then give (6.7). $\square$

Proof of Proposition 6.8. Let $\gamma, \kappa, L, \varepsilon$ be as in Lemma 6.7, decreasing κ if necessary. For $q > 0$ define

$$h_q(g) := \frac{\tanh^2(A\sqrt{q}\,g)}{q} = A^2 g^2 \left(\frac{\tanh(A\sqrt{q}\,g)}{A\sqrt{q}\,g}\right)^2,$$

with the continuous interpretation at $g = 0$. Since $x \mapsto \tanh x/x$ is decreasing on $(0, \infty)$, the map $q \mapsto h_q(g)$ is decreasing for each fixed g.

Choose $R_0 \in (0, \inf I_*)$ so small that

$$\left(\frac{\tanh(A\sqrt{R_0}u)}{A\sqrt{R_0}u}\right)^2 \geqslant 1 - \varepsilon \qquad \text{for every } |u| \leqslant L. \tag{B.2}$$

For $|g| \leqslant L$, (B.2) gives

$$h_{R_0}(g) \geqslant (1 - \varepsilon)A^2 g^2.$$

For $|g| > L$, the map $|g| \mapsto h_{R_0}(g) = R_0^{-1}\tanh^2(A\sqrt{R_0}g)$ is increasing, and hence

$$h_{R_0}(g) \geqslant h_{R_0}(L) \geqslant (1 - \varepsilon)A^2 L^2.$$

Therefore

$$h_{R_0}(g) \geqslant (1 - \varepsilon)A^2 \min(g^2, L^2) \qquad \text{for every } g \in \mathbb{R}. \tag{B.3}$$

Hence, for $0 < q \leqslant R_0$,

$$\frac{F_{A,N}(q, G)}{q} = \frac{1}{N}\sum_{i=1}^{N} h_q(G_i) \geqslant \frac{1}{N}\sum_{i=1}^{N} h_{R_0}(G_i) \geqslant (1 - \varepsilon)A^2 \frac{1}{N}\sum_{i=1}^{N} \min(G_i^2, L^2).$$

Since $x \mapsto x^{-\gamma N}$ is decreasing on $(0, \infty)$, applying Lemma 6.7 gives

$$\sup_{0<q\leqslant R_0} \mathbb{E}\left[\left(\frac{F_{A,N}(q, G)}{q}\right)^{-\gamma N}\right] \leqslant e^{-\kappa N}$$

for all sufficiently large N. □

Proof of Lemma 6.9. There is $c_{A,R_0} > 0$ such that, for every $q \in [R_0, 1]$ and every $g \in \mathbb{R}$,

$$\tanh^2(A\sqrt{q}\,g) \geqslant c_{A,R_0} \min(g^2, 1). \tag{B.4}$$

Indeed, for $|g| \leqslant 1$ this follows from the monotonicity of $\tanh x/x$ on $[0, A]$, while for $|g| \geqslant 1$ it follows from $\tanh^2(A\sqrt{q}\,|g|) \geqslant \tanh^2(A\sqrt{R_0})$. Thus

$$F_{A,N}(q, G) \geqslant c_{A,R_0} \frac{1}{N}\sum_{i=1}^{N} \min(G_i^2, 1).$$

The bound follows from (B.1), after absorbing $c_{A,R_0}^{-\gamma N}$ into e^{bN}. □

Proof of Proposition 6.11. Let R_0, γ, κ be as in Proposition 6.8, and let b be the constant in Lemma 6.9 for this choice of R_0, γ. Define

$$V_N(q) := q^{-\gamma N}, \qquad q > 0.$$

For $q > 0$, the two negative-moment estimates give the Foster inequality

$$K_{A,N} V_N(q) \leqslant e^{-\kappa N} V_N(q) + e^{bN}. \tag{B.5}$$

Indeed, if $q \leqslant R_0$ then

$$K_{A,N}V_N(q) = q^{-\gamma N}\mathbb{E}\Big[(F_{A,N}(q,G)/q)^{-\gamma N}\Big] \leqslant e^{-\kappa N}V_N(q),$$

while for $q \in [R_0, 1]$ the outside estimate gives $K_{A,N}V_N(q) \leqslant e^{bN}$.

Let $(Z_t^{(N)})_{t\geqslant 0}$ be the unrounded squared-radius chain started from 1. Iterating (B.5) yields

$$\mathbb{E}V_N(Z_t^{(N)}) \leqslant e^{-\kappa N t} + \frac{e^{bN}}{1-e^{-\kappa N}}. \tag{B.6}$$

Hence

$$\int V_N(q)\,\overline{\nu}_T^1(\mathrm{d}q) \leqslant \frac{1}{T(1-e^{-\kappa N})} + \frac{e^{bN}}{1-e^{-\kappa N}}. \tag{B.7}$$

Here $\overline{\nu}_T^1$ is the Cesàro average started from $q = 1$ in (2.46).

Take a weakly convergent subsequence $\overline{\nu}_{T_j}^1 \Rightarrow \overline{\nu}$ with $T_j \to \infty$. For $M < \infty$ put $V_N^M(q) := \min\{V_N(q), M\}$ for $q > 0$ and $V_N^M(0) := M$. Then V_N^M is continuous on $[0,1]$. Applying (B.7) with $T = T_j$ and then letting $j \to \infty$, the term containing T_j^{-1} vanishes and

$$\int V_N^M(q)\,\overline{\nu}(\mathrm{d}q) \leqslant \frac{e^{bN}}{1-e^{-\kappa N}}.$$

Letting $M \to \infty$ and using monotone convergence, we obtain $\int_{(0,1]} q^{-\gamma N}\,\overline{\nu}(\mathrm{d}q) < \infty$ and $\overline{\nu}(\{0\}) = 0$. By Proposition 2.5, $\overline{\nu}$ is a nonzero invariant law. By Proposition 6.1, $\overline{\nu} = \nu_{A,N}$. The same monotone-convergence passage gives (6.15). ☐

B.2. Localization, total variation, and concentration.

Proof of Proposition 6.12. Let R_0, γ be as in Proposition 6.8, and let b be the constant from Lemma 6.9 for this choice of R_0, γ. Set

$$V_N(q) := q^{-\gamma N}, \qquad q > 0.$$

By Proposition 6.11, after increasing C if necessary,

$$\int_0^1 V_N(q)\,\nu_{A,N}(\mathrm{d}q) \leqslant Ce^{bN}. \tag{B.8}$$

Now choose a second threshold $r \in (0, R_0)$ so small that

$$b + \gamma \log r < 0. \tag{B.9}$$

Since $V_N(q) \geqslant r^{-\gamma N}$ on $(0, r]$,

$$\nu_{A,N}((0,r]) \leqslant r^{\gamma N}\int_0^1 V_N(q)\,\nu_{A,N}(\mathrm{d}q) \leqslant Ce^{-cN}. \tag{B.10}$$

For the fixed-time entrance from $[r,1]$ into I_*, note that by Lemma 2.1, every unrounded deterministic orbit started from $[r,1]$ converges monotonically to q_* without crossing it. Hence

$$\big|V_A^{t+1}(q) - q_*\big| \leqslant \big|V_A^t(q) - q_*\big|, \qquad q \in [r,1].$$

Dini's theorem shows that this convergence is uniform on $[r,1]$. Therefore there are $m \in \mathbb{N}$ and $\eta > 0$ such that

$$V_A^m([r,1]) \subset I_{*,-\eta} := \{q \in I_* : \mathrm{dist}(q, I_*^c) \geqslant \eta\}. \tag{B.11}$$

Let $L < \infty$ be a Lipschitz constant for V_A on $[0,1]$ and set

$$\eta_m := \frac{\eta}{2(1 + L + \cdots + L^{m-1})}.$$

If the noise variables in (2.14) have absolute value at most η_m during the first m steps, then the stochastic radius at time m is within $\eta/2$ of its deterministic counterpart and hence belongs to I_*. Hoeffding's inequality (2.16) and a union bound over these fixed m steps give

$$\sup_{q \in [r,1]} K^m_{A,N}(q, I_*^c) \leqslant Ce^{-cN}. \tag{B.12}$$

Since $\nu_{A,N}$ is invariant,

$$\nu_{A,N}(I_*^c) = \int_0^1 K^m_{A,N}(q, I_*^c)\, \nu_{A,N}(\mathrm{d}q) \leqslant \nu_{A,N}((0,r]) + \sup_{q\in[r,1]} K^m_{A,N}(q, I_*^c).$$

Combining (B.10) and (B.12) proves (6.16), and the weaker estimate (6.17) follows immediately. $\square$

Proof of Lemma 6.13. Let μ_q be the law of $Y_q := \tanh^2(A\sqrt{q}\,G)$ on $(0,1)$, and let f_q be the density in (6.1). Its q-score is

$$\ell_q(y) := \partial_q \log f_q(y) = -\frac{1}{2q} + \frac{\operatorname{arctanh}(\sqrt{y})^2}{2A^2q^2}. \tag{B.13}$$

Since $\operatorname{arctanh}(\sqrt{Y_q}) = \left|A\sqrt{q}\,G\right|$, we have

$$\ell_q(Y_q) = \frac{G^2 - 1}{2q}, \qquad \mathbb{E}\ell_q(Y_q) = 0, \qquad \mathbb{E}\ell_q(Y_q)^2 = \frac{1}{2q^2}. \tag{B.14}$$

Fix a compact subinterval $I = [q_-, q_+] \Subset (0,1)$. On this interval the displayed density is a C^1 map into $L^1((0,1))$: after writing $u = \operatorname{arctanh}(\sqrt{y})$, the substitution $y = \tanh^2 u$ gives a uniform integrable Gaussian envelope for $\partial_q f_q = \ell_q f_q$. Thus $q \mapsto \mu_q$ is differentiable in total variation on I, with signed derivative measure $\partial_q \mu_q = \ell_q \mu_q$. Here and below $|\lambda|$ denotes the total variation measure of a signed measure λ; with our convention for total variation distance between probability measures, one divides the signed variation norm by two.

Let $Y_{1,q}, \ldots, Y_{N,q}$ be independent copies of Y_q and set

$$M_{N,q} := \frac{1}{N}\sum_{i=1}^N Y_{i,q}, \qquad S_{N,q} := \sum_{i=1}^N \ell_q(Y_{i,q}).$$

The product family is therefore differentiable in total variation by the tensorized score calculation:

$$\partial_q \mu_q^{\otimes N} = S_{N,q}\, \mu_q^{\otimes N}; \tag{B.15}$$

equivalently, for every bounded measurable $\varphi : (0,1)^N \to \mathbb{R}$,

$$\partial_q \mathbb{E}\varphi(Y_{1,q}, \ldots, Y_{N,q}) = \mathbb{E}[\varphi(Y_{1,q}, \ldots, Y_{N,q}) S_{N,q}]. \tag{B.16}$$

Since $S_{N,q}$ is the density of $\partial_q \mu_q^{\otimes N}$ against $\mu_q^{\otimes N}$, its total variation norm is

$$\left|\partial_q \mu_q^{\otimes N}\right|((0,1)^N) = \mathbb{E}|S_{N,q}| \leqslant \left(\mathbb{E}S_{N,q}^2\right)^{1/2} = \frac{\sqrt{N}}{\sqrt{2}\,q}, \tag{B.17}$$

where the last identity uses (B.14) and the independence of the $Y_{i,q}$.

The squared-radius transition kernel $K_{A,N}(q,\cdot)$ is the law of $M_{N,q}$, that is, the pushforward of $\mu_q^{\otimes N}$ under the averaging map $(y_1,\dots,y_N)\mapsto N^{-1}\sum_i y_i$; pushforward does not increase the total variation distance of probability measures. Since $q\mapsto\mu_q^{\otimes N}$ is continuously differentiable in total variation, $\mu_q^{\otimes N}-\mu_{q'}^{\otimes N}=\int_{q'}^{q}\partial_s\mu_s^{\otimes N}\,\mathrm{d}s$ as signed measures, so integrating (B.17) and halving,

$$\|K_{A,N}(q,\cdot)-K_{A,N}(q',\cdot)\|_{\mathrm{TV}}\leqslant\tfrac12\Big|\mu_q^{\otimes N}-\mu_{q'}^{\otimes N}\Big|((0,1)^N)\leqslant\frac12\int_{q\wedge q'}^{q\vee q'}\frac{\sqrt{N}}{\sqrt{2}\,s}\,\mathrm{d}s\leqslant\frac{\sqrt{N}}{2\sqrt{2}\,q_-}|q-q'|.$$

Since $q,q'\in I\Subset(0,1)$, this proves (6.18) with $C_I=(2\sqrt{2}\,q_-)^{-1}$. $\square$

Proof of Proposition 6.14. Invariance of $\nu_{A,N}$ and (2.15) give

$$\begin{aligned}\int_0^1(q-q_*)^2\,\nu_{A,N}(\mathrm{d}q)&=\int_0^1\mathbb{E}\Big[\big(F_{A,N}(q,G)-q_*\big)^2\Big]\nu_{A,N}(\mathrm{d}q)\\&=\int_0^1(V_A(q)-q_*)^2\nu_{A,N}(\mathrm{d}q)+O(N^{-1}).\end{aligned}\tag{B.18}$$

On I_*, (6.6) gives $|V_A(q)-q_*|\leqslant\kappa|q-q_*|$. On I_*^c, we use the trivial bound $|V_A(q)-q_*|\leqslant 1$ and (6.17). Hence, if

$$V_N:=\int_0^1(q-q_*)^2\,\nu_{A,N}(\mathrm{d}q),$$

then $V_N\leqslant\kappa^2V_N+\frac{C}{N}$. Since $\kappa<1$, this gives $V_N\leqslant\frac{C}{(1-\kappa^2)N}$. The claim follows after enlarging C. $\square$

Proof of Lemma 6.15. Step 1: the finite-time entrance. Put $e_t^{(N)}:=Z_t^{(N)}-q_{t,N}$. Since the comparison orbit starts from the same finite-N radius $q_{0,N}$, one has $e_0^{(N)}=0$. Before s_*, no stability is needed: V_A is Lipschitz on $[0,1]$, and (2.15) and (2.16) give, after finitely many iterations,

$$\mathbb{E}[(e_{s_*}^{(N)})^2]\leqslant\frac{C}{N},\qquad\mathbb{P}\Big(\Big|e_{s_*}^{(N)}\Big|>\eta\Big)\leqslant C\exp(-c_\eta N)\tag{B.19}$$

for every fixed $\eta>0$.

Step 2: The stopped second moment. Use the one-step noise $\zeta_{t+1}^{(N)}$ and natural filtration $\mathcal{F}_t$ from (2.14) and (2.15). Then

$$e_{t+1}^{(N)}=Z_{t+1}^{(N)}-q_{t+1,N}=V_A(Z_t^{(N)})-V_A(q_{t,N})+\zeta_{t+1}^{(N)}.$$

Moreover,

$$\mathbb{E}[\zeta_{t+1}^{(N)}\mid\mathcal{F}_t]=0,\qquad\mathbb{E}[(\zeta_{t+1}^{(N)})^2\mid\mathcal{F}_t]\leqslant\frac{C}{N}.$$

On $\{t<\tau_*^{(N)}\}$, both $Z_t^{(N)}$ and $q_{t,N}$ lie in the stable interval I_*. Hence the mean value theorem and (6.6) give

$$\Big|V_A(Z_t^{(N)})-V_A(q_{t,N})\Big|\leqslant\kappa\Big|e_t^{(N)}\Big|.$$

Therefore, using the preceding decomposition and the martingale property of $\zeta_{t+1}^{(N)}$,

$$\mathbb{E}\Big[(e_{t+1}^{(N)})^2\,\Big|\,\mathcal{F}_t\Big]=\Big(V_A(Z_t^{(N)})-V_A(q_{t,N})\Big)^2+\mathbb{E}\Big[(\zeta_{t+1}^{(N)})^2\,\Big|\,\mathcal{F}_t\Big]\leqslant\kappa^2(e_t^{(N)})^2+\frac{C}{N}$$

on $\{t<\tau_*^{(N)}\}$. Apply (2.37) to the time-shifted process $(e_{s_*+j}^{(N)})_{j\geqslant0}$, with $\alpha_{j,N}=\kappa$ and $\sigma_{j,N}=C$. The geometric sum in that estimate is bounded by $(1-\kappa^2)^{-1}$, while (B.19) supplies the $O(N^{-1})$ initial term. Hence

$$\mathbb{E}\Big[(e_t^{(N)})^2\mathbb{1}_{\{\tau_*^{(N)}>t\}}\Big]\leqslant\frac{C}{N}.\tag{B.20}$$

Step 3: No exit and removal of stopping. Choose $\eta > 0$ so small that

$$\kappa\delta_* + \eta < \delta_*.$$

If

$$\left|e_{s_*}^{(N)}\right| \leqslant \eta \qquad \text{and} \qquad \left|\zeta_j^{(N)}\right| \leqslant \eta \quad \text{for every } s_* < j \leqslant T_N,$$

then the recursion

$$\left|e_{t+1}^{(N)}\right| \leqslant \kappa\left|e_t^{(N)}\right| + \left|\zeta_{t+1}^{(N)}\right|$$

and the margin in (6.5) imply $Z_t^{(N)} \in I_*$ for every $s_* \leqslant t \leqslant T_N$. Hence (2.16) and (B.19) and a union bound give

$$\mathbb{P}(\tau_*^{(N)} \leqslant T_N) \leqslant CT_N e^{-c_0 N}.$$

Since $\left|e_t^{(N)}\right| \leqslant 1$, splitting according to whether the squared-radius chain has exited before time t gives

$$\mathbb{E}[(e_t^{(N)})^2] = \mathbb{E}\Big[(e_t^{(N)})^2 \mathbb{1}_{\{\tau_*^{(N)} > t\}}\Big] + \mathbb{E}\Big[(e_t^{(N)})^2 \mathbb{1}_{\{\tau_*^{(N)} \leqslant t\}}\Big] \leqslant \mathbb{E}\Big[(e_t^{(N)})^2 \mathbb{1}_{\{\tau_*^{(N)} > t\}}\Big] + \mathbb{P}(\tau_*^{(N)} \leqslant t).$$

Combining this with (6.20) and (B.20) and $T_N = O(\log N)$ gives, uniformly for $t \leqslant T_N$,

$$\mathbb{E}[(e_t^{(N)})^2] \leqslant \frac{C}{N} + C(\log N)e^{-c_0 N}.$$

Multiplying by N and increasing C for finitely many small values of N proves (6.21). $\square$

B.3. Contraction at the cutoff scale.

Proof of Lemma 6.16. Step 1: Setup. Take C_0 as in (6.22) and fix $b > 0$. All constants below are independent of c; the lower bound on N and the $o(1)$ terms may depend on its fixed value, which is harmless because the limit in N is taken first. Fix $c \geqslant 0$. By (6.23),

$$n_N(c) \leqslant C_0 \log N$$

for all large N. The estimates from Lemma 6.15 will be used only up to this time. For all sufficiently large N, put

$$\ell_N := \max\{0, \lfloor b \log\log N \rfloor\}, \qquad s_N := \max\{0, n_N(c) - 1 - \ell_N\}.$$

Since $n_N(c) - 1 - \ell_N > 0$ eventually, $s_N + \ell_N = n_N(c) - 1$ for all large N, and $\ell_N = o(\log N)$. Because $n_N(c)$ is of order $\log N$ while $\ell_N = o(\log N)$, we have $s_N \to \infty$. Hence, after increasing N if necessary,

$$s_N \geqslant s_*, \qquad 0 \leqslant s_N \leqslant s_N + \ell_N = n_N(c) - 1 \leqslant C_0 \log N.$$

The orbit estimates from Lemma 6.15 therefore apply at every time used in the last block.

Step 2: Block-starting distance. By the deterministic asymptotic in (2.32), together with the definition of $t_N(q_0)$ and the bounded integer-time rounding in $n_N(c)$,

$$|q_{s_N,N} - q_*| \leqslant CN^{-1/2}\mu_A^{c-\ell_N} \tag{B.21}$$

for all large N, with C independent of c. Indeed,

$$s_N - \big(t_N(q_0) + c - \ell_N\big) \in [-2, -1],$$

so the rounding contributes at most the fixed factor μ_A^{-2}; only the lower bound on N in the locally uniform deterministic asymptotic may depend on c. Since $\widetilde{Z}_0^{(N)} \sim \nu_{A,N}$ and the law $\nu_{A,N}$

is invariant, also $\widetilde{Z}^{(N)}_{s_N} \sim \nu_{A,N}$. Therefore, by the triangle inequality, Cauchy's inequality, (6.21), and (6.19),

$$\mathbb{E}\Big|Z^{(N)}_{s_N} - \widetilde{Z}^{(N)}_{s_N}\Big| \leqslant \mathbb{E}\Big|Z^{(N)}_{s_N} - q_{s_N,N}\Big| + |q_{s_N,N} - q_*| + \mathbb{E}\Big|\widetilde{Z}^{(N)}_{s_N} - q_*\Big|$$
$$\leqslant CN^{-1/2} + CN^{-1/2}\mu_A^{c-\ell_N} + CN^{-1/2}.$$

Hence

$$\mathbb{E}\Big|Z^{(N)}_{s_N} - \widetilde{Z}^{(N)}_{s_N}\Big| \leqslant CN^{-1/2}\Big(\mu_A^{c-\ell_N} + 1\Big). \tag{B.22}$$

Step 3: Localizing the Block. During the last ℓ_N steps, both squared-radius chains remain in a small neighborhood of q_* with high probability. This localization is used only for the local Taylor contraction; the sharper size estimate is (B.22). Let

$$\rho_N := (\log N)^{-2}, \qquad J_N := [q_* - \rho_N, q_* + \rho_N] \cap [0,1].$$

For every $0 \leqslant j \leqslant \ell_N$, the unrounded deterministic orbit remains closer to q_* as time increases. Hence (B.21) implies

$$q_{s_N+j,N} \in [q_* - \rho_N/2, q_* + \rho_N/2] \tag{B.23}$$

for all large N, because $N^{-1/2}\mu_A^{c-\ell_N} = o(\rho_N)$. Chebyshev's inequality and (6.21) therefore give

$$\mathbb{P}\Big(Z^{(N)}_{s_N+j} \notin J_N\Big) \leqslant \mathbb{P}\Big(\Big|Z^{(N)}_{s_N+j} - q_{s_N+j,N}\Big| > \rho_N/2\Big) \leqslant \frac{C}{N\rho_N^2}.$$

Similarly, stationarity of $\widetilde{Z}^{(N)}$ and (6.19) imply

$$\mathbb{P}\Big(\widetilde{Z}^{(N)}_{s_N+j} \notin J_N\Big) \leqslant \frac{C}{N\rho_N^2}.$$

Thus, with

$$E_j := \Big\{Z^{(N)}_{s_N+j} \in J_N,\, \widetilde{Z}^{(N)}_{s_N+j} \in J_N\Big\},$$

we have

$$\mathbb{P}(E_j^c) \leqslant \frac{C}{N\rho_N^2}. \tag{B.24}$$

Step 4: Local contraction. On J_N the deterministic map has derivative close to its derivative at q_*. Since $V_A'(q_*) = \mu_A$ and V_A'' is bounded in a neighborhood of q_*, the mean value theorem gives, for all large N and all $q, q' \in J_N$,

$$\big|V_A(q) - V_A(q')\big| \leqslant (\mu_A + C\rho_N)\big|q - q'\big|. \tag{B.25}$$

We choose N large enough that $a_N := \mu_A + C\rho_N < 1$.

Let $D_j := \mathbb{E}|Z^{(N)}_{s_N+j} - \widetilde{Z}^{(N)}_{s_N+j}|$. *Synchronous recursion.* Conditional on the two radii at time $s_N + j$, the next step uses the same Gaussian vector in both squared-radius chains. For fixed $g \in \mathbb{R}^N$, the map $q \mapsto F_{A,N}(q,g)$ is nondecreasing. Indeed, for $q > 0$,

$$\partial_q \tanh^2(A\sqrt{q}\, g_i) = \frac{Ag_i}{\sqrt{q}} \tanh(A\sqrt{q}\, g_i)\, \mathrm{sech}^2(A\sqrt{q}\, g_i) \geqslant 0,$$

and the endpoint $q = 0$ follows by continuity. Monotonicity therefore removes the absolute value inside the expectation, and for every $q, q' \in [0,1]$,

$$\mathbb{E}\big|F_{A,N}(q,G) - F_{A,N}(q',G)\big| = \big|V_A(q) - V_A(q')\big|.$$

Applying this identity conditionally gives

$$\mathbb{E}\Big[\Big|Z^{(N)}_{s_N+j+1}-\widetilde{Z}^{(N)}_{s_N+j+1}\Big|\;\Big|\;Z^{(N)}_{s_N+j},\widetilde{Z}^{(N)}_{s_N+j}\Big]=\Big|V_A(Z^{(N)}_{s_N+j})-V_A(\widetilde{Z}^{(N)}_{s_N+j})\Big|.$$

On E_j, (B.25) bounds this by $a_N\Big|Z^{(N)}_{s_N+j}-\widetilde{Z}^{(N)}_{s_N+j}\Big|$. On E_j^c we use only the trivial bound by one. Taking expectations and using (B.24) gives

$$D_{j+1}\leqslant a_N D_j+\frac{C}{N\rho_N^2}\qquad 0\leqslant j<\ell_N. \tag{B.26}$$

Iterating the recursion and using $a_N<1$ gives

$$D_{\ell_N}\leqslant a_N^{\ell_N}D_0+\frac{C}{N\rho_N^2}\sum_{k=0}^{\ell_N-1}a_N^k\leqslant a_N^{\ell_N}D_0+\frac{C}{N\rho_N^2}.$$

Since $\rho_N=(\log N)^{-2}$,

$$\frac{1}{N\rho_N^2}=\frac{(\log N)^4}{N}=o(N^{-1/2}).$$

Hence

$$D_{\ell_N}\leqslant a_N^{\ell_N}D_0+o(N^{-1/2}). \tag{B.27}$$

Step 5: Terminal estimate. The block length recovers the factor $\mu_A^{\ell_N}$, while the perturbation $C\rho_N$ has no asymptotic effect. Indeed,

$$a_N^{\ell_N}=\mu_A^{\ell_N}\left(1+\frac{C\rho_N}{\mu_A}\right)^{\ell_N},$$

and $\ell_N\rho_N\to 0$. Therefore

$$a_N^{\ell_N}=\mu_A^{\ell_N}(1+o(1)). \tag{B.28}$$

Combining (B.22), (B.27) and (B.28) gives

$$\sqrt{N}\,D_{\ell_N}\leqslant C\mu_A^{\ell_N}\left(\mu_A^{c-\ell_N}+1\right)+o(1)\leqslant C\mu_A^c+C\mu_A^{\ell_N}+o(1)=C\mu_A^c+o(1).$$

Here $\mu_A^{\ell_N}\to 0$ because $\mu_A\in(0,1)$ and $\ell_N\to\infty$. The constants in the preceding estimate are independent of c, while the $o(1)$ term may depend on its fixed value. Since D_{ℓ_N} is the expectation in (6.24), the claimed bound follows. □

Acknowledgements

The author used OpenAI's ChatGPT 5.5 and Anthropic's Claude Opus 4.8 for drafting, editing, and producing the accompanying Lean development through agentic coding harnesses. The author supervised this work, reviewed the resulting manuscript and formalization, and takes responsibility for their contents.

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